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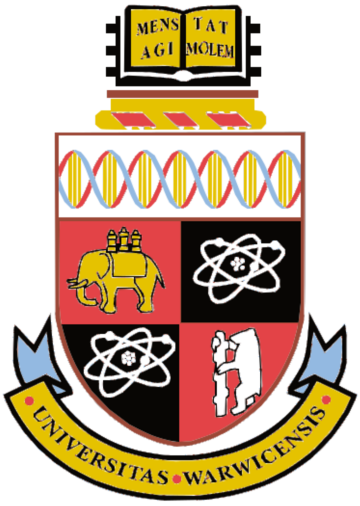
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L^p convergence of eigenfunction expansions for the Laplacian on planar domains

by

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Thesis

To be submitted to the University of Warwick

in partial fulfilment of the requirements for the degree of

Doctor of Philosophy in Mathematics

Warwick Mathematics Institute

July 2024

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Contents

List of figures iii

Acknowledgments iv

Declarations vi

Abstract vii

Chapter 1 Introduction 1

Notation . . . 9

Chapter 2 Preliminaries of Fourier analysis on the torus 11

2.1 Recap of 1-dimensional Fourier analysis . . . 11

2.1.1 Fourier series on the circle . . . 11

2.1.2 Summation of Fourier series . . . 14

2.1.3 L^p convergence . . . 15

2.1.4 Hausdorff–Young inequalities . . . 16

2.2 Fourier series on the n-torus . . . 25

2.2.1 Appendix: Cesàro summation and Bochner–Riesz means . . . 29

2.3 Decay of Fourier coefficients . . . 30

Chapter 3 Triangular domains 32

3.1 The 45-90-45 triangle . . . 33

3.2 Antisymmetric eigenfunctions on the equilateral triangle . . . 37

3.3 L^p convergence of the Fourier series of f_a . . . 40

3.4 Symmetric eigenfunctions on the equilateral triangle . . . 43

3.5 L^p convergence of the Fourier series of f_s . . . 45

3.6 Main results for the equilateral triangle . . . 47

3.7 Concluding remarks . . . 48

3.8 Appendix: Non-convergence for spherical cutoffs . . . 49

Chapter 4 Bessel–Fourier series on a disc **54**
4.1 The Poisson problem and Bessel–Fourier series 56
4.2 Convergence in L^2 58
4.3 Asymptotics for Bessel functions and their zeros 59
4.4 L^p convergence of 1-dimensional Bessel–Fourier series 62
4.5 Mixed norm convergence of the 2-dimensional Bessel–Fourier series . 64
4.6 Concluding remarks 68

Chapter 5 Decay of Bessel–Fourier coefficients **69**
5.1 Two useful identities 74
5.2 'Strong' decay theorems 75
5.3 'Weak' decay theorems 80
5.4 Bounds on $x^{\alpha+1}K_M^\alpha(x)$ 82
5.5 Comments for further research 88

Chapter 6 Domain perturbation **91**
6.1 Preliminaries 92
6.1.1 The inverse of the Laplacian on L^p 95
6.2 Continuity and splitting of eigenvalues 98
6.3 Bounds on $T - T_\varphi$ 100
6.4 Resolvent bounds 106
6.5 Continuity of projections under domain perturbations 109
6.6 Conclusions and open problems 114

Chapter 7 Conclusion **116**

Appendix A Translations of the epigraphs **118**

Bibliography **120**

List of figures

1.1 Truncation of Fourier series 2

3.1 45-90-45 to square transformation 33
3.2 Hemiequilateral to rectangle transformation 38
3.3 Partitioning of $\mathbb{N}_0^2$ 42
3.4 The sign arrangements for $\mathscr{P}_s$ 43
3.5 Limitations of the triangle-to-rectangle constructions 49
3.6 Duplication of the rectangles in the Besicovitch set 53

5.1 Plots of $K_M^{1/2}(x)$ 76
5.2 Values of $|J_1(j_m)|$ 79
5.3 The function $\frac{\cos v}{v^{1/2-\alpha}}$ 87

6.1 The 'bad' region of a cubic truncation 115

Acknowledgments

Nihil est quod malim quam me et esse gratum et videri. Haec enim est una virtus non solum maxima sed etiam mater virtutum omnium reliquarum.

—Cicero

The list of acknowledgments for such a large project is, in itself, a huge undertaking. Thus, before I can say 'thank you', I must say 'I am sorry' to all those who I have not been able to include.

First of all, I owe an enormous debt of gratitude to my supervisor James Robinson, for introducing me to this problem and kick-starting this exciting, sometimes frustrating, but hugely rewarding journey. For all our conversations at the board and, when necessary, on Teams; for your ever constructive criticism of every draft; for your kindest support in the low moments, and the laughs we've shared along the way: thank you, James.

I also wish to thank José Rodrigo, for our helpful discussions and his support throughout the PhD (and beyond!); and Aníbal Rodríguez Bernal, whose pointers on resolvents of the Laplacian were invaluable for the completion of Chapter 6. I also owe them an apology for inflicting the next hundred pages upon their summer.

For hosting me at the Acoustics Research Institute in Vienna, I am grateful to Peter Balazs and Dimitri Bytchenkoff, as well as the great research group at ARI.

I am also grateful to EPSRC for providing the funding that has enabled this project, and the Warwick Mathematics Institute for providing all the support to make this happen. I especially wish to thank Alison Humphries for her support from the MRC; Rachel Mason, for helping me to leave Warwick every once in a

while; and Leah McGuigan, who always had the answers.

Over the last four years—COVID notwithstanding!—I have come to appreciate just how important it is to have a strong community to do good Mathematics. I could not have wished for a friendlier, more welcoming community than Warwick's. I especially wish to thank Lucas Lavoyer, Arjun Sobnack, Jakub Takáč, Andrew Rout, Andreas Koller, Sotiris Kotitsas, Andreas Mountakis, Georgios Athanasopoulos, Jesús Fernández, Milos Tasic, Irene Gil and Phoebe Valentine not only for stimulating Mathematics, but for all the other great times and conversations that have enriched my—and I hope their!—PhD experience.

I am immensely grateful to my mother, Kate, for always being there with constant love and support.

E igualmente de agradecido le estoy a mi padre, Fernando. No he olvidado tu magistral teorema sobre los triángulos, y por el cual te dedico, a cambio, los míos del capítulo 3:

> *¿En qué se parece un caballo a un triángulo?*
>
> *Un caballo es una animal; un animal es una bestia; una bestia es un bruto; Bruto mató a César; César no dijo nada; el que nada no se ahoga; el que no se ahoga, flota; flota es un conjunto de barcos; un conjunto de barcos es una escuadra; y una escuadra es un triángulo.*

Tampoco me olvido de las tres influencias y amistades que me impulsaron a iniciar mis caminos por las matemáticas, y han permanecido una fuente de inspiración todos esto años: Rafael Payá, Jorge Losada y, por supuesto, cómplice dionisiaco de cerebro inquieto, José Luis Abián.

Por último, y más importante, quiero dedicar esta tesis a Patricia, el *summum bonum* del que participan todas las demás alegrías que he tenido durante estos últimos años. Gracias por tu amor y tu apoyo incondicional; gracias por compartir las ilusiones de este camino, y por levantarme cuantas veces he tropezado; gracias, en suma, porque me lo has dado todo. Te quiero.

Declarations

This thesis is submitted to the University of Warwick in support of my application for the degree of Doctor of Philosophy in Mathematics. It has been composed by myself and has not been submitted in any previous application for any degree.

Parts of this thesis have been published by the author (or in collaboration with their supervisor):

(i) Chapter 3 has appeared in *Proceedings of the Edinburgh Mathematical Society*; see Acosta Babb (2023b).

(ii) A summary of Section 4.5 has appeared in *Comptes Rendus – Mathématique*; see Acosta Babb (2023a).

(iii) Chapter 5 is being prepared for submission.

(iv) Chapter 6 is available on the arXiv; see Acosta Babb and Robinson (2024).

Abstract

Fourier expansions are a key tool in the analysis of partial differential equations. Irregular functions can still be expanded as Fourier series, and then by truncating the Fourier series one obtains a sequence of smooth functions that converges to the original irregular function. This idea—and related techniques—forms a key part of many proofs in the area. However, truncated Fourier series—and, more generally, truncated eigenfunction expansions—may not converge in L^p for $p \neq 2$, and this can prove problematic in many situations. Based on the observation that carefully chosen truncations of Fourier series do converge in L^p when $p \neq 2$, we have investigated how much this can be carried over the eigenfunction expansions in other domains (the Fourier series case corresponding to the Laplacian in a square domain).

We have obtained new convergence results for certain types of triangular domains, as well as divergence results for 45-90-45 triangles.

We have analysed the situation in the disc, where the eigenfunctions are products of trigonometric and Bessel functions, giving rise to Bessel–Fourier series. We have constructed integrable functions whose Bessel–Fourier coefficients decay slower than $m^{-\alpha}$ for $0 \leqslant \alpha < 1/2$, subject to the condition $x^{\alpha+1} f(x) \in L^1[0,1]$. We provide numerical evidence to suggest that the same should hold when $\alpha = 1/2$, and show how such a result could be used to construct an integrable function on the disc whose Bessel–Fourier series diverges for any truncation method.

Finally, we consider the general setting in which we perturb a planar domain, Ω, into another, $\widetilde{\Omega}$, in a controlled way. Under suitable assumptions on the perturbation, we obtained L^p continuity results for the partial sums of the eigenfunction expansions for the Dirichlet Laplacian as Ω deforms into $\widetilde{\Omega}$.

Chapter 1

Introduction

> *Haec igitur incommoda, hasque apparentes contradictiones penitus evitabimus, si voci* summae *aliam notionem, atque vulgo fieri solet, tribuamus.*
>
> —Euler

In one dimension, there is only one natural way to truncate the partial sums of a Fourier series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n)\mathrm{e}^{2\pi \mathrm{i} n x}, \tag{1.0.1}$$

namely

$$S_N f(x) := \sum_{n=-N}^{N} \widehat{f}(n)\mathrm{e}^{2\pi \mathrm{i} n x}. \tag{1.0.2}$$

We then say that the Fourier series (1.0.1) converges if, and only if, $S_N f \to f$ as $N \to \infty$. The sense of 'converges' that will concern us in this thesis is L^p norm covergence:

$$\|S_N f - f\|_{L^p} \to 0 \quad \text{as} \quad N \to \infty. \tag{1.0.3}$$

It is a classic result of Riesz (1928) that, for all $1 < p < \infty$, Eq. (1.0.3) holds for all $f \in L^p$.

When moving to higher dimensions, we have a 2-dimensional Fourier series,

$$\sum_{m,n} \widehat{f}(m,n)\mathrm{e}^{2\pi \mathrm{i}(mx+ny)}, \tag{1.0.4}$$

and an ambiguity arises: how must we truncate the partial sums to ensure convergence? (One could also wonder about the order of the sums, but we will not consider this question.)

Two fairly simple approaches immediately suggest themselves:

(i) *Cubic truncations*, whereby we define the partial sums to be

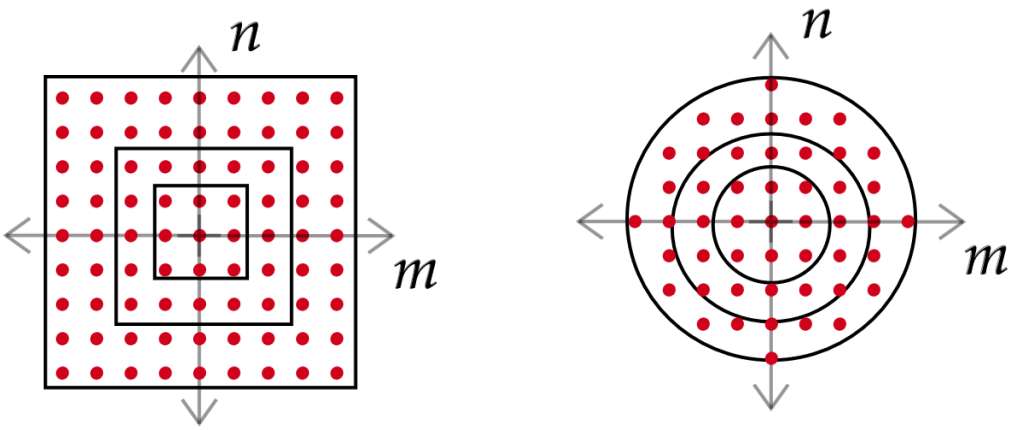
$$S_{[N]}f(x,y) := \sum_{|m|,|n|\leqslant N} \widehat{f}(m,n)\mathrm{e}^{2\pi\mathrm{i}(mx+ny)}, \tag{1.0.5}$$

generalising Eq. (1.0.2) 'in each dimension', so to speak—see Fig. 1.1a.

(ii) *Spherical truncations*, defined by

$$S_N f(x,y) := \sum_{m^2+n^2\leqslant N^2} \widehat{f}(m,n)\mathrm{e}^{2\pi\mathrm{i}(mx+ny)}, \tag{1.0.6}$$

which generalise the idea of cutting-off at a fixed distance from the origin. The name should be clear from Fig. 1.1b



(a) Cubic truncations on the lattice $\mathbb{Z}^2$.

(b) Spherical truncations on the lattice $\mathbb{Z}^2$.

Figure 1.1: Some methods of truncation of Fourier series in two dimensions.

Happily, the cubic truncations from Eq. (1.0.5) converge in L^p in the range $1 < p < \infty$ as a straightforward generalisation of Riesz's results just mentioned. Essentially, the factorisation of the exponentials for separate variables allows us to consider $S_{[N]}f(x,y)$ as multiple copies of the 1-dimensional sums from (Eq. (1.0.2)), and apply induction. The proof is given in detail in 2.2.

In order to tackle the convergence of the spherical truncations in higher dimensions, we need other, more subtle methods. We will work in $\mathbb{R}^2$ and write $x \in \mathbb{R}^2$ and $n \in \mathbb{Z}^2$. Their inner product is denoted by $x \cdot n$, as usual.

By the Uniform Boundedness Principle, L^p convergence of the (spherical) partial sums S_N is equivalent to the uniform boundedness over $N \in \mathbb{N}$ of the operators $S_N \colon L^p \to L^p$. Now, let us re-write the spherical partial sums as

$$S_N f(x) = \sum_{n \in \mathbb{Z}^2} \chi_{B(0,1)}\left(\frac{n}{N}\right) \widehat{f}(n) \mathrm{e}^{2\pi \mathrm{i} n \cdot x}.$$

Here, $\chi_{B(0,1)}$ is the indicator function of the unit ball in $\mathbb{R}^2$.

The easiest way to bound the operators S_N is to compute

$$\int_{\mathbb{T}^2} S_N f(x) g(x) \, \mathrm{d}x$$

and take a supremum over g with $\|g\|_{L^q(\mathbb{T}^2)} \leqslant 1$ (where $\frac{1}{p} + \frac{1}{q} = 1$).

We are going to do this in a round-about way. Consider functions f, g with compact support in $\mathbb{R}^2$, and choose N large enough so that the functions

$$f_N(x) := f(Nx) \quad \text{and} \quad g_N(x) := g(Nx)$$

are supported on $[-1/2, 1/2]^2$ (which we think of as the torus $\mathbb{T}^2$). It is not hard to see that

$$\widehat{f_N}(\xi) = \frac{1}{N^2} \widehat{f}\left(\frac{\xi}{N}\right) \quad \text{and} \quad \|f_N\|_{L^p(\mathbb{T}^2)} = \frac{1}{N^{2/p}} \|f\|_{L^p(\mathbb{R}^2)}.$$

Thus, on the one hand, we have

$$\int_{\mathbb{T}^2} S_N f_N(x) g_N(x) \, \mathrm{d}x \leqslant \frac{1}{N^2} \|S_N\|_{L^p \to L^p} \|f\|_{L^p(\mathbb{R}^2)} \|g\|_{L^q(\mathbb{R}^2)}. \tag{1.0.7}$$

(Here $\frac{1}{p} + \frac{1}{q} = 1$.) On the other hand,

$$\begin{aligned} \int_{\mathbb{T}^2} S_N f_N(x) g_N(x) \, \mathrm{d}x &= \sum_{n \in \mathbb{Z}^2} \chi_{B(0,1)}\left(\frac{n}{N}\right) \widehat{f_N}(n) \widehat{g_N}(n) \\ &= \frac{1}{N^4} \sum_{n \in \mathbb{Z}^2} \chi_{B(0,1)}\left(\frac{n}{N}\right) \widehat{f}\left(\frac{n}{N}\right) \widehat{g}\left(\frac{n}{N}\right). \end{aligned} \tag{1.0.8}$$

Combining Eqs. (1.0.7) and (1.0.8), and cancelling a factor of N^{-2} from both sides, we have

$$\frac{1}{N^2} \sum_{n \in \mathbb{Z}^2} \chi_{B(0,1)}\left(\frac{n}{N}\right) \widehat{f}\left(\frac{n}{N}\right) \widehat{g}\left(\frac{n}{N}\right) \leqslant \|S_N\|_{L^p \to L^p} \|f\|_{L^p(\mathbb{R}^2)} \|g\|_{L^q(\mathbb{R}^2)},$$

where $\widehat{f}$ and $\widehat{g}$ denote the *Fourier transforms* of f and g on $\mathbb{R}^2$. The sum on the left is a Riemann sum, so taking $N \to \infty$ we obtain

$$\int_{\mathbb{R}^2} \chi_{B(0,1)}(\xi)\widehat{f}(\xi)\widehat{g}(\xi)\,\mathrm{d}\xi \leqslant \sup_{N\geqslant 1} \|S_N\|_{L^p\to L^p}\,\|f\|_{L^p(\mathbb{R}^2)}\,\|g\|_{L^q(\mathbb{R}^2)}\,.$$

The operator T defined by setting $\widehat{Tf} := \chi_{B(0,1)}\widehat{f}$ is called the *ball multiplier*, and so, taking a supremum over $\|g\|_{L^q(\mathbb{R}^2)} \leqslant 1$, we have shown that

$$\|Tf\|_{L^p(\mathbb{R}^2)} \leqslant \sup_{N\geqslant 1} \|S_N\|_{L^p\to L^p}\,\|f\|_{L^p(\mathbb{R}^2)}\,.$$

To sum up (and generalising to $\mathbb{R}^n$): L^p convergence of the spherical partial sums S_N in $\mathbb{R}^n$ implies the L^p boundedness of the ball multiplier T in $\mathbb{R}^n$.

Unfortunately, nobody expected the spherical partial sums in higher dimensions to be as well-behaved as their cubic counterparts. The following is not hard to prove.

Theorem 1.0.1. *If $n \geqslant 2$ and $T\colon L^p(\mathbb{R}^n) \to L^p(\mathbb{R}^n)$ is bounded, then necessarily $\frac{2n}{n+1} < p < \frac{2n}{n-1}$.*

Proof. We do the case $n = 2$ for simplicity; the argument is adapted from Chapter 5 in Grafakos (2014a).

Let f be a Schwartz function such that $\widehat{f} \equiv 1$ on $B(0,2)$ and $\widehat{f} \equiv 0$ outside $B(0,3)$. Then,

$$Tf(x) = \int_{|\xi|\leqslant 1} \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\cdot\xi}\,\mathrm{d}\xi = \int_{\mathbb{R}^2} \chi_{B(0,1)}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\cdot\xi}\,\mathrm{d}\xi = \frac{J_1(2\pi\,|x|)}{|x|},$$

using the formula for $\widehat{\chi_{B(0,1)}}$ in Appendix B.5 of Grafakos (2014b); here J_1 is a Bessel function of the first kind, for which see Chapter 4. From the asymptotics for J_1 (to be discussed more fully in Lemma 4.3.1 below), we have that

$$Tf(x) \asymp \frac{\cos\left(2\pi\,|x| - 3\pi/4\right)}{|x|^{3/2}} + \frac{1}{|x|^{5/2}} \quad \text{for} \quad |x| \geqslant 1.$$

Note that, on each annulus

$$A_k := \left\{x \in \mathbb{R}^2 : k + \frac{1}{4} \leqslant |x| \leqslant k + \frac{1}{2}\right\},$$

we have

$$\frac{\sqrt{2}}{2} \leqslant \cos\left(2\pi\,|x| - 3\pi/4\right) \leqslant 1.$$

Now, we consider the range

$$\frac{4}{5} < p \leqslant \frac{4}{3}.$$

If T is unbounded for every p in this range, then it is also unbounded on every $1 < p \leqslant \frac{4}{3}$. Indeed, suppose that T were bounded for some $p \leqslant \frac{4}{5}$. Since T is bounded on L^2, it follows by interpolation that T is bounded on $L^{4/3}$, which we will show that it is not. Finally, a duality argument shows that T cannot be bounded on any L^p with $4 \leqslant p < \infty$.

Let us therefore consider $\frac{4}{5} < p \leqslant \frac{4}{3}$ and compute

$$\begin{aligned}\|Tf\|^p_{L^p(\mathbb{R}^2)} &= \int_{|x|\leqslant 1} |Tf(x)|^p \, \mathrm{d}x + \int_{|x|\geqslant 1} |Tf(x)|^p \, \mathrm{d}x \\ &\gtrsim 1 + \sum_{k=2}^{\infty} \int_{A_k} |x|^{-\frac{3p}{2}} \, \mathrm{d}x - \int_{|x|\geqslant 1} |x|^{-\frac{5p}{2}} \, \mathrm{d}x.\end{aligned}$$

The integral over $|x| \leqslant 1$ is finite, since, near 0, $|J_1(2\pi\,|x|)| \leqslant 1$ and $\frac{1}{|x|}$ is integrable. The last integral is

$$\int_{|x|\geqslant 1} |x|^{-\frac{5p}{2}} \, \mathrm{d}x = 2\pi \int_1^{\infty} r^{1-\frac{5p}{2}} \, \mathrm{d}r < \infty,$$

since $p > \frac{4}{5}$. Finally, integrating and applying the Mean Value Theorem, we have

$$\sum_{k=2}^{\infty} \int_{A_k} |x|^{-\frac{3p}{2}} \, \mathrm{d}x = 2\pi \sum_{k=2}^{\infty} \int_{k+\frac{1}{4}}^{k+\frac{1}{2}} r^{1-\frac{3p}{2}} \, \mathrm{d}r \gtrsim \sum_{k=2}^{\infty} k^{1-\frac{3p}{2}},$$

which diverges when $p \leqslant \frac{4}{3}$. □

The L^p boundedness of T in the range $\frac{2n}{n+1} < p < \frac{2n}{n-1}$ was dubbed the 'disc conjecture'. The failure of this conjecture was shown by Fefferman in his groundbreaking paper of 1971.

Theorem 1.0.2 (The Ball Multiplier Theorem)**.** *If $n \geqslant 2$ and $T\colon L^p(\mathbb{R}^2) \to L^p(\mathbb{R}^n)$ is bounded, then $p = 2$.*

And along with it fell the hopes of proving convergence for spherical truncations of Fourier series.

This result is concerning, since the spherical truncations are the most natural choice in other domains, where the convenient device of labelling eigenfunctions

by pairs $(m, n) \in \mathbb{Z}^2$ is absent. As Bochner remarks (1936, pp. 179–180, updated notation and omitting equation numbering),

> The elementary exponentials
>
> $$u(x) = \mathrm{e}^{\mathrm{i}(n_1 x_1 + \cdots + n_k x_k)}$$
>
> (all $n_1, \ldots, n_k$ integers) are a complete set of regular solutions of the characteristic value problem
>
> $$\Delta u(x) = -\lambda u(x),$$
>
> if this equation is being considered on the (closed) torus
>
> $$0 \leqslant x_1 < 2\pi, \ldots, 0 \leqslant x_k < 2\pi,$$
>
> and $\Delta u(x)$ is the Laplace operator with respect to the Euclidean metric on the torus, namely,
>
> $$\frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_k^2}.$$
>
> Since $\lambda = n_1^2 + \cdots n_k^2$, our way of writing series [1.0.4] in the form [1.0.6] satisfies the very natural principle of ordering the terms in series [1.0.4] according to the magnitude of the characteristic values λ.

(Note that references in square brackets are to *our* equations, not Bochner's!) Once we move away from the comfort of the torus (or the square), we may only be able to truncate an eigenfunction expansion by the size of the eigenvalues, and this is precisely the spherical cut-off that Fefferman tells us cannot work for Fourier series! For further discussion of Fefferman's result, we invite the readers to consult Davis and Chang (1987) for a very readable exposition. A summary of Fefferman's proof is offered below in the Appendix to Chapter 3, in the context of Fourier series on a 45-90-45 triangle. For a comprehensive survey of the literature on the Laplacian and its eigenfunctions, see Grebenkov and Nguyen (2013)

Recently, Fefferman et al. (2022) have raised the issue anew, formulating the question which animates our research in this thesis: given a differential operator with an orthonormal family w_k of eigenfunctions, is there a choice $\{w \in E_N : N \in \mathbb{N}\}$ of eigenfunctions such that the truncations

$$S_{E_N} f = \sum_{w \in E_N} \langle f, w \rangle \, w$$

are 'well-behaved' in L^p for some values of $p \neq 2$?

After a review of the classic theory of Fourier series (Chapter 2), we will begin (in Chapter 3) by investigating triangular domains. Following early work of Lamé (1833), other authors such as Práger (1998) and McCartin (2003) have derived explicit trigonometric expressions for Dirichlet eigenfunctions on the equilateral triangle. Neither of these authors, however, consider questions of convergence for the associated eigenfunction series, and subsequent work by Adcock (2009), Huybrechs et al. (2011), and Sun and Li (2005), restricts attention to numerical methods or convergence in L^2-based spaces such as H^k. To our knowledge, this is the first treatment of the L^p convergence of series of eigenfunctions on the triangle for $p \neq 2$.

We will next turn our attention to the disc in Chapters 4 and 5. The eigenfunctions for the disc are products of trigonometric and Bessel functions and so share a similar product structure to the classic Fourier series on $\mathbb{T}^2$. Specifically, the eigenfunctions are of the form

$$\mathrm{e}^{2\pi \mathrm{i} n\theta} J_{|n|}(j_{|n|,m} r),$$

giving rise to the multidimensional Bessel–Fourier series

$$\sum_{n=-\infty}^{\infty} \sum_{m=1}^{\infty} a_{m,n} \mathrm{e}^{2\pi \mathrm{i} n\theta} J_{|n|}(j_{|n|,m} r). \tag{1.0.9}$$

(Here, $J_{|n|}$ denotes a Bessel function of the first kind and $j_{|n|,m}$ its non-negative zeros.) The best result we were able to find in the literature is due to Balodis and Córdoba (1999), who proved norm convergence in the mixed norm space $L^p_{\mathrm{rad}}(L^2_{\mathrm{ang}})$ defined by the condition

$$\|f\|_{p,2} := \left[\int_0^1 \left(\sum_k |f_k(r)|^2 \right)^{p/2} r \, \mathrm{d}r \right]^{1/p} < \infty,$$

where $f_k(r)$ is the kth Fourier coefficient of the angular function $f(r, \cdot)$ for fixed r:

$$f_k(r) := \int_0^1 f(r, \theta) \mathrm{e}^{2\pi \mathrm{i} k\theta} \, \mathrm{d}\theta.$$

By truncating the series (1.0.9) in the ranges $|n| \leqslant N$, $m \leqslant M$, they were able to show that there is a constant $A > 0$ such that the Bessel–Fourier series of a function $f \in L^p_{\mathrm{rad}}(L^2_{\mathrm{ang}})$ converges to f in the $\|\cdot\|_{p,2}$ norm provided that $M \geqslant AN + 1$ and $\frac{4}{3} < p < 4$. Furthermore, the endpoints for the range of p are sharp. (See Balodis

and Córdoba, 1999, Theorem 2, of which our discussion is a special case when $d = 2$.)

By modifying their proof, we were able to improve the result to L^p convergence with respect to the usual measure $r\,\mathrm{d}r\,\mathrm{d}\theta$ on the disc, provided that $2 \leqslant p < 4$ and

$$\|f\|_{p,q} := \left[\int_0^1 \left(\sum_k |f_k(r)|^q\right)^{p/q} r\,\mathrm{d}r\right]^{1/p} < \infty, \quad \text{where} \quad \frac{1}{p} + \frac{1}{q} = 1.$$

As far as we know, the problem of L^p convergence for functions $f \in L^p(r\,\mathrm{d}r\,\mathrm{d}\theta)$ in the range $\frac{4}{3} < p < 4$ is still open. This problem will be discussed in Chapter 4.

In Chapter 5 we will be concerned with the (L^1) *divergence* of Bessel–Fourier series on the disc. By considering radial functions, we reduce the 2-dimensional Bessel–Fourier series (1.0.9) to a 1-dimensional expansion

$$f(r) \sim \sum_{m=1}^{\infty} f_m J_0(j_{0,m} r).$$

Our starting point is the 'Riemann–Lebesgue lemma' for Bessel–Fourier coefficients, which is stronger than the classic Riemann–Lebesgue lemma for Fourier coefficients.

Theorem 1.0.3 (Young 1920)**.** *If* $\sqrt{x} f(x) \in L^1([0,1])$*, then*

$$\frac{f_m}{\sqrt{m}} \to 0 \quad \textit{as} \quad m \to \infty.$$

We show that, for every $0 \leqslant \alpha < \frac{1}{2}$, we can construct a function f with $x^{1+\alpha} f(x) \in L^1([0,1])$ such that $f_m = o(m^\alpha)$. While we have not managed to show that the same is true when $\alpha = \frac{1}{2}$, we have numerical reasons to suggest that this end-point case should also hold, and we discuss how it could be applied to construct a function $f \in L^1(r\,\mathrm{d}r\,\mathrm{d}\theta)$ whose 2-dimensional Bessel–Fourier series diverges in L^1 *regardless of the method of truncation.*

We have been unable to locate much literature in this area, besides the article of Young (1920) and treatise of Watson (1995). More recently, however, Saadi and Daher (2022, 2023) have exploited the decay of Bessel–Fourier coefficients to study when a function belongs to a Lipschitz class.

Finally, in Chapter 6 we set our sights on the general case. Given a bounded domain Ω in the plane, we deform it by a suitable map φ into another domain $\widetilde{\Omega}$. Our aim is to compare the eigenfunction expansions for the Dirichlet Laplacian on each of Ω and $\widetilde{\Omega}$. Under suitable assumptions on φ, we obtain an L^p continuity results for the 'spherical partial sums' as we perturb Ω into $\widetilde{\Omega}$.

This approach is inspired by Lamberti and Lanza de Cristoforis (2005), who obtained similar results in the H_0^1 setting. The main difficulty in generalising their results to L^p was obtaining bounds on the L^p norms of resolvents for the *inverse* Laplacian. We believe these results to be new.

The (elliptic) PDE litterature on domain deformation is vast, as the survey article by Henrot (1994) and the chapter by Daners (2008) show. Most authors deal with convergence of solutions: e.g., Savaré and Schimperna (2002) consider L^2 energy estimates on differences of solutions; Arendt and Daners (2007) deal with uniform convergence; while Daners (2003) also considers L^p convergence. Kalmykov and Kovalev (2019) study the converges of Green's functions as the domain varies. There are comparatively fewer results on the the behaviour of eigenfunction expansions.

Notation

> Die Mathematiker ſind eine Art Franzoſen: redet man zu ihnen, ſo überſetzen ſie es in ihre Sprache, und dann iſt es alſobald ganz etwas anders.
>
> —Goethe

Before we begin, we briefly record some notation that is used throughout the thesis.

Recall the *big-O* notation

$$f = O(g) \quad \text{as} \quad x \to \infty,$$

which means that there is a constant $C > 0$ such that

$$|f(x)| \leqslant C\,|g(x)| \quad \text{for sufficiently large } x.$$

We will also sometimes write

$$A \lesssim B$$

to mean that there is a constant $C > 0$ such that

$$A \leqslant CB \quad \text{for all 'appropriate values' of } A \text{ and } B$$

in such a way that C does not depend on A or B. In particular,

$$A \lesssim 1$$

means that A is uniformly bounded.

We write $A \asymp B$ to mean that $A \lesssim B$ and $B \lesssim A$.

Little-o notation is also helpful:

$$f(x) = o(g(x)) \quad \text{means} \quad \frac{f(x)}{g(x)} \to 0 \quad \text{as} \quad x \to \infty.$$

We will take the *natural* numbers, $\mathbb{N}$, to comprise the *positive* integers, and, where necessary, write $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$.

Chapter 2

Preliminaries of Fourier analysis on the torus

Forsan et haec olim meminisse iuvabit.

—Virgil

2.1 Recap of 1-dimensional Fourier analysis

In this section we develop some standard results and examples from one-dimensional Fourier analysis to be used in later sections. Standard references for this material are Grafakos (2014b) and Duoandikoetxea Zuazo (2001). For the results on measures and Hausdorff–Young inequalities in Section 2.1.4, we follow Katznelson (2004), specifically Chap. 1, §7 and Chap. 4, §2, adding appropriate details.

2.1.1 Fourier series on the circle

We let $\mathbb{T} := \mathbb{R}/\mathbb{Z}$ be the additive torus, identified with the interval $[0, 1]$ under periodic boundary conditions. Thus, any function $f\colon \mathbb{T} \to \mathbb{C}$ may be regarded as a $\mathbb{Z}$-periodic function on $\mathbb{R}$:

$$f(x + n) = f(x) \quad \text{for all} \quad x \in \mathbb{T} \text{ and } n \in \mathbb{Z}.$$

Since $\mathbb{T}$ is compact, $C^0(\mathbb{T})$ is the set of *uniformly continuous* functions from $\mathbb{T}$ to $\mathbb{C}$, and its dual space is $M(\mathbb{T})$, the signed Radon measures on $\mathbb{T}$, equipped with the *total variation norm*

$$\|\mu\|_{M(\mathbb{T})} = |\mu|(\mathbb{T}) \equiv \int_{\mathbb{T}} \mathrm{d}|\mu|.$$

(See Conway, 2007, Appendix C.)

In Section 2.1.4, we will need to consider the Fourier series of measures. Given a measure $\mu \in M(\mathbb{T})$, we define its *Fourier coefficients* by

$$\widehat{\mu}(n) := \overline{\langle \mu, \mathrm{e}^{2\pi \mathrm{i} n t} \rangle} \equiv \int_{\mathbb{T}} \mathrm{e}^{-2\pi \mathrm{i} n t} \, \mathrm{d}\mu(t) \quad \text{for all} \quad n \in \mathbb{Z}.$$

(Here $\langle \cdot, \cdot \rangle$ denotes the duality pairing between $M(\mathbb{T})$ and $C^0(\mathbb{T})$.) In particular, if $f \in L^1(\mathbb{T})$, then we have the familiar formula

$$\widehat{f}(n) = \int_{\mathbb{T}} f(t) \mathrm{e}^{-2\pi \mathrm{i} n t} \, \mathrm{d}t \quad \text{for all} \quad n \in \mathbb{Z}.$$

Remark on notation: 2π or not 2π? Another common convention identifies $\mathbb{T} \cong [0, 2\pi]$ and defines Fourier coefficients as either

$$\frac{1}{2\pi} \int_0^{2\pi} f(t) \mathrm{e}^{-\mathrm{i} n t} \, \mathrm{d}t \quad \text{or} \quad \frac{1}{\sqrt{2\pi}} \int_0^{2\pi} f(t) \mathrm{e}^{-\mathrm{i} n t} \, \mathrm{d}t.$$

Both of these choices introduce cumbersome prefactors in the coefficients. The former alternative has the further unpleasant side effect of inserting additional factors of 2π in the convolution theorem below. In any case, we prefer the symmetric ranges $[0, 1]$ for both the radial and angular variables used on the disc, as well as having $\mathbb{Z}$ for the additive group as opposed to $2\pi\mathbb{Z}$. We thus follow in the footsteps of Duoandikoetxea Zuazo (2001) and Grafakos (2014a,b).

Formally, then, to each measure $\mu \in M(\mathbb{T})$ we associate a *Fourier series*

$$\sum_{n \in \mathbb{Z}} \widehat{\mu}(n) \mathrm{e}^{2\pi \mathrm{i} n t}. \tag{2.1.1}$$

The convergence of such series, for various classes of functions, is the backbone of classical Fourier analysis.

Of special interest to us will be the *trigonometric polynomials*, i.e. finite sums of the form

$$g(t) = \sum_{|n| \leqslant N} a_n \mathrm{e}^{2\pi \mathrm{i} n t} \quad \text{for some} \quad a_n \in \mathbb{C}.$$

Clearly $a_n = \widehat{g}(n)$ for $|n| \leqslant N$; we call N the *degree* of g.

For $f, g \in L^1(\mathbb{T})$, recall that their *convolution* $f * g$ is defined by

$$f * g(t) := \int_{\mathbb{T}} f(t - s) g(s) \, \mathrm{d}s,$$

which is the unique[1] operation on $L^1(\mathbb{T})$ such that

$$\widehat{f * g}(n) = \widehat{f}(n)\widehat{g}(n) \quad \text{for all} \quad n \in \mathbb{Z}.$$

(This identity is sometimes referred to as the *convolution theorem*.) Hence, we can express weighted sums of Fourier coefficients as convolutions with appropriate kernels:

(i) *Partial sums*

$$S_N\mu(t) := \sum_{|n|\leqslant N} \widehat{\mu}(n)\mathrm{e}^{2\pi \mathrm{i} nt} = D_N * \mu(t),$$

where

$$D_N(t) := \sum_{|n|\leqslant N} \mathrm{e}^{2\pi \mathrm{i} nt}$$

is the *Dirichlet kernel*. Indeed, when f is a function, we have

$$\begin{aligned} D_N * f(t) &= \int_{\mathbb{T}} D_N(t-s)f(s)\,\mathrm{d}s \\ &= \sum_{|n|\leqslant N} \left(\int_{\mathbb{T}} f(s)\mathrm{e}^{-2\pi \mathrm{i} ns}\,\mathrm{d}s \right) \mathrm{e}^{2\pi \mathrm{i} nt} \\ &= \sum_{|n|\leqslant N} \widehat{f}(n)\mathrm{e}^{2\pi \mathrm{i} nt} \equiv S_N f(t). \end{aligned}$$

For convolution of measures, see (2.1.4) the following aside, below.

(ii) *Cesàro sums*

$$\sigma_N\mu(t) = \frac{1}{N+1}\sum_{|n|\leqslant N} S_n\mu(t) = \sum_{|n|\leqslant N} \left(1 - \frac{|n|}{N+1}\right) \widehat{\mu}(n)\mathrm{e}^{2\pi \mathrm{i} nt} = F_N * \mu(t),$$

where

$$F_N(t) := \sum_{|n|\leqslant N} \left(1 - \frac{|n|}{N+1}\right) \mathrm{e}^{2\pi \mathrm{i} nt}$$

is the *Fejér kernel*.

(iii) *Vallée Poussin kernels*

$$V_N(t) := 2F_{2N+1}(t) - F_N(t)$$

[1] That is: if $\widehat{f \odot g}(k) = \widehat{f}(k)\widehat{g}(k)$ for all $f, g \in L^1(\mathbb{T})$ and $k \in \mathbb{Z}$, then $f \odot g \equiv f * g$. This can be verified by applying the Fourier inversion formula to $f \odot g$ and essentially running the proof of the convolution theorem 'in reverse'.

$$\equiv \sum_{|n|\leqslant N} \mathrm{e}^{2\pi i n t} + \sum_{N+1\leqslant|n|\leqslant 2N+1} \left(2 - \frac{|n|}{N+1}\right) \mathrm{e}^{2\pi i n t}.$$

Note that, for trigonometric polynomials g of degree N, $g * V_N(t) = g(t)$ holds, since $\widehat{V_N}(n) = 1$ for $|n| \leqslant N$ and g has no Fourier modes for $|n| > N$. Such kernels will be useful in the proof of Proposition 2.1.13 below.

2.1.2 Summation of Fourier series

Kernels such as the above allow us to study the convergence in various senses of the Fourier series (2.1.1).

Definition 2.1.1. A *summability kernel* is a family $\{K_N : N \geqslant 1\} \subset L^1(\mathbb{T})$ such that

(i) $\sup_{N\geqslant 1} \|K_N\|_{L^1(\mathbb{T})} < \infty$;

(ii) for all $N \geqslant 1$, $\int_{\mathbb{T}} K_N = 1$; and

(iii) for any neighborhood V of 0 in $\mathbb{T}$ we have $\lim_{N\to\infty} \int_V |K_N| = 0$.

Summability kernels are especially useful in light of the next theorem (Grafakos, 2014b, Theorem 1.2.19).

Theorem 2.1.2. *Let $\{K_N\}_N$ be a summability kernel. Then,*

(i) If $f \in L^p(\mathbb{T})$ with $1 \leqslant p < \infty$, then $\|K_N * f - f\|_{L^p(\mathbb{T})} \to 0$.

(ii) If $f \in C^0(\mathbb{T})$, then $K_N * f \to f$ uniformly.

The Fejér kernels F_N form a summability kernel, as is easily deduced from the expression

$$F_N(t) \equiv \frac{1}{N+1} \left(\frac{\sin(\pi(N+1)t)}{\sin(\pi t)}\right)^2, \tag{2.1.2}$$

which can be obtained from trigonometric formulae (Grafakos, 2014b, see Proposition 3.1.7).

Remark. In Chapter 5, we will need a similar kernel for the Cesàro means of Bessel–Fourier series. Unfortunately, in that case we have not found a nice, closed form for the kernel. In particular, while numerics suggest that these kernels are also non-negative, we were unable to show this analytically.

Corollary 2.1.3. *If $f \in L^p(\mathbb{T})$ for $1 \leqslant p < \infty$ (respectively, $f \in C^0(\mathbb{T})$), then $\sigma_N f \to f$ in L^p (respectively, uniformly).*

In particular, since $\sigma_N f$ is a trigonometric polynomial, we have the following result.

Corollary 2.1.4. *Trigonometric polynomials are dense in $L^p(\mathbb{T})$ for $1 \leqslant p < \infty$ and in $C^0(\mathbb{T})$.*

Unfortunately, $\|D_N\|_{L^1(\mathbb{T})}$ grows logarithmically in N, so D_N fails to be a summability kernel. (See Zygmund, 2003, p. 67.) Thus, investigation of the L^p partial sums must proceed by other, more subtle, arguments.

2.1.3 L^p convergence

The simplest case is L^2: since $\{e^{2\pi i n}\}_{n\in\mathbb{Z}}$ is an orthonormal basis for $L^2(\mathbb{T})$ (e.g. by standard Sturm–Liouville Theory, see Conway, 2007, II.§6), we obtain the L^2 theory of Fourier series directly from Hilbert space results.

Theorem 2.1.5. *Let $f \in L^2(\mathbb{T})$. Then*

(i) $S_N f \to f$ in L^2; and

(ii) Plancherel's Identity holds:

$$\|f\|^2_{L^2(\mathbb{T})} = \sum_{z\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2.$$

To study the more general case, i.e. $p \neq 2$, we make the following observation.

Proposition 2.1.6. *$S_N f$ converges to f in the L^p norm (for $1 \leqslant p < \infty$) if, and only if,*

$$\sup_{N\geqslant 1} \|S_N\|_{L^p(\mathbb{T})\to L^p(\mathbb{T})} < \infty. \tag{2.1.3}$$

Proof. Suppose that $S_N f \to f$ in L^p for each $f \in L^p(\mathbb{T})$. Then the sequence $(\|S_N f\|_{L^p(\mathbb{T})})_N$ is bounded for each f. Since each of the operators S_N is bounded on $L^p(\mathbb{T})$, Eq. (2.1.3) follows from the Uniform Boundedness Principle.

Conversely, pick $f \in L^p(\mathbb{T})$ and approximate it by a trigonometric polynomial, g such that $\|f - g\|_{L^p(\mathbb{T})} < \varepsilon$. Note that, for all large enough N, $S_N g = g$, so we have

$$\begin{aligned}\|S_N f - f\|_{L^p(\mathbb{T})} &\leqslant \|S_N f - S_N g\|_{L^p(\mathbb{T})} + \|f - g\|_{L^p(\mathbb{T})} \\ &\leqslant \sup_{N\geqslant 1} \|S_N\|_{L^p(\mathbb{T})\to L^p(\mathbb{T})} \|f - g\|_{L^p(\mathbb{T})} + \|f - g\|_{L^p(\mathbb{T})} \lesssim \varepsilon.\end{aligned}$$

Taking $\varepsilon \to 0$ shows that $S_N f \to f$ in L^p.[2] □

The uniform bound (2.1.3) can be obtained from the boundedness of the *Hilbert Transform*

$$Hf(x) := \text{p. v.} \int_{\mathbb{R}} \frac{f(y)}{x-y}\,\mathrm{d}y \quad \text{for all} \quad f \in \mathscr{S}(\mathbb{R}).$$

The details are developed, e.g., in Duoandikoetxea Zuazo (2001). For a historical discussion, see the notes to Chapter 4 in Grafakos (2014b); it suffices here to record the result.

Theorem 2.1.7. *For $f \in L^p(\mathbb{T})$ and $1 < p < \infty$, we have $\|S_N f - f\|_{L^p(\mathbb{T})} \to 0$.*

One can explicitly construct a counterexample in L^1; see the opening of Chapter 5.

2.1.4 Hausdorff–Young inequalities

In the L^2 setting, we had a useful and elegant relationship between the L^2 norm of f and the ℓ^2 norm of its Fourier coefficients. For $p \neq 2$, such formulae are not available. The best we can do is the following.

Theorem 2.1.8 (Hausdorff–Young Inequalities)**.** *Let $1 \leqslant p \leqslant 2$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$\left(\sum_{z \in \mathbb{Z}} \left| \widehat{f}(n) \right|^q \right)^{1/q} \leqslant \|f\|_{L^p(\mathbb{T})}.$$

Furthermore, the best possible constant above is 1. If $2 \leqslant p < \infty$, then we have the 'Reverse Hausdorff–Young' inequality:

$$\|f\|_{L^p(\mathbb{T})} \leqslant \left(\sum_{n \in \mathbb{Z}} \left| \widehat{f}(n) \right|^q \right)^{1/q}.$$

Proof. We prove the reverse inequality; the proof of the classical inequality is similar.

Consider the linear operator $T\colon c_{00}(\mathbb{Z}) \to L^\infty(\mathbb{T})$ defined by

$$T(c_n)_n := \sum_n c_n \mathrm{e}^{2\pi \mathrm{i} n t}$$

[2] This proof is taken from Duoandikoetxea Zuazo (2001, Lemma 1.8). A more general result is found in Grafakos (2014b, Prop. 4.1.1), including Proposition 2.1.6 as Corollary 4.1.5.

where $c_{00}(\mathbb{Z})$ is the space of sequences on $\mathbb{Z}$ that are eventually zero. This operator is clearly bounded into $L^\infty(\mathbb{T})$:

$$\|T(c_n)\|_{L^\infty(\mathbb{T})} \leqslant \sum_n |c_n| = \|(c_n)\|_{\ell^1}\,.$$

Since c_{00} is dense in ℓ^1, it follows that T admits a bounded extension as an operator $\ell^1(\mathbb{Z}) \to L^\infty(\mathbb{T})$ with norm at most 1. The result follows by interpolation with Plancherel's identity for $p = 2$. □

The rest of this section is concerned with the failure of the above inequalities for other values of q—these results will become important in Chapter 4, as we discuss the mixed-norm convergence of Bessel–Fourier series on the disc. But first, we need to develop further the Fourier series of measures.

We extend Cesàro summation to $\mu \in M(\mathbb{T})$ by duality:

$$\langle \sigma_N \mu, f\rangle := \langle \mu, \sigma_N f\rangle \quad \text{for all} \quad f \in C^0(\mathbb{T}).$$

In other words, we may write

$$\sigma_N \mu = \sum_{|n|\leqslant N} \left(1 - \frac{|n|}{N+1}\right) \widehat{\mu}(n) \mathrm{e}^{2\pi \mathrm{i} n t},$$

with equality understood in $M(\mathbb{T})$, i.e.

$$\langle \sigma_N \mu, f\rangle = \sum_{|n|\leqslant N} \left(1 - \frac{|n|}{N+1}\right) \widehat{\mu}(n)\widehat{f}(n) \mathrm{e}^{2\pi \mathrm{i} n t} \quad \text{for all} \quad f \in C^0(\mathbb{T}).$$

Note that $\|\sigma_N\|_{M(\mathbb{T})\to M(\mathbb{T})} = 1$. Indeed, since the operators σ_N as defined on both $C^0(\mathbb{T})$ and $M(\mathbb{T})$ are adjoints of each other, they have the same norm, and $\|\sigma_N\|_{C^0\to C^0} = 1$ as follows from

$$\|\sigma_N f\|_{C^0(\mathbb{T})} = \|F_N * f\|_{C^0(\mathbb{T})} \leqslant \|F_N\|_{L^1(\mathbb{T})}\, \|f\|_{C^0(\mathbb{T})} = \|f\|_{C^0(\mathbb{T})}\,.$$

The supremum is achieved for $f(t) \equiv 1$.

Proposition 2.1.9. *For $f \in C^0(\mathbb{T})$ and $\mu \in M(\mathbb{T})$, we have*

$$\langle \mu, f\rangle = \lim_{N\to\infty} \sum_{|n|\leqslant N} \left(1 - \frac{|n|}{N+1}\right) \widehat{f}(n)\overline{\widehat{\mu}(n)}.$$

In other words, $\sigma_N \mu \overset{*}{\rightharpoonup} \mu$ in the weak- topology on $(C^0(\mathbb{T}))^*$.*

Proof. If g is a trigonometric polynomial, then

$$\langle \mu, g \rangle = \int_{\mathbb{T}} \sum_{|n| \leqslant N} \widehat{g}(n) \mathrm{e}^{2\pi i n t} \, \mathrm{d}\mu(t) = \sum_{|n| \leqslant N} \widehat{g}(n) \overline{\widehat{\mu}(n)}$$

and we are done.

In general, uniformly approximate $f \in C^0(\mathbb{T})$ by trigonometric polynomials and apply the continuity of $\mu \in (C^0(\mathbb{T}))^*$. □

Using this proposition, we can characterise the Fourier coefficients of measures.

Theorem 2.1.10. *For $\{a_n\}_{n \in \mathbb{Z}} \subset \mathbb{C}$, the following are equivalent:*

(i) There is a measure $\mu \in M(\mathbb{T})$ such that $\widehat{\mu}(n) = a_n$ for all $n \in \mathbb{Z}$, and a constant $C > 0$ such that $\|\mu\|_{M(\mathbb{T})} \leqslant C$.

(ii) For every trigonometric polynomial g, $|\sum \widehat{g}(n)\overline{a_n}| \leqslant C \|g\|_{C^0(\mathbb{T})}$.

Proof. Note that $S_N g \to g$ for trigonometric polynomials, so $\sigma_N g \to g$ too. Thus, assuming (i), we have

$$\begin{aligned} \left| \sum \widehat{g}(n)\overline{a_n} \right| = \left| \sum \widehat{g}(n)\overline{\widehat{\mu}(n)} \right| &\leqslant |\langle \mu, g \rangle| \\ &\leqslant \|\mu\|_{M(\mathbb{T})} \|g\|_{C^0(\mathbb{T})} \leqslant C \|g\|_{C^0(\mathbb{T})}, \end{aligned}$$

where we used Proposition 2.1.9 to get

$$\lim \langle \mu, S_N g \rangle = \lim \langle \mu, \sigma_N g \rangle = \langle \mu, g \rangle .$$

Conversely, assume (ii) and define the functional $\Phi(g) := \sum \widehat{g}(n)\overline{a_n}$, whose norm is at most C. By the Hahn–Banach Theorem, Φ admits a unique extension to a functional in $(C^0(\mathbb{T}))^*$, which we identify with a measure $\mu \in M(\mathbb{T})$ with $\|\mu\|_{M(\mathbb{T})} \leqslant C$. Evaluating μ at $\mathrm{e}^{-2\pi i n t}$ we have

$$\widehat{\mu}(n) = \left\langle \Phi, \mathrm{e}^{-2\pi i n t} \right\rangle = a_n.$$ □

This theorem gives us a way of showing that some sequences cannot be the Fourier series of any measure, and will be used in the proof of Proposition 2.1.13 below.

We can also define L^1 operators by 'convolution' with measures.

Theorem 2.1.11. *If $\mu \in M(\mathbb{T})$, then there is a unique bounded linear operator $T_\mu \colon L^1(\mathbb{T}) \to L^1(\mathbb{T})$ such that*

$$\|T_\mu\|_{L^1 \to L^1} \leqslant \|\mu\|_{M(\mathbb{T})} \quad \text{and} \quad \widehat{T_\mu f}(n) = \widehat{\mu}(n)\widehat{f}(n).$$

Proof. If g is a trigonometric polynomial, then define

$$T_\mu g(t) := \sum_n \widehat{\mu}(n)\widehat{g}(n)\mathrm{e}^{2\pi \mathrm{i} n t}.$$

Suppose first that $\mu = h\,\mathrm{d}t$ for some $h \in C^0(\mathbb{T})$. Then $T_\mu g = g * h$ and $\|\mu\|_{M(\mathbb{T})} = |\mu|(\mathbb{T}) = \|h\|_{L^1(\mathbb{T})}$. Hence,

$$\|T_\mu g\|_{L^1(\mathbb{T})} \leqslant \|\mu\|_{M(\mathbb{T})} \|g\|_{L^1(\mathbb{T})}.$$

More generally, if we define h_N by setting

$$\sigma_N \mu(t) = \sum_{|n| \leqslant N} \left(1 - \frac{|n|}{N+1}\right) \widehat{\mu}(n)\mathrm{e}^{2\pi \mathrm{i} n t} \equiv h_N(t)\,\mathrm{d}t,$$

then

$$\int_{\mathbb{T}} |h_N|\,\mathrm{d}t = \|\sigma_N \mu\|_{M(\mathbb{T})} \leqslant \|\mu\|_{M(\mathbb{T})},$$

since, as noted above, σ_N has norm 1 on $M(\mathbb{T})$. Thus, for any trigonometric polynomial g,

$$\|T_\mu g\| = \left\| \lim_{N \to \infty} h_N * g \right\|_{L^1(\mathbb{T})} \leqslant \|\mu\|_{M(\mathbb{T})} \|g\|_{L^1(\mathbb{T})}$$

by Proposition 2.1.9.

Thus, we have successfully defined a (unique) L^1-bounded linear map T_μ on trigonometric polynomials satisfying $\widehat{T_\mu g}(n) = \widehat{\mu}(n)\widehat{g}(n)$ and with norm at most $\|\mu\|_{M(\mathbb{T})}$, which therefore admits a bounded extension to $L^1(\mathbb{T})$ (by Corollary 2.1.4) with the same norm. □

Although we have not formally introduced convolution with measures, we will use the notation

$$\mu * f := T_\mu f \quad \text{for all} \quad f \in L^1(\mathbb{T}). \tag{2.1.4}$$

Aside: convolution of measures. The notation can be formally justified by the fact that

$$\widehat{\mu * f}(n) = \widehat{\mu}(n)\widehat{f}(n).$$

On the other hand, we can introduce convolution of measure $\mu, \nu \in M(\mathbb{T})$ as follows.

Define the linear functional

$$Tf := \int_{\mathbb{T}} \int_{\mathbb{T}} f(t+s)\, \mathrm{d}\mu(t)\, \mathrm{d}\nu(s) \quad \text{for all} \quad f \in C^0(\mathbb{T}).$$

Since the measures are finite, the functional is well-defined and

$$|Tf| \leqslant \|\mu\|_{M(\mathbb{T})} \|\nu\|_{M(\mathbb{T})} \|f\|_{C^0(\mathbb{T})}.$$

Hence, $T \in (C^0(\mathbb{T}))^* \cong M(\mathbb{T})$ and we *define* $\mu * \nu$ as the unique measure such that

$$Tf = \int_{\mathbb{T}} f\, \mathrm{d}(\mu * \nu) \quad \text{for all} \quad f \in C^0(\mathbb{T}).$$

When $f(t) = \mathrm{e}^{-2\pi i n t}$ we obtain

$$\widehat{(\mu * \nu)}(n) = \widehat{\mu}(n)\widehat{\nu}(n).$$

Hence, if $\nu = f\, \mathrm{d}t$, then $\widehat{\nu}(n) = \widehat{f}(n)$ and we recover $\mu * \nu = \mu * f\, \mathrm{d}t$ by the uniqueness parts of Theorem 2.1.11 and the Riesz–Markov–Kakutami Representation Theorem.

Now, if we can construct a sequence $\{a_n\} \subset \mathbb{C}$ which is ℓ^p summable but such that the norms

$$\left\| \sum_n a_n \widehat{g}(n) \mathrm{e}^{2\pi i n t} \right\|_{L^1(\mathbb{T})}$$

are not uniformly bounded over all trigonometric polynomials g, then Theorems 2.1.10 and 2.1.11 imply that the a_n are not the Fourier coefficients of any measure, nor therefore an $L^q(\mathbb{T})$ function for any $1 \leqslant q < \infty$.

To construct such a sequence, we need a technical lemma.

Lemma 2.1.12. *Let $\varepsilon > 0$. There is a trigonometric polynomial g such that*

$$\|g\|_{L^\infty(\mathbb{T})} \leqslant 1 \quad \textit{and} \quad \|g\|_{L^2(\mathbb{T})} \geqslant \frac{1}{2} \quad \textit{and} \quad \|\widehat{g}(n)\|_{\ell^\infty(\mathbb{Z})} \leqslant \varepsilon.$$

Proof. Let $\psi(t) := \mathrm{e}^{\mathrm{i}\cos(t)}$, so that $|\psi(t)| \equiv 1$. Since $\sum \left|\widehat{\psi}(n)\right|^2 = \|\psi\|^2_{L^2(\mathbb{T})} = 1$ and the sum has more than one *non-zero* term, it follows that $\rho := \left\|\widehat{\psi}(n)\right\|_{\ell^\infty} < 1$. So choose an integer M such that $\rho^M < \varepsilon$ and $\eta < 1$ such that $\eta^M > 1/2$.

Setting $\varphi(t) := \sigma_N \psi(t)$, recall that $\|\varphi\|_{C^0(\mathbb{T})} \leqslant \|\psi\|_{C^0(\mathbb{T})} = 1$. Furthermore,

by the reverse triangle inequality, we can choose N large enough that

$$\| |\varphi| - |\psi| \|_{C^0(\mathbb{T})} < 1 - \eta,$$

whence

$$\eta < |\varphi(t)| \leqslant 1 \quad \text{for all} \quad t \in \mathbb{T}.$$

Finally, set

$$g(t) := \prod_{j=1}^{M} \varphi((3N)^j t).$$

We now compute the Fourier coefficients of g. To do so, note that we can write $g(t) = \varphi(Kt)P(t)$ where $K = 3N$ and P is a trigonometric polynomial.

Claim: $\widehat{g}(n) = \sum_{n=Km+k} \widehat{P}(k)\varphi(m)$.

To prove the claim, write

$$\widehat{g}(n) = \sum_k \widehat{P}(k) \int_0^1 \varphi(Kt) \mathrm{e}^{-2\pi \mathrm{i}(n-k)t} \, \mathrm{d}t.$$

(Note that the sum is finite since P is a trigonometric polynomial.) Writing $Km = n - k$ for $|k| \leqslant N < K/2$, we see that the decomposition $n = Km + k$ is unique for each such k. Furthermore, we can exploit the periodicity of the integrand to get

$$\int_0^1 \varphi(Kt)\mathrm{e}^{2\pi \mathrm{i} Kmt} \, \mathrm{d}t = \sum_{j=1}^{K} \int_{j-1}^{j} \varphi(s)\mathrm{e}^{-2\pi \mathrm{i} ms} \frac{\mathrm{d}s}{K} = \int_0^1 \varphi(s)\mathrm{e}^{-2\pi \mathrm{i} ms} \, \mathrm{d}s,$$

and the claim follows.

We can now repeat the argument for $P(t) = \varphi((3N)^2 t)G(t)$, where G is again a trigonometric polynomial, and so on for M steps. It follows from this induction that the Fourier coefficients of g are products of M coefficients of φ. Thus,

$$|\widehat{g}(n)| \leqslant \rho^M \leqslant \varepsilon \quad \text{and} \quad \frac{1}{2} < \eta^M < |g(t)| \leqslant 1.$$

Note that, as $|\mathbb{T}| = 1$,

$$\|g\|_{L^2(\mathbb{T})}^2 = \int_{\mathbb{T}} |g(t)|^2 \, \mathrm{d}t \geqslant \frac{1}{4},$$

and we have established all the claimed bounds. □

Proposition 2.1.13. *For any $p > 2$, there exists a sequence $\{a_n\} \subset \mathbb{C}$ such that $\sum |a_n|^p < \infty$, but a_n are not the Fourier coefficients of any measure in $M(\mathbb{T})$.*

Proof. For each $j \geqslant 1$, let g_j be a trigonometric polynomial as in Lemma 2.1.12 with $\varepsilon = 2^{-j}$. Then

$$\sum_n |\widehat{g_j}(n)|^p \leqslant 2^{-j(p-2)} \sum_n |\widehat{g_j}(n)|^2 \leqslant 2^{-j(p-2)} \|g_j\|^2_{L^2(\mathbb{T})} \leqslant 2^{-j(p-2)} \tag{2.1.5}$$

as $\|g_j\|_{L^2(\mathbb{T})} \leqslant \|g_j\|_{L^\infty(\mathbb{T})} \leqslant 1$ on $\mathbb{T}$.

Now select integers m_j such that $m_j - m_{j-1} > 3(N_j + N_{j-1})$, where $N_j = \deg g_j$. This ensures that the modulated polynomials $\mathrm{e}^{2\pi \mathrm{i} m_j t} g_j(t)$ have no frequencies in common. In particular, choosing a_n such that $\sum_n a_n \mathrm{e}^{2\pi \mathrm{i} n t} = \sum_j j g_j(t) \mathrm{e}^{2\pi \mathrm{i} m_j}$, satisfies

$$\sum_n |a_n|^p = \sum_j \sum_{|n| \leqslant N_j} j^p \, |\widehat{g_j}(n)|^p < \sum_j j^p 2^{-j(p-2)} < \infty$$

by Eq. (2.1.5).

To check that the frequencies are indeed disjoint, note that a simple induction yields

$$m_j - m_l > 3(N_j + N_l) \quad \text{whenever} \quad j > l.$$

Thus, if $\mathrm{e}^{2\pi \mathrm{i} m_j t} g_j$ and $\mathrm{e}^{2\pi \mathrm{i} m_l t} g_l$ had a common frequency n, we would have

$$n = m_j + k_j = m_l + k_l \quad \text{with} \quad |k_j| \leqslant N_j, |k_l| \leqslant N_l,$$

implying the impossible

$$N_j + N_l \geqslant k_l - k_j = m_j - m_l > 3(N_j + N_l).$$

This is where the Vallée Poussin kernel comes in handy: Suppose there were a measure μ such that $\mu = \sum a_n \mathrm{e}^{2\pi \mathrm{i} n t}$ in $M(\mathbb{T})$. Then, since for the range of frequencies of $m_j - N_j \leqslant n \leqslant m_j + N_j$, $(\mathrm{e}^{2\pi \mathrm{i} m_j t} V_{N_j})\widehat{\ }(n) = 1$ and $\mathrm{e}^{2\pi \mathrm{i} m_j t} g_j$ has no frequencies outside of that range, we have

$$\mu * (\mathrm{e}^{2\pi \mathrm{i} m_j t} V_{N_j})(t) = j \mathrm{e}^{2\pi \mathrm{i} m_j t} g_j(t).$$

Thus, by Theorem 2.1.11, we have

$$\left\| \mu * (\mathrm{e}^{2\pi \mathrm{i} m_j t} V_{N_j}) \right\|_{L^1(\mathbb{T})} \leqslant \|\mu\|_{M(\mathbb{T})} (\mathbb{T}) \left\| V_{N_j} \right\|_{L^1(\mathbb{T})} \leqslant 3 \|\mu\|_{M(\mathbb{T})},$$

since $V_N = 2F_{2N+1} - F_N$ and $\|F_N\|_{L^1(\mathbb{T})} = 1$ for all N. On the other hand, since $\|g_j\|_{L^\infty(\mathbb{T})} \leqslant 1$,

$$\frac{1}{4} \leqslant \|g_j\|^2_{L^2(\mathbb{T})} \leqslant \|g_j\|_{L^1(\mathbb{T})};$$

whence

$$\frac{j}{12} \leqslant \|\mu\|_{M(\mathbb{T})} \quad \text{for all} \quad j \geqslant 1,$$

so Theorem 2.1.10 implies $\mu \notin M(\mathbb{T})$, contradicting our assumption. □

As a consequence, we deduce that the ℓ^p norm of the the Fourier coefficients of f offers no control over the L^p norm of f.

Corollary 2.1.14. *For $p > 2$, there is no constant $C_p > 0$ such that*

$$\|f\|_{L^p(\mathbb{T})} \leqslant C_p \left(\sum_{n \in \mathbb{Z}} \left| \widehat{f}(n) \right|^p \right)^{1/p}.$$

Proof. Choose $f_N(t) := \sum_{|n| \leqslant N} a_n \mathrm{e}^{2\pi i n t}$ and note that $\|f_N\|_{L^p(\mathbb{T})} \to \infty$ as $N \to \infty$, while $\left\| \widehat{f_N}(n) \right\|_{\ell^p(\mathbb{Z})} < \infty$ uniformly in N. □

We now refute the 'dual inequality' to that of Corollary 2.1.14, i.e. when $p < 2$, there is no constant $c_p > 0$ such that

$$\left(\sum_{n \in \mathbb{Z}} \left| \widehat{f}(n) \right|^p \right)^{1/p} \leqslant c_p \|f\|_{L^p(\mathbb{T})}.$$

To this end, consider the function

$$f(t) := \sum_{n=2}^{\infty} \frac{\mathrm{e}^{in \log n}}{\sqrt{n} (\log n)^2} \mathrm{e}^{2\pi i n t}. \tag{2.1.6}$$

Lemma 2.1.15 (Zygmund 2003, Theorem 4.7)**.** *Uniformly in $t \in \mathbb{R}$, we have the estimate*

$$\sum_{n=2}^{N} \mathrm{e}^{in \log n} \mathrm{e}^{int} = O(\sqrt{N}).$$

Proposition 2.1.16. *The series defined in Eq. (2.1.6) converges uniformly. Therefore, $f \in C^0(\mathbb{T})$.*

Proof. We use a summation by parts formula (Grafakos, 2014b, Appendix F):

$$\sum_{n=0}^{N} a_n b_n = A_N b_N - \sum_{n=0}^{N-1} A_n (b_{n+1} - b_n), \quad \text{where} \quad A_n := \sum_{k=0}^{n} a_k.$$

Thus,

$$\sum_{n=2}^{N} \frac{\mathrm{e}^{in\log n}}{\sqrt{n}(\log n)^2}\mathrm{e}^{int} = \sum_{n=2}^{N-1}\left(\sum_{k=2}^{N}\mathrm{e}^{in\log n}\mathrm{e}^{int}\right)\left\{\frac{1}{\sqrt{n}(\log n)^2} - \frac{1}{\sqrt{n+1}(\log(n+1))^2}\right\} + \frac{1}{\sqrt{N}(\log N)^2}\sum_{n=2}^{N}\mathrm{e}^{in\log n}\mathrm{e}^{int}.$$

Focus on the expression in braces, call it L_n. Then

$$(\log n)^2 n^{3/2} L_n = n - n\left(\frac{n}{n+1}\right)^{1/2}\left(\frac{\log n}{\log(n+1)}\right)^2.$$

We now note that

$$\left(\frac{n}{n+1}\right)^{1/2} = \left(1-\frac{1}{n+1}\right)^{1/2} = 1 - \frac{1}{2(n+1)} + O\left(\frac{1}{n}\right),$$

and so, upon writing $\log n = \log(n+1) + \log(\frac{n}{n+1})$,

$$\frac{\log n}{\log(n+1)} = 1 + \frac{\log\left(1-\frac{1}{n+1}\right)}{\log(n+1)} \geqslant 1 - O\left(\frac{1}{n\log n}\right),$$

where we have used the Taylor expansion $\log(1-x) = -x + O(x^2)$ as $x \to 0^+$. Multiplying these terms together, we see that

$$\begin{aligned}(\log n)^2 n^{3/2} L_n &\leqslant n - n\left(1 - \frac{1}{2(n+1)} + O\left(\frac{1}{n^n}\right)\right)\left(1 - O\left(\frac{1}{n\log n}\right)\right)^2 \\ &\leqslant \frac{1}{2} + O\left(\frac{1}{n}\right) \quad \text{as} \quad n \to \infty,\end{aligned}$$

i.e. $L_n = O(n^{-3/2}(\log n)^{-2})$. Hence, by the previous lemma, we obtain

$$\left|\sum_{n=2}^{N} \frac{\mathrm{e}^{in\log n}}{\sqrt{n}(\log n)^2}\mathrm{e}^{int}\right| \lesssim \sum_{n=2}^{N-1}\frac{1}{n^2(\log n)^2} + \frac{1}{N(\log N)^2},$$

and the series on the right clearly converges. □

Corollary 2.1.17. *For $1 \leqslant p < 2$, there is no constant $C_p > 0$ such that*

$$\left(\sum_{n\in\mathbb{Z}}\left|\widehat{f}(n)\right|^p\right)^{1/p} \leqslant C_p \left\|f\right\|_{L^p(\mathbb{T})}.$$

Proof. Consider the function f from Eq. (2.1.6). Since $\mathbb{T}$ is bounded and f is continuous, clearly $f \in L^p(\mathbb{T})$. Its Fourier coefficients are not ℓ^q-summable for any $q < 2$. Indeed, set

$$a_k := 2^k \left|\widehat{f}(2^k)\right|^q = \frac{2^{k(1-q/2)}}{k^{2q}(\log 2)^{2q}}$$

and observe that

$$\frac{a_{k+1}}{a_k} = 2^{1-q/2}\left(\frac{k}{k+1}\right)^{2q} \to 2^{1-q/2} > 1$$

as $k \to \infty$, since $1 - q/2 > 0$. Thus, $\sum_k a_k$ does not converge, so $\sum_k \left|\widehat{f}(k)\right|^q$ diverges by Cauchy's Condensation Test. □

We sum up the results in Corolaries 2.1.14 and 2.1.17.

Proposition 2.1.18. *If $p \neq 2$, then there are no constants $c_p, C_p > 0$ such that*

$$c_p \|f\|_{L^p(\mathbb{T})} \leqslant \left(\sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^p\right)^{1/p} \leqslant C_p \|f\|_{L^p(\mathbb{T})}$$

for all $f \in L^p(\mathbb{T})$.

2.2 Fourier series on the n-torus

Recall that in one dimension, and for $1 < p < \infty$, we have

$$\|S_N f - f\|_{L^p(\mathbb{T})} \to 0 \quad \text{and} \quad \|S_N f\|_{L^p(\mathbb{T})} \lesssim \|f\|_{L^p(\mathbb{T})}. \tag{2.2.1}$$

In fact, both statements are equivalent as per Proposition 2.1.6.

In higher dimensions, i.e. $\mathbb{T}^n = [0,1]^n$, we encounter an ambiguity when truncating the partial sums. The two natural choices alluded to in the introduction—*viz.* spherical truncations, S_N, and cubic truncations, $S_{[N]}$—can still be expressed as a convolution with a Dirichlet kernel. Correspondingly, we introduce the *spherical n-Dirichlet kernel*

$$\overset{\circ}{D}{}^n_N(x) := \sum_{|k|\leqslant N} \mathrm{e}^{2\pi \mathrm{i} x\cdot k}, \quad \text{so} \quad S_N f(x) \equiv \overset{\circ}{D}{}^n_N * f(x),$$

and the *cubic n-Dirichlet kernel*

$$D^n_{[N]}(x) := \sum_{|k_j|\leqslant N} \mathrm{e}^{2\pi \mathrm{i} x\cdot k}, \quad \text{so} \quad S_{[N]} f(x) \equiv D^n_{[N]} * f(x). \tag{2.2.2}$$

The analogues of Eq. (2.2.1) hold for the cubic truncations, but *not* the spherical ones, as we will discuss shortly.

The reason behind the success of the cubic truncations is that we can factor the square n-Dirichlet kernel as follows

$$D^n_{[N]}(x_1, \dots, x_n) = D_N(x_1) \cdots D_N(x_n), \tag{2.2.3}$$

(here D_N is the 1-dimensional Dirichlet kernel) allowing us to apply the 1-dimensional result to deduce the n-dimensional one:

$$\left\|S_{[N]}f - f\right\|_{L^p(\mathbb{T}^n)} \to 0 \quad \text{and} \quad \left\|S_{[N]}f\right\|_{L^p(\mathbb{T}^n)} \lesssim \|f\|_{L^p(\mathbb{T}^n)} \tag{2.2.4}$$

for all $f \in L^p(\mathbb{T}^n)$, $1 < p < \infty$.

By considering products of Fejér kernels, which are again summability kernels, we can reproduce the proof of Corollary 2.1.4.

Lemma 2.2.1. *Trigonometric polynomials, i.e. expressions of the form*

$$g(x) = \sum_{|k| \leqslant d} \widehat{g}(k) \mathrm{e}^{2\pi \mathrm{i} x \cdot k}, \quad \widehat{g}(k) \in \mathbb{C},$$

are dense in $L^p(\mathbb{T}^n)$.

Note that trigonometric polynomials are finite sums, so the 'truncation' type does not matter. This observation lends some generality to the following proposition, which is just a generalisation of Proposition 2.1.6.

Proposition 2.2.2. $S_{[N]}f$ *converges to* f *in the* L^p *norm (for* $1 \leqslant p < \infty$*) if, and only if,*

$$\sup_{N \geqslant 1} \left\|S_{[N]}\right\|_{L^p(\mathbb{T}^n) \to L^p(\mathbb{T}^n)} < \infty. \tag{2.2.5}$$

Proof. Suppose that $S_{[N]}f \to f$ in L^p for each $f \in L^p(\mathbb{T}^n)$. Then the sequence $(\left\|S_{[N]}f\right\|_{L^p(\mathbb{T}^n)})_N$ is bounded for each f. Since the operators $f \mapsto S_{[N]}f$ are bounded on $L^p(\mathbb{T}^n)$, Eq. (2.1.3) follows from the Uniform Boundedness Principle.

Conversely, pick $f \in L^p(\mathbb{T}^n)$ and approximate by a trigonometric polynomial, g such that $\|f - g\|_{L^p(\mathbb{T}^n)} < \varepsilon$. Note that, for large enough N, $S_{[N]} \equiv g$, so we have

$$\left\|S_{[N]}f - f\right\|_{L^p(\mathbb{T}^n)} \leqslant \left\|S_{[N]}f - S_{[N]}g\right\|_{L^p(\mathbb{T}^n)} + \|f - g\|_{L^p(\mathbb{T}^n)} \lesssim \varepsilon.$$

Taking $\varepsilon \to 0$ shows that $S_{[N]}f \to f$ in L^p. □

As remarked earlier, this result applies equally to the spherical truncations. The issue for convergence is that the circular truncation operators are not uniformly bounded, while their cubic counterparts are, as we will now show.

Theorem 2.2.3. *For any $f \in L^p(\mathbb{T}^n)$, the cubic truncations $S_{[N]}f$ converge to f in L^p.*

Proof. It suffices to assume that f is a trigonometric polynomial, which will allow us to treat all sums as being finite. By Proposition 2.2.2 and Eq. (2.2.2), we need only check that

$$\sup_{N>0} \int_{\mathbb{T}^n} \left| D^n_{[N]} * f(x) \right|^p \, \mathrm{d}x \leqslant K^{pn} \, \|f\|^p_{L^p(\mathbb{T}^n)} \tag{2.2.6}$$

for some constant $K > 0$ independent of f.

The proof proceeds by induction, having already disposed of the 1-dimensional base case in Section 2.1. In order to exhaust neither our alphabet nor our reader, we will illustrate the inductive step for $n = 2$ with variables $(x, y) \in \mathbb{T}^2$ and $(j, k) \in \mathbb{Z}^2$.

Since f is a trigonometric polynomial, so is

$$g_{N,y}(x) := \sum_{j\in\mathbb{Z}} \left(\sum_{|k|\leqslant N} \widehat{f}(j,k) \mathrm{e}^{2\pi \mathrm{i} k y} \right) \mathrm{e}^{2\pi \mathrm{i} j x}.$$

We want to relate the sum over $j \in \mathbb{Z}$ to a Fourier series for $f(x, y)$ when x is held fixed. To do this, set $f^s(x) := f(x, s)$, and note that

$$f(x,s) = f^s(x) = \sum_{j\in\mathbb{Z}} \widehat{f^s}(j) \mathrm{e}^{2\pi \mathrm{i} j x}, \tag{2.2.7}$$

where

$$\widehat{f^s}(j) = \int_0^1 f^s(t) \mathrm{e}^{-2\pi \mathrm{i} j t} \, \mathrm{d}t = \int_0^1 f(t,s) \mathrm{e}^{-2\pi \mathrm{i} j t} \, \mathrm{d}t. \tag{2.2.8}$$

Now, using Eqs. (2.2.7) and (2.2.8) in the fourth and third equalities below, respectively, we have

$$\begin{aligned}
\sum_{j\in\mathbb{Z}} \widehat{f}(j,k) \mathrm{e}^{2\pi \mathrm{i} j x} &= \sum_{j\in\mathbb{Z}} \left(\int_0^1 \int_0^1 f(t,s) \mathrm{e}^{-2\pi \mathrm{i}(jt+ks)} \, \mathrm{d}t \, \mathrm{d}s \right) \mathrm{e}^{2\pi \mathrm{i} j x} \\
&= \int_0^1 \sum_{j\in\mathbb{Z}} \left(\int_0^1 f(t,s) \mathrm{e}^{-2\pi \mathrm{i} j t} \, \mathrm{d}t \right) \mathrm{e}^{2\pi \mathrm{i} j x} \mathrm{e}^{-2\pi \mathrm{i} k s} \, \mathrm{d}s \\
&= \int_0^1 \left(\sum_{j\in\mathbb{Z}} \widehat{f^s}(j) \mathrm{e}^{2\pi \mathrm{i} j x} \right) \mathrm{e}^{-2\pi \mathrm{i} k s} \, \mathrm{d}s
\end{aligned}$$

$$= \int_0^1 f^s(x)\mathrm{e}^{-2\pi\mathrm{i}ks}\,\mathrm{d}s$$
$$= \int_0^1 f(x,s)\mathrm{e}^{-2\pi\mathrm{i}ks}\,\mathrm{d}s = \widehat{f_x}(k), \tag{2.2.9}$$

as desired, where we now take $f_x(y) := f(x,y)$. With these notational gymnastics out of the way, we can proceed to re-write $g_{N,y}$ as follows:

$$\begin{aligned} g_{N,y}(x) &= \sum_{|k|\leqslant N}\left(\sum_{j\in\mathbb{Z}}\widehat{f}(j,k)\mathrm{e}^{2\pi\mathrm{i}jx}\right)\mathrm{e}^{2\pi\mathrm{i}ky} \\ &= \sum_{|k|\leqslant N}\widehat{f_x}(k)\mathrm{e}^{2\pi\mathrm{i}ky} && \text{by Eq. (2.2.9)} \\ &= (D_N * f_x)(y). \end{aligned}$$

At this point we use the factorisation property Eq. (2.2.3):

$$\begin{aligned} D_N * g_{N,y}(x) &= \int_0^1 D_N(t)g_{N,y}(x-t)\,\mathrm{d}t \\ &= \int_0^1 D_N(t)(D_N * f_{x-t})(y)\,\mathrm{d}t \\ &= \int_0^1\int_0^1 D_N(t)D_N(s)f_{x-t}(y-s)\,\mathrm{d}s\,\mathrm{d}t \\ &= \int_0^1\int_0^1 D^2_{[N]}(t,s)f(x-t,y-s)\,\mathrm{d}t\,\mathrm{d}s \\ &= D^2_{[N]} * f(x,y). \end{aligned}$$

From these equations we can conclude, using Eq. (2.2.6) for $n=1$,

$$\begin{aligned} \int_0^1\int_0^1 \left|D^2_{[N]} * f(x,y)\right|^p \,\mathrm{d}x\,\mathrm{d}y &= \int_0^1\int_0^1 |D_N * g_{N,y}(x)|\,\mathrm{d}x\,\mathrm{d}y \\ &\leqslant K^p\int_0^1\int_0^1 |g_{N,y}(x)|\,\mathrm{d}x\,\mathrm{d}y \\ &= K^p\int_0^1\int_0^1 |D_N * f_x(y)|\,\mathrm{d}y\,\mathrm{d}x \\ &\leqslant K^{2p}\int_0^1\int_0^1 |f_x(y)|\,\mathrm{d}x\,\mathrm{d}y \\ &= K^{2p}\left\|f\right\|_{L^p(\mathbb{T}^2)}. \end{aligned}$$

In the more general case, the first inequality uses the base case $n=1$ to obtain a

reduction

$$\left\|D_N * g_{N,x'}\right\|_{L^p(\mathbb{T})} \leqslant K \left\|g_{N,x'}\right\|_{L^p(\mathbb{T})},$$

while the second applies the inductive hypothesis on $n-1$ for the bound

$$\left\|D_{[N]}^{n-1} * f_{x_1}\right\|_{L^p(\mathbb{T}^{n-1})} \leqslant K^{n-1} \left\|f_{x_1}\right\|_{L^p(\mathbb{T}^{n-1})},$$

where $x = (x_1, x') \in \mathbb{T} \times \mathbb{T}^{n-1}$. (See Grafakos, 2014b, Theorem 4.1.8, whose proof we slighly altered here.) □

As can be seen in the proof, the crucial step is the reduction of dimension afforded by the function $g_{N,x'}$ which allows us to use the convergence in $L^p(\mathbb{T}^{n-1})$ to conclude convergence in $L^p(\mathbb{T}^n)$. Once again, this in made possible by the factorisation of the cubic n-Dirichlet kernel, a property not enjoyed by the spherical kernel.

The failure of convergence of the spherical means is a consequence of Ferfferman's celebrated Ball Multiplier Theorem from 1971. In-depth treatments may be found in Grafakos (2014a); Stein (1993). We offer a summary of the proof in Section 3.8, as we adapt it to Fourier series on a 45-90-45 triangle.

2.2.1 Appendix: Cesàro summation and Bochner–Riesz means

One can also ask about the Cesàro sums of the n-dimensional Fourier series, i.e.

$$\sigma_{[N]}^n f(x) = \frac{1}{N+1} \sum_{k=0}^{N} S_{[k]} f(x) = \sum_{|k_j| \leqslant N} \left(1 - \frac{|k|}{N+1}\right) \widehat{f}(k) \mathrm{e}^{2\pi \mathrm{i} x \cdot k},$$

which can again be expressed as a convolution with the n-dimensional Fejér kernel

$$F_N^n(x) := \prod_{j=1}^{n} F_N(x_j) \equiv \frac{1}{(N+1)^n} \prod_{j=1}^{n} \left(\frac{\sin(\pi(N+1)x_j)}{\sin(\pi x_j)}\right)^2.$$

Theorem 2.1.2 applies and we conclude that, for all $1 \leqslant p < \infty$,

$$\left\|\sigma_{[N]}^n f - f\right\|_{L^p(\mathbb{T}^n)} \to 0 \quad \text{for all} \quad f \in L^p(\mathbb{T}^n).$$

This analysis is possible since the set of n-dimensional Fejér kernels is still a summability kernel (Grafakos, 2014b, Prop. 3.3.10). However, consider now the

Cesàro means of the *spherical sums*, i.e.

$$\sum_{|k|\leqslant N}\left(1-\frac{|k|}{N+1}\right)\widehat{f}(k)\mathrm{e}^{2\pi\mathrm{i}x\cdot k}$$

(note that we now sum over k with $k_1^2+\ldots+k_n^2\leqslant N^2$). Then, there is a function $f\in L^1(\mathbb{T}^3)$ whose spherical Cesàro means diverge a.e.! (See Grafakos, 2014b, Theorem 4.2.5.)

One way to recover convergence is to study the modified expression

$$\sum_{|k|\leqslant N}\left(1-\frac{|k|}{N+1}\right)^{1+\varepsilon}\widehat{f}(k)\mathrm{e}^{2\pi\mathrm{i}x\cdot k}\quad\text{for}\quad\varepsilon>0.$$

Equivalently, one can consider the so-called *Bochner–Riesz means*

$$B_N^\alpha f(x):=\sum_{|k|\leqslant N}\left(1-\frac{|k|^2}{N^2}\right)^{\alpha}\widehat{f}(k)\mathrm{e}^{2\pi\mathrm{i}x\cdot k}\quad\text{for}\quad\mathrm{Re}(\alpha)\geqslant 0.$$

The following theorem is known.

Theorem 2.2.4. *Let $1\leqslant p<\infty$ and $f\in L^p(\mathbb{T}^n)$. If $\alpha>(n-1)\left|\frac{1}{p}-\frac{1}{2}\right|$, then*

$$\|B_N^\alpha f-f\|_{L^p(\mathbb{T}^n)}\to 0.$$

For further discussion and proofs, see Sect. 4.1.3 and following in Grafakos (2014b). For the Bochner–Riesz multiplier and its relationship to the restriction problem for the Fourier transform, the reader is invited to consult Chapter 5 of Grafakos (2014a).

2.3 Decay of Fourier coefficients

In this section we record some results on the decay of Fourier coefficients, with an eye to generalising them in Chapter 5 to the case of Bessel–Fourier coefficients.

The basic result is the Riemann–Lebesgue lemma.

Theorem 2.3.1. *Given $f\in L^1(\mathbb{T})$, we have $\widehat{f}(k)\to 0$ as $k\to\infty$.*

If we assume higher regularity of the coefficients, e.g. $f\in C^m(\mathbb{T})$ for some $m\geqslant 1$, then the formula

$$\widehat{f^{(m)}}(k)=(-2\pi\mathrm{i})^m\,|k|^m\,\widehat{f}(k)$$

implies that the coefficients decay like $|k|^{-m}$.

For a merely integrable f, however, we cannot say anything more about the decay of $\widehat{f}(k)$, as the following theorem shows (see, e.g., Grafakos, 2014b, Theorem 3.3.4).

Theorem 2.3.2. *Let $a_k \geqslant 0$ be a sequence such that $a_k \to 0$ as $|k| \to \infty$. Then, there is a function $f \in L^1(\mathbb{T})$ whose Fourier coefficients satisfy $\widehat{f}(k) \geqslant a_k$ for all $k \in \mathbb{Z}$.*

We sketch the main ingredients of the proof, which can be found in §3.3 of Grafakos (2014b), since we will use reuse some of these ideas in Chapter 5.

Recall that a sequence $\{c_m\}_m$ is *convex* if

$$c_m \geqslant 0 \quad \text{and} \quad c_m + c_{m+2} \geqslant 2c_{m+1} \quad \text{for all} \quad m \in \mathbb{N}.$$

Lemma 2.3.3. *Given a sequence $a_m \geqslant 0$ tending to zero as $m \to \infty$, there is a convex sequence $c_m \geqslant a_m$ also tending to zero.*

Lemma 2.3.4. *Given a convex decreasing sequence $\{c_m\}_{m\geqslant 0}$ of positive numbers which tend to zero, and a fixed integer $k \geqslant 0$,*

$$\sum_{l=0}^{\infty}(l+1)(c_{l+k} + c_{l+k+2} - 2c_{l+k+1}) = c_k.$$

The main idea behind the proof of Theorem 2.3.2 is to use Lemma 2.3.3 to find a convex sequence $c_m \geqslant a_m \geqslant 0$ and use the Fejér kernels to define the function

$$f(x) := \sum_{k=1}^{\infty}(k+1)(c_k + c_{k+2} - 2c_{k+1})F_k(x).$$

Since, by Eq. (2.1.2), $F_k(x) \geqslant 0$, we can easily compute

$$\|F_k\|_{L^1(\mathbb{T})} = \int_{\mathbb{T}} F_k(x)\,\mathrm{d}x = 1,$$

and so, by Lemma 2.3.4,

$$\sum_{k=0}^{\infty}(k+1)(c_k + c_{k+2} - 2c_{k+1})\,\|F_k\|_{L^1(\mathbb{T})} = c_0 < \infty.$$

Thus, f is a well-defined L^1 function, and it is straightforward (if tedious) to check that $\widehat{f}(k) = c_{|k|}$ for all $k \in \mathbb{Z}$.

We provide fuller details in Chapter 5.

Chapter 3

Triangular domains

> τὸ δὲ τρίγωνον ἁπάντων ἡγεμονοῦν τῶν
> εὐθυγράμμων παντί που δῆλον. ὅτι τῇ τριάδι
> κατέχεται καὶ κατ'ἐχείνην μεμόρφωται.
>
> —Proclus

In this chapter we establish L^p convergence for trigonometric series of eigenfunctions for the Dirichlet Laplacian on three types of triangular domains: (i) the 45-90-45 triangle, (ii) the equilateral triangle, and (iii) the hemiequilateral triangle (i.e. half an equilateral triangle cut along its height).

Eigenfunctions for the latter types, (ii) and (iii), were obtained by Práger (1998) and McCartin (2003), as mentioned in Chapter 1. Using their insights, we prove that, on each of these domains, any L^p function can be written as a norm-convergent series of eigenfunctions. We also give a complete treatment of these issues for the simpler case (i). As far as we know, the result for type-(i) triangles is new.

In Section 3.1 we derive the eigenfunctions for the type-(i) triangles, prove their completeness, and establish L^p convergence of the associated series. This serves as a useful prototype for the arguments that we will develop in the following sections.

The result for equilateral domains is obtained by a decomposition of a function into a symmetric and antisymmetric part with respect to the line $x = 0$. This reduces the problem to type-(iii) domains, which, once solved, easily yields the equilateral case.

Thus, in Sections 3.2 and 3.4 we separately derive the antisymmetric and symmetric modes, following the ingenious approach of Práger (1998) (see Fig. 3.2). The hemiequilateral triangle T_1 is tiled into the rectangle $R = [0, \sqrt{3}] \times [0, 1]$, where we exploit the classical L^p convergence of 'double-sine' and 'cosine-sine' series on R to derive convergence on T_1. This is our main contribution, which is worked out

in Sections 3.3 and 3.5. These results are combined in Section 3.6 to obtain L^p convergence on the equilateral triangle.

In Section 3.7 we conclude with some remarks about the limitations of this argument: owing to a theorem of Lamé, the three domains (i)–(iii) listed above are the 'only' ones amenable to this procedure, based as it is on expansions in terms of trigonometric functions.

The appendix in Section 3.8 outlines the proof of the failure of convergence of the 'spherical' partial sums for the 45-90-45 triangle, which is a straightforward adaptation of Fefferman's argument from 1971. We thus take this opportunity to give a summary of the main ideas behind his classical paper on the ball multiplier.

This chapter has appeared in *Proceedings of the Edinburgh Mathematical Society* as Acosta Babb (2023b).

3.1 The 45-90-45 triangle

Taking our cue from Práger's analysis of the equilateral case, we reflect the 45-90-45 triangle along the hypotenuse to obtain a square; see Fig. 3.1.

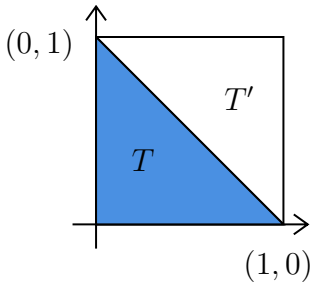


Figure 3.1: Reflecting T into the square $[0,1]^2$ along the diagonal $y = 1 - x$.

Placing the triangle, T, at the vertices $(1,0)$, $(0,1)$, and $(0,0)$, the reflections have the simple from

$$x = 1 - \eta \quad \text{and} \quad y = 1 - \xi, \quad \text{where} \quad (\xi, \eta) \in T.$$

Given a function $f : T \to \mathbb{R}$ we define the *prolongation* of f

$$\mathscr{P}f(x,y) := \begin{cases} f(x,y), & \text{if } (x,y) \in T; \\ -f(1-y, 1-x), & \text{if } (x,y) \in T'; \end{cases}$$

where T' denotes the reflected triangle, so that $[0,1]^2 = T \cup T'$.

We now expand $\mathscr{P}f$ as a double-sine series on $[0,1]^2$:

$$\begin{aligned}\widehat{\mathscr{P}f}(m,n) &:= 4\int_0^1\int_0^1 \mathscr{P}f(x,y)\sin(\pi mx)\sin(\pi ny)\,\mathrm{d}x\,\mathrm{d}y \\ &= 4\int_T f(\xi,\eta)\Big[\sin(\pi m\xi)\sin(\pi n\eta) \\ &\qquad -\sin(\pi m(1-\eta))\sin(\pi n(1-\xi))\Big]\,\mathrm{d}\xi\,\mathrm{d}\eta \\ &= 4\int_T f(\xi,\eta)u_{m,n}(\xi,\eta)\,\mathrm{d}\xi\,\mathrm{d}\eta, \qquad (3.1.1)\end{aligned}$$

where we defined the functions

$$u_{m,n}(\xi,\eta) := \sin(\pi m\xi)\sin(\pi n\eta) - \sin(\pi m(1-\eta))\sin(\pi n(1-\xi))$$

for all points $(\xi,\eta)\in T$.

Lemma 3.1.1. *We record the following facts about the functions $u_{m,n}$ (indexed by $m,n\in\mathbb{N}$).*

(i) For all $m\neq n$:

$$u_{m,n} = \sin(\pi m\xi)\sin(\pi n\eta) - (-1)^{m+n}\sin(\pi n\xi)\sin(\pi m\eta),$$

and clearly $u_{m,n}\equiv 0$ when $n=m$.

(ii) Symmetry of the indices:

$$u_{n,m} = -(-1)^{m+n}u_{m,n} \quad \text{for all} \quad m\neq n;$$

therefore, it suffices to consider the set with $0<m<n$.

(iii) Letting (x,y) range over $[0,1]^2$,

$$\mathscr{P}\left[u_{m,n}|_T\right](x,y) = u_{m,n}(x,y) \quad \text{for all} \quad m\neq n.$$

In other words: restricting $u_{m,n}$ from $[0,1]^2$ to T and prolonging by $\mathscr{P}$ yields $u_{m,n}$ once more, as a function on $[0,1]^2$.

(iv) The set $\{u_{m,n} : 0<m<n\}$ is a complete, orthogonal set in $L^2(T)$ and

$$\|u_{m,n}\|_{L^2(T)} = \frac{1}{2}.$$

Proof. Statements (i)–(iii) are easily verified. For the completeness claim in (iv), suppose that $f \in L^2(T)$ is such that

$$\int_T f u_{m,n} = 0 \quad \text{for all} \quad 0 < m < n.$$

It follows from (i), (iii), and (3.1.1) that the Fourier coefficients of $\mathscr{P}f$ on $[0,1]^2$ vanish, so $\mathscr{P}f = 0$ in $L^2([0,1]^2)$, and therefore $f = \mathscr{P}f|_T = 0$. This proves the completeness of the set.

From (iii) and (3.1.1), it is clear that

$$\int_T u_{m,n} u_{k,l} = \int_0^1 \int_0^1 u_{m,n}(x,y) \sin(\pi k x) \sin(\pi l y) \, \mathrm{d}x \, \mathrm{d}y.$$

Ordering the indices as $0 < m < n$ and $0 < k < l$, it follows from the formula in (i) that the functions $u_{m,n}$ and $u_{k,l}$ are pairwise orthogonal; setting $m = k$ and $n = l$ easily yields the norm of $u_{m,n}$. □

Corollary 3.1.2. *The function $u_{m,n}$ with $0 < m < n$ is an eigenfunction of (minus) the Dirichlet Laplacian on T with eigenvalue $\pi^2(m^2 + n^2)$.*

Proof. It is immediate from (i) and (iii) in Lemma 3.1.1 that

$$-\Delta u_{m,n}(x,y) = \pi^2(m^2 + n^2) u_{m,n}(x,y)$$

pointwise for all $(x,y) \in [0,1]^2$, and $u_{m,n} \equiv 0$ on the boundary of the square. Thus, restricting to $T \subset [0,1]^2$ yields eigenfunctions of the Laplacian on T that clearly vanish along the edges parallel to the axes. It remains to show that $u_{m,n}$ also vanishes on the hypotenuse $y = 1 - x$, but this is immediately verified on substitution into the formula from (i) above. □

We thus have a complete, orthogonal set of eigenfunctions for the Dirichlet Laplacian on T. We can define the Fourier coefficients as usual:

$$f^{\triangle}(m,n) := \frac{1}{\|u_{m,n}\|^2_{L^2(T)}} \int_T f u_{m,n} = \widehat{\mathscr{P}f}(m,n);$$

the last equality holds by Eq. (3.1.1).

The partial sum operators, truncated by indices, for the Fourier series on T and $[0,1]^2$ are

$$S_N^T f := \sum_{0<m<n\leqslant N} f^{\triangle}(m,n) u_{m,n}, \quad \text{and}$$

$$S_N^{[0,1]^2} := \sum_{0<m,n\leqslant N} \widehat{\mathscr{P}f}(m,n)\sin(\pi mx)\sin(\pi ny),$$

respectively.

The next proposition tells us that the prolongation operation commutes with the Fourier partial sums, allowing us to 'push forward' the L^p convergence results from the square to the triangle.

Proposition 3.1.3. *Let $f \in L^p(T)$. Then:*

$$\mathscr{P}\left[S_N^T f\right](x,y) = S_N^{[0,1]^2}\left[\mathscr{P}f\right](x,y) \quad \textit{for all} \quad (x,y) \in [0,1]^2.$$

Proof. Break up the lattice $[1,N]^2 \cap \mathbb{N}^2$ into the following regions:

$$I : 0 < m < n \leqslant N \quad \text{and} \quad II : 0 < n < m \leqslant N.$$

(Recall that, by (i) in Lemma 3.1.1, the diagonal $m = n$ yields no eigenfunctions.) Hence:

$$\begin{aligned}
S_N^{[0,1]^2}\left[\mathscr{P}f\right](x,y) &= \sum_I \widehat{\mathscr{P}f}(m,n)\sin(\pi mx)\sin(\pi ny) \\
&\quad + \sum_{II} \widehat{\mathscr{P}f}(m,n)\sin(\pi mx)\sin(\pi ny) \\
&= \sum_I \widehat{\mathscr{P}f}(m,n)\Big[\sin(\pi mx)\sin(\pi ny) \\
&\quad - (-1)^{m+n}\sin(\pi nx)\sin(\pi my)\Big] && \text{Lemma 3.1.1 (ii)} \\
&= \sum_I \widehat{\mathscr{P}f}(m,n)u_{m,n}(x,y) && \text{Lemma 3.1.1 (i)} \\
&= \sum_I \widehat{\mathscr{P}f}(m,n)\mathscr{P}u_{m,n}(x,y) && \text{Lemma 3.1.1 (iii)} \\
&= \mathscr{P}\left[S_N^T f\right](x,y),
\end{aligned}$$

where in the last line we make use of the linearity of $\mathscr{P}$ and the identity

$$f^{\triangle}(m,n) = \widehat{\mathscr{P}f}(m,n)$$

for all (m,n) in I. □

We now have everything we need to prove our result.

Theorem 3.1.4. *Let $f \in L^p(T)$ with $1 < p < \infty$. Then, $S_N^T f \to f$ in $L^p(T)$.*

Proof. Note that, for $f \in L^p$, we have

$$\|\mathscr{P} f\|^p_{L^p([0,1]^2)} = 2\,\|f\|^p_{L^p(T)}\,.$$

Thus, by successively applying this equality and Proposition 3.1.3, we have

$$\begin{aligned}\left\|S_N^T f - f\right\|^p_{L^p(T)} &= \frac{1}{2}\left\|\mathscr{P}\left[S_N^T f\right] - \mathscr{P} f\right\|^p_{L^p([0,1]^2)} \\ &= \frac{1}{2}\left\|S_N^{[0,1]^2}\mathscr{P} f - \mathscr{P} f\right\|^p_{L^p([0,1]^2)} \to 0,\end{aligned}$$

by the L^p convergence of Fourier double-sine series on $[0,1]^2$. □

3.2 Antisymmetric eigenfunctions on the equilateral triangle

To tackle convergence on the equilateral triangle T, we divide it along its midline $x = 0$ into two congruent hemiequilateral triangles, and call the right-hand one T_1 (see Fig. 3.2). Any function $f : T \to \mathbb{R}$ may be decomposed into a symmetric and antisymmetric part with respect to x, i.e. $f = f_a + f_s$ where

$$f_a(x,y) := \frac{f(x,y) - f(-x,y)}{2} \quad \text{and} \quad f_s(x,y) := \frac{f(x,y) + f(-x,y)}{2}$$

for all $(x,y) \in T$. Clearly each of f_a and f_s is determined by its values on $T_1 \subset T$.

Let us begin with the L^p convergence of the Fourier series of f_a on the triangle T_1. Denote by (ξ,η) the coordinates of a point in T_1. Following Práger (1998), we will define a function on the rectangle $R = [0,\sqrt{3}] \times [0,1]$ by successively reflecting the point $(\xi,\eta) \in T_1$ across various boundary lines, according to the pattern in Fig. 3.2. Each time we perform a reflection, we must chang the sign of the function, in order to preserve smoothness along the vanishing lines shown inside R. Thus, for each coordinate pair (x,y) in the rectangle, we write

$$(x,y) = (x_i(\xi,\eta), y_i(\xi,\eta)) \quad \text{for exactly one} \quad (x_i,y_i) \in T_i.$$

Note that the i subscript serves just as a reminder that the point (x_i,y_i) lies in the region T_i of the Cartesian plane.

The *antisymmetric prolongation* of $u\colon T_1 \to \mathbb{R}$ to the rectangle R is defined

by

$$\mathscr{P}_a u(x,y) := \begin{cases} u(\xi,\eta) & \text{if } (x,y) = (x_1,y_1) \in T_1, \\ -u(\xi,\eta) & \text{if } (x,y) = (x_2,y_2) \in T_2, \\ u(\xi,\eta) & \text{if } (x,y) = (x_3,y_3) \in T_3, \\ u(\xi,\eta) & \text{if } (x,y) = (x_4,y_4) \in T_4, \\ -u(\xi,\eta) & \text{if } (x,y) = (x_5,y_5) \in T_5, \\ u(\xi,\eta) & \text{if } (x,y) = (x_6,y_6) \in T_6. \end{cases}$$

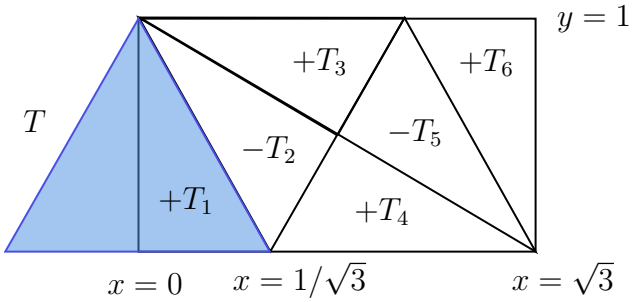


Figure 3.2: The equilateral triangle T (shaded), the right-angled triangle T_1, and six congruent copies T_i arranged into the rectangle $R = [0,\sqrt{3}] \times [0,1]$. The '$\pm$' signs indicate a symmetric or antisymmetric reflection, respectively, in the definition of $\mathscr{P}_a$. Assuming zero boundary conditions on T_1, the extension by $\mathscr{P}_a$ vanishes on all lines drawn in R and its boundary. See Práger (1998).

More succinctly (albeit less precisely) we express this relation as

$$\mathscr{P}_a u(x,y) = \sum_{i=1}^{6} c_i u(\xi(x_i,y_i),\eta(x_i,y_i)), \tag{3.2.1}$$

where c_i are the appropriate '$\pm$' signs.

From Fig. 3.2 it is clear that the prolongation $\mathscr{P}_a u$ vanishes along the sides of the rectangle R. We therefore express $\mathscr{P}_a f_a$ as a double-sine series on R:

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \widehat{\mathscr{P}_a f_a}(m,n) \sin\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y)\,.$$

The Fourier coefficients of $\mathscr{P}_a f_a$ are:

$$\widehat{\mathscr{P}_a f_a}(m,n) = \frac{4}{\sqrt{3}} \int_R \mathscr{P}_a f_a(x,y) \sin\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y)\ \mathrm{d}x\,\mathrm{d}y$$

$$= \frac{4}{\sqrt{3}} \int_{T_1} f_a(\xi,\eta) \left[\sum_{i=1} c_i \sin\left(\frac{\pi m x_i}{\sqrt{3}}\right) \sin(\pi n y_i) \right] \mathrm{d}\xi\, \mathrm{d}\eta \quad \text{by (3.2.1)}$$
$$= \frac{4}{\sqrt{3}} \int_{T_1} f_a(\xi,\eta) u_{m,n}(\xi,\eta)\, \mathrm{d}\xi\, \mathrm{d}\eta, \tag{3.2.2}$$

where we defined the antisymmetric eigenfunctions as

$$u_{m,n}(\xi,\eta) := \sum_{i=1} c_i \sin\left(\frac{\pi m x_i}{\sqrt{3}}\right) \sin(\pi n y_i)\,.$$

(Recall that each pair $(x_i, y_i) \in R$ is a function of $(\xi,\eta) \in T_1$.)

By explicit computation using the parametrisations of the T_i, Práger obtained the following.

Lemma 3.2.1. *For integers $0 < m < n$ with the same parity (i.e. both odd or both even),*

$$\begin{aligned} u_{m,n}(x,y) = {} & 2\sin\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y) \\ & - 2(-1)^{(m+n)/2} \sin\left(\frac{\pi x}{2\sqrt{3}}(m+3n)\right) \sin\left(\frac{\pi y}{2}(m-n)\right) \\ & + 2(-1)^{(m-n)/2} \sin\left(\frac{\pi x}{2\sqrt{3}}(m-3n)\right) \sin\left(\frac{\pi y}{2}(m+n)\right). \end{aligned} \tag{3.2.3}$$

Furthermore, $u_{m,n} \equiv 0$ whenever n and m have different parity, or *$n = m$,* or *$m = 3n$. Finally, with $0 < m < n$ as above, define the pairs*

$$\begin{cases} m' = \frac{1}{2}(3n - m), \\ n' = \frac{1}{2}(m + n); \end{cases} \quad \textit{and} \quad \begin{cases} m'' = \frac{1}{2}(m + 3n), \\ n'' = \frac{1}{2}(m - n). \end{cases}$$

Then,

$$u_{m',n'} = -(-1)^{(m-n)/2} u_{m,n} \quad \textit{and} \quad u_{m'',n''} = (-1)^{(m+n)/2} u_{m,n}. \tag{3.2.4}$$

The next technical result is all that stands between us and the L^p convergence of a $u_{m,n}$-expansion of f_a on T_1.

Lemma 3.2.2. *The functions $\{u_{m,n} : 0 < m < n$ and $m \equiv n \mod 2\}$ are pairwise orthogonal (in both T and R). Furthermore,*

$$\mathscr{P}_a(u_{m,n}|_T(x,y)) = u_{m,n}(x,y) \quad \textit{for all} \quad (x,y) \in R, \tag{3.2.5}$$

from which it follows that

$$\|u_{m,n}\|^2_{L^2(T_1)} = \frac{\sqrt{3}}{2}. \tag{3.2.6}$$

Corollary 3.2.3. *The functions $u_{m,n}$ are eigenfunctions of (minus) the Dirichlet Laplacian on T_1, with eigenvalue $\pi^2(\frac{m^2}{3} + n^2)$.*

Proof. Since each $\mathscr{P}_a u_{m,n}$ is a finite combination of eigenfunctions on R, we have

$$-\Delta \mathscr{P}_a u_{m,n}(x,y) = \pi^2 \left(\frac{m^2}{3} + n^2\right) \mathscr{P}_a u_{m,n}(x,y) \quad \text{for all} \quad (x,y) \in R$$

by direct calculation. This equation holds pointwise owing to the smoothness of the functions; therefore, it holds when restricted to T_1. But $\mathscr{P}_a u_{m,n} \equiv u_{m,n}$ on T_1, whence

$$-\Delta u_{m,n}(\xi,\eta) = \pi^2 \left(\frac{m^2}{3} + n^2\right) u_{m,n}(\xi,\eta) \quad \text{for all} \quad (\xi,\eta) \in T.$$

These functions clearly vanish on the boundary of the triangle by construction. □

So far, our discussion is a recap of the results in Práger (1998), although we have added a more explicit proof of Corollary 3.2.3 than can be found there. We will now use these results to obtain L^p convergence of the eigenfunction expansions.

3.3 L^p convergence of the Fourier series of f_a

Equation Eq. (3.2.6) tells us that the 'Fourier coefficients' of f_a (with respect to $u_{m,n}$) are

$$f_a^{\triangle}(m,n) := \frac{2}{\sqrt{3}} \int_{T_1} f_a(\xi,\eta) u_{m,n}(\xi,\eta)\, \mathrm{d}\xi\, \mathrm{d}\eta,$$

and it follows from (3.2.2) that

$$\widehat{\mathscr{P}_a f_a}(m,n) = 2 f_a^{\triangle}(m,n). \tag{3.3.1}$$

To help keep track of our indices, we will introduce the sets

$$\mathscr{U}_N := \{(m,n) : 0 < m < n \leqslant N \text{ and } m \equiv n \mod 2\}.$$

Denote the N-th 'component-wise' partial sums on the triangle T_1 and rect-

angle R, respectively, by

$$S_N^{T_1} f_a(\xi,\eta) := \sum_{(m,n)\in\mathscr{U}_N} f_a^{\triangle}(m,n) u_{m,n}(\xi,\eta),$$
$$S_N^{R} \mathscr{P}_a f_a(x,y) := \sum_{0<m,n\leqslant N} \widehat{\mathscr{P}_a f_a}(m,n) \sin\left(\frac{\pi m}{\sqrt{3}}x\right) \sin(\pi n y)\,.$$

The next theorem is a key ingredient in the proof of our main result.

Theorem 3.3.1. *Let $f \in L^p(T)$ with $1 < p < \infty$, and denote its antisymmetric part by f_a. Then $f_a \in L^p(T_1)$ and $S_N^{T_1} f_a \to f_a$ in $L^p(T_1)$.*

Proof. Since $f \in L^p(T)$ and f_a is a linear combination of f and its reflections, clearly $f_a \in L^p(T)$ and, by symmetry (and abuse of notation), $f_a = f_a|_{T_1} \in L^p(T_1)$.

Next, we break up the set of indices with $0 < m, n \leqslant N$ into three sets:

$$0 < m < n, \quad n < m < 3n, \quad \text{and} \quad 3n < m,$$

noting that the coefficients when $m = n$ and $m = 3n$ vanish. From Lemma 3.2.1, we may label these groups as (m,n), (m',n'), and (m'',n''), respectively. Note that such labelling divides $[1,N]\times[1,N]\cap\mathbb{Z}^2$ into three disjoint regions, which we label as $I = \mathscr{U}_N$, II, and III; see Fig. 3.3.

It follows from (3.2.4) that

$$\widehat{\mathscr{P}_a f_a}(m,n) = -(-1)^{(m-n)/2}\widehat{\mathscr{P}_a f_a}(m',n') = (-1)^{(m+n)/2}\widehat{\mathscr{P}_a f_a}(m'',n'').$$

Hence,

$$\begin{aligned}
S_N^{R} \mathscr{P}_a f_a(x,y) = &\sum_{(m,n)\in\mathscr{U}_N} \widehat{\mathscr{P}_a f_a}(m,n) \sin\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y) \\
&+ \sum_{(m',n')\in II} \widehat{\mathscr{P}_a f_a}(m',n') \sin\left(\frac{\pi m' x}{\sqrt{3}}\right) \sin\left(\pi n' y\right) \\
&+ \sum_{(m'',n'')\in III} \widehat{\mathscr{P}_a f_a}(m'',n'') \sin\left(\frac{\pi m'' x}{\sqrt{3}}\right) \sin\left(\pi n'' y\right) \\
= &\sum_{(m,n)\in\mathscr{U}_N} \widehat{\mathscr{P}_a f_a}(m,n) \Bigg[\sin\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y) \\
&- (-1)^{(m+n)/2} \sin\left(\frac{\pi x}{2\sqrt{3}(m+3n)}\right) \sin\left(\frac{\pi y}{2}(m-n)\right) \\
&+ (-1)^{(m-n)/2} \sin\left(\frac{\pi x}{2\sqrt{3}}(m-3n)\right) \sin\left(\frac{\pi y}{2}(m+n)\right)\Bigg]
\end{aligned}$$

$$= \sum_{(m,n)\in\mathscr{U}_N} \frac{\widehat{\mathscr{P}_a f_a}(m,n)}{2} u_{m,n}(x,y) \quad \text{by (3.2.3)}$$
$$= \sum_{(m,n)\in\mathscr{U}_N} f_a^{\triangle}(m,n)\mathscr{P}_a u_{m,n}(x,y).$$

The last line follows from (3.2.5) and (3.3.1). Since $\mathscr{P}_a$ is a linear operator, we have therefore proved that

$$\mathscr{P}_a\left[S_N^{T_1} f_a\right](x,y) = S_N^R \mathscr{P}_a f_a(x,y).$$

Furthermore, since each transformation $T_1 \to T_i$ is an isometry, we have

$$\|\mathscr{P}_a u\|_{L^p(R)} = 6^{1/p}\,\|u\|_{L^p(T_1)} \quad \text{for all} \quad u \in L^p(T_1).$$

Combining the last two displayed equations with the L^p convergence of double-sine series on R, we obtain

$$\begin{aligned}\left\|S_N^{T_1} f_a - f_a\right\|_{L^p(T_1)}^p &= \frac{1}{6}\left\|\mathscr{P}_a\left[S_N^{T_1} f_a\right] - \mathscr{P}_a f_a\right\|_{L^p(R)}^p \\ &= \frac{1}{6}\left\|S_N^R \mathscr{P}_a f_a - \mathscr{P}_a f_a\right\|_{L^p(R)}^p \to 0\end{aligned}$$

as $N \to \infty$. □

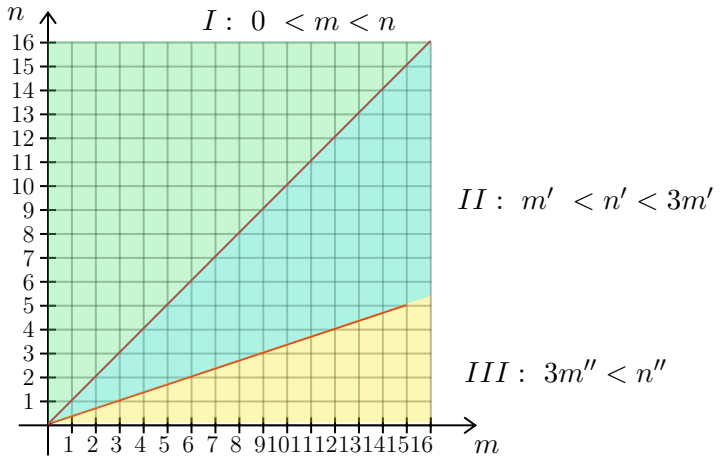


Figure 3.3: The three regions in the lattice $[1,16] \times [1,16]$. The lines correspond to the 'degenerate cases' $m = n$ and $m = 3n$ in which $u_{m,n} \equiv 0$, and thus the Fourier coefficients vanish. Note that all points where m and n have opposite parity will also be excluded.

3.4 Symmetric eigenfunctions on the equilateral triangle

It is now time to deal with the symmetric part: f_s. To do so, we introduce a *symmetric prolongation* of u to R:

$$\mathscr{P}_s u(x,y) := \sum_{i=1}^{6} d_i u(\xi(x_i),\eta(y_i)) \quad \text{where} \quad d_i := \begin{cases} 1 & \text{if } i = 1,4,5; \\ -1 & \text{if } i = 2,3,6. \end{cases} \tag{3.4.1}$$

This notation uses the same 'shorthand' as in (3.2.1), but uses a different combination of reflections and anti-reflections; see Fig. 3.4.

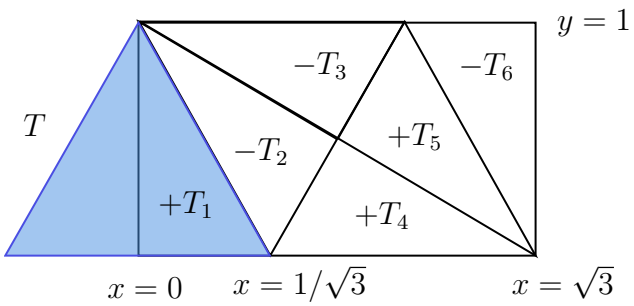


Figure 3.4: The sign arrangements for the symmetric prolongation $\mathscr{P}_s$. See Práger (1998).

From Fig. 3.4 it is clear that the prolongation $\mathscr{P}_s u$ only vanishes on the horizontal sides $y = 0$ and $y = 1$ of the rectangle, since for an x-symmetric function we cannot assume that $u(x,0) \equiv 0$. We therefore express $\mathscr{P}_s f_s$ as a cosine-sine series on R:

$$\sum_{m=0}^{\infty}\sum_{n=1}^{\infty} \widehat{\mathscr{P}_s f_s}(m,n) \cos\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y)\,.$$

When $m, n > 0$, the Fourier coefficients of $\widehat{\mathscr{P}_s f_s}$ are given by

$$\begin{aligned} \widehat{\mathscr{P}_s f_s}(m,n) &= \frac{4}{\sqrt{3}} \int_R \mathscr{P}_s f_s(x,y) \cos\left(\frac{\pi m x}{\sqrt{3}}\right) \sin(\pi n y)\ \mathrm{d}x\,\mathrm{d}y \\ &= \frac{4}{\sqrt{3}} \int_{T_1} f_s(\xi,\eta) \left[\sum_{i=1} d_i \cos\left(\frac{\pi m x_i}{\sqrt{3}}\right) \sin(\pi n y_i)\right] \mathrm{d}\xi\,\mathrm{d}\eta \quad \text{by (3.2.1)} \\ &= \frac{4}{\sqrt{3}} \int_{T_1} f_s(\xi,\eta) v_{m,n}(\xi,\eta)\,\mathrm{d}\xi\,\mathrm{d}\eta; \end{aligned} \tag{3.4.2}$$

when $m = 0$ and $n > 0$, they are

$$\widehat{\mathscr{P}_s f_s}(0,n) = \frac{2}{\sqrt{3}} \int_{T_1} f_s(\xi,\eta) v_{0,n}(\xi,\eta)\,\mathrm{d}\xi\,\mathrm{d}\eta. \tag{3.4.3}$$

Here we have defined the symmetric eigenfunctions as

$$v_{m,n}(\xi,\eta) := \sum_{i=1} d_i \cos\left(\frac{\pi m x_i}{\sqrt{3}}\right) \sin\left(\pi n y_i\right).$$

(Recall that each pair $(x_i, y_i) \in R$ is a function of $(\xi,\eta) \in T_1$.)

Note that these are no longer eigenfunctions for the Dirichlet Laplacian on the *hemi*-equilateral triangle, but their (symmetric) extensions to the *equilateral* triangle, T, are eigenfunctions of $-\Delta$ on T.

By explicit computation, Práger obtained the following.

Lemma 3.4.1. *For integers $0 \leqslant m < n$ with the same parity,*

$$\begin{aligned} v_{m,n}(x,y) = {} & 2\cos\left(\frac{\pi m x}{\sqrt{3}}\right)\sin\left(\pi n y\right) \\ & + 2(-1)^{(m+n)/2}\cos\left(\frac{\pi x}{2\sqrt{3}}(m+3n)\right)\sin\left(\frac{\pi y}{2}(m-n)\right) \\ & - 2(-1)^{(m-n)/2}\cos\left(\frac{\pi x}{2\sqrt{3}}(3n-m)\right)\sin\left(\frac{\pi y}{2}(m+n)\right). \end{aligned} \tag{3.4.4}$$

Furthermore, $v_{m,n} \equiv 0$ whenever n and m have different parity. Finally, with $0 < m < n$ as above, define the pairs

$$\begin{cases} m' = \frac{1}{2}(3n-m), \\ n' = \frac{1}{2}(m+n); \end{cases} \quad \textit{and} \quad \begin{cases} m'' = \frac{1}{2}(m+3n), \\ n'' = \frac{1}{2}(m-n). \end{cases}$$

Then,

$$v_{m',n'} = -(-1)^{(m-n)/2} v_{m,n}, \quad v_{m'',n''} = -(-1)^{(m+n)/2} v_{m,n}, \tag{3.4.5}$$

and, for all $n > 0$,

$$v_{0,n} = -(-1)^{n/2} v_{3n/2,n/2}. \tag{3.4.6}$$

Remark. Note that we no longer exclude the cases $m = 0$ and $m = 3n$, since the corresponding $v_{m,n}$ do not vanish, and must therefore be carefully accounted for when we compute the partial sums. Note also that m and n must have the same parity, so that n is even when $m = 0$, and, consequently, the right-hand side of (3.4.6) is well-defined. We can thus regard $(m', n') = (3n/2, n/2)$ when $m = 0$.

It is also clear that $v_{m,n}$ are the symmetric eigenfunctions of the Dirichlet Laplacian on T, restricted to T_1; the proof is identical to that of Corollary 3.2.3. (See McCartin, 2003, for a fuller treatment.)

Lemma 3.4.2. *The eigenfunctions* $\{v_{m,n} : 0 \leqslant m < n, n > 0, m \equiv n \mod 2\}$ *are pairwise orthogonal in* T *and* R. *Furthermore,*

$$\mathscr{P}_s(v_{m,n}|_T(x,y)) = v_{m,n}(x,y) \quad \textit{for all} \quad (x,y) \in R, \tag{3.4.7}$$

from which it follows that

$$\|v_{m,n}\|^2_{L^2(T_1)} = \frac{\sqrt{3}}{2} \qquad \textit{for all} \quad n, m > 0; \tag{3.4.8}$$

$$\|v_{0,n}\|^2_{L^2(T_1)} = \sqrt{3} \qquad \textit{for all} \quad n > 0. \tag{3.4.9}$$

Once again, these results are containted in Práger (1998). They are crucial to our proof of L^p convergence in the next section.

3.5 L^p convergence of the Fourier series of f_s

Writing

$$f_s^{\triangle}(m,n) := \frac{1}{\|v_{m,n}\|^2_{L^2(T_1)}} \int_{T_1} f_s v_{m,n},$$

it immediately follows from (3.4.2),(3.4.8) and (3.4.3),(3.4.9), that

$$\widehat{\mathscr{P}_s f_s}(m,n) = 2 f_s^{\triangle}(m,n) \quad \text{for all} \quad m \geqslant 0, n > 0. \tag{3.5.1}$$

The next theorem establishes the L^p convergence of the eigenfunction expansions for the hemi-equilateral triangles, and is the other key ingredient of our main result.

Theorem 3.5.1. *Let* $f \in L^p(T)$ *with* $1 < p < \infty$, *and denote its symmetric part by* f_s. *Then* $f_s \in L^p(T_1)$ *and* $S_N^{T_1} f_s \to f_s$ *in* $L^p(T_1)$.

Proof. The proof is essentially the same as that of Theorem 3.3.1, the only modifications arising from the 'degenerate' indices $(0,n)$ and $(3n,n)$.

As before, we break up the set of indices with $0 \leqslant m \leqslant N$, $0 < n \leqslant N$ into three sets:

$$0 < m < n, \quad n < m < 3n \quad \text{and} \quad 3n < m$$

sparing the cases $m = 0$ and $m = 3n$. From Lemma 3.4.1, we may label these groups as (m,n), (m',n'), and (m'',n''), respectively, dividing the lattice $[0,N] \times [1,N] \cap \mathbb{Z}^2$ into three disjoint regions; see once again Fig. 3.3.

It follows from (3.4.5) that, for all positive indices,

$$\widehat{\mathscr{P}_s f_s}(m,n) = -(-1)^{(m-n)/2}\widehat{\mathscr{P}_s f_s}(m',n') = (-1)^{(m+n)/2}\widehat{\mathscr{P}_s f_s}(m'',n''),$$

and from (3.4.6) that

$$\widehat{\mathscr{P}_s f_s}(0,n) = -(-1)^{n/2}\widehat{\mathscr{P}_s f_s}(3n/2,n/2) \quad \text{for all} \quad n>0. \tag{3.5.2}$$

Recall that we identify $(0',n') = (3n/2,n/2)$ and that this is precisely the case $m' = 3n'$; see Section 3.4.

We begin by grouping the partial sums on R as follows:

$$S_N^R \mathscr{P}_s f_s(x,y) = \sum_{\substack{0<m,n\leqslant N\\ m\neq n, m\neq 3n}} \widehat{\mathscr{P}_s f_s}(m,n)\cos\left(\frac{\pi m x}{\sqrt{3}}\right)\sin(\pi n y) \tag{3.5.3}$$

$$+\left[\sum_{(0,n)} \widehat{\mathscr{P}_s f_s}(0,n)\sin(\pi n y)\right.$$

$$\left.+\sum_{(0',n')} \widehat{\mathscr{P}_s f_s}(0',n')\cos\left(\frac{0'\pi x}{\sqrt{3}}\right)\sin\left(\pi n' y\right)\right]. \tag{3.5.4}$$

The calculations for the term (3.5.3) proceed in the same way as for the antisymmetric part $S_N^R \mathscr{P}_a f_a$, using (3.4.4) instead of (3.2.3) after grouping the various cosine-sine terms. We will therefore only consider the bracketed term (3.5.4) in detail.

The sums in (3.5.4) are taken over even $n \leqslant N$ and pairs $(0',n') = (3n/2,n/2)$ corresponding to $m = 3n$. Using (3.5.2) we have

$$\sum_{(0,n)} \widehat{\mathscr{P}_s f_s}(0,n)\sin(\pi n y) + \sum_{(0',n')} \widehat{\mathscr{P}_s f_s}(0',n')\cos\left(\frac{0'\pi x}{\sqrt{3}}\right)\sin\left(\pi n' y\right)$$

$$= \sum_{(0,n)} \widehat{\mathscr{P}_s f_s}(0,n)\left[\sin(\pi n y) - \mathbf{2}(-1)^{n/2}\cos\left(\frac{2n\pi x}{2\sqrt{3}}\right)\sin\left(\frac{3\pi y}{2}\right)\right]$$

$$= \sum_{(0,n)} \widehat{\mathscr{P}_s f_s}(0,n)\left[\cos\left(\frac{\pi\cdot 0\cdot x}{\sqrt{3}}\right)\sin(\pi n y)\right.$$

$$+(-1)^{(0+n)/2}\cos\left(\frac{\pi x}{2\sqrt{3}}(0+3n)\right)\sin\left(\frac{\pi m}{2}(0-n)\right)$$

$$\left.-(-1)^{(0-n)/2}\cos\left(\frac{\pi x}{2\sqrt{3}}(3n-0)\right)\sin\left(\frac{\pi m}{2}(n-0)\right)\right]$$

$$= \sum_{(0,n)} \frac{\widehat{\mathscr{P}_s f_s}(0,n)}{2} v_{0,n}(x,y) \quad \text{by (3.4.4)}$$
$$= \sum_{(0,n)} f_s^{\triangle}(0,n) \mathscr{P}_s v_{0,n}(x,y).$$

The bold factor of 2 in the second line arises from the fact that the normalisation constant for $\widehat{\mathscr{P}_s f_s}(m,n)$, $m \neq 0$, is twice that of $\widehat{\mathscr{P}_s f_s}(0,n)$; see (3.4.2) and (3.4.3). The last line follows from (3.4.7) and (3.5.1). Once again we have

$$\mathscr{P}_s \left[S_N^{T_1} f_s \right](x,y) = S_N^R \mathscr{P}_s f_s(x,y),$$

and we can conclude that $S_N^{T_1} f_s \to f_s$ in $L^p(T_1)$. □

3.6 Main results for the equilateral triangle

We now wrap up with the main results as they apply to the original equilateral triangle, T.

Theorem 3.6.1 (Práger, 1998)**.** *The functions*

$$\begin{aligned} u_{m,n}: &\quad m,n = 1,2,\ldots, && m \equiv n \mod 2, \quad 0 < m < n; \\ v_{m,n}: &\quad m = 0,1,2,\ldots, n = 1,2,\ldots, && m \equiv n \mod 2, \quad 0 \leqslant m < n, \end{aligned}$$

form a complete, orthogonal system on T, consisting of eigenfunctions of the Dirichlet Laplacian. Furthermore, each eigenfunction, $u_{m,n}$ or $v_{m,n}$, corresponds to the eigenvalue $\pi^2 \left(\frac{m^2}{3} + n^2 \right)$.

As for the L^p theory, we can combine our Theorems 3.3.1 and 3.5.1 to obtain the full result on $L^p(T)$. As before, we use the following notation to keep track of our indices in the asymmetric and symmetric parts of the sums:

$$\mathscr{U}_N := \{(m,n) : 0 < m < n \leqslant N \text{ and } m \equiv n \mod 2\},$$
$$\mathscr{V}_N := \{(m,n) : 0 \leqslant m < n \leqslant N \text{ and } m \equiv n \mod 2\}.$$

Theorem 3.6.2. *Let $f \in L^p(T)$ with $1 < p < \infty$. With the notation of the previous sections, let*

$$S_N^T f := \sum_{(m,n)\in\mathscr{U}_N} f_a^{\triangle}(m,n) u_{m,n} + \sum_{(m,n)\in\mathscr{V}_N} f_s^{\triangle}(m,n) v_{m,n},$$

Then, $S_N^T f \to f$ *in* $L^p(T)$.

Proof. For any function f on T, write its decomposition into a symmetric and antisymmetric part with respect to the y-axis:

$$f = f_a + f_s.$$

Then clearly

$$\|f_\nu\|_{L^p(T)}^p = 2\,\|f_\nu\|_{L^p(T_1)}^p \quad \text{for} \quad \nu = a, s,$$

whence

$$\|f\|_{L^p(T)} \leqslant 2^{1/p} \left(\|f_a\|_{L^p(T_1)} + \|f_s\|_{L^p(T_1)} \right).$$

Hence, since

$$\left[S_N^T f\right]_\nu = S_N^{T_1} f_\nu \quad \text{for} \quad \nu = a, s,$$

it follows that

$$\left\|S_N^T f - f\right\|_{L^p(T)} \leqslant 2^{1/p} \left(\left\|S_N^{T_1} f_a - f_a\right\|_{L^p(T_1)} + \left\|S_N^{T_1} f_s - f_s\right\|_{L^p(T_1)} \right) \to 0$$

by Theorems 3.3.1 and 3.5.1. □

3.7 Concluding remarks

Our argument made use of Práger's cunning triangle-to-rectangle transformation in order to reduce the convergence problem on the triangle to the well-known convergence on the rectangle. In a similar, but more direct way, we were able to obtain these results for the 45-90-45 triangle.

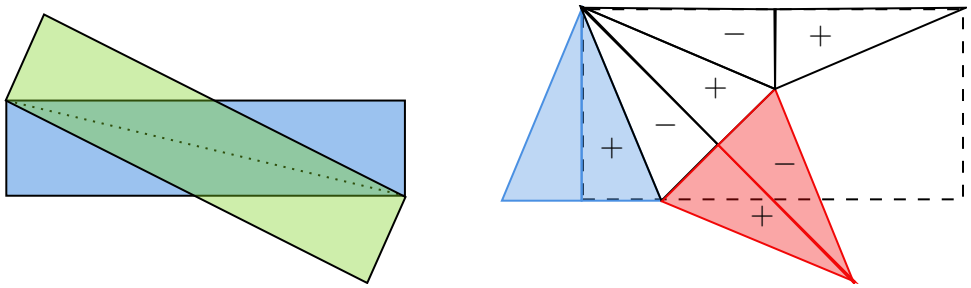
There are, however, limitations to this approach, as a consequence of the following theorem due to Lamé (1833) and reported as Theorem 3.1 in McCartin (2003).

Theorem 3.7.1 (Lamé's Fundamental Theorem)**.** *Suppose that* $f(x, y)$ *can be represented by the trigonometric series*

$$f(x, y) = \sum_i A_i \sin(\lambda_i x + \mu_i y + \alpha_i) + B_i \cos(\lambda_i x + \mu_i y + \beta_i)$$

with $\lambda_i^2 + \mu_i^2 = k^2$*. Then* f *is antisymmetric about any line about which it vanishes.*

The implication for our argument is that, assuming Dirichlet boundary conditions on the hypothenuse of T_1, we will generate a nodal (i.e. vanishing) line along



(a) In general, the diagonal is not a line of symmetry, so it cannot be a nodal line.

(b) Mimicking the Práger construction with an angle of $\pi/2n$ ($n \neq 2, 3$) does not tile a rectangle.

Figure 3.5: Limitations of the triangle-to-rectangle constructions.

the diagonal of the rectangle (see, e.g., Fig. 3.2). Lamé's Theorem then requires the eigenfunctions to have a line of anti-symmetry along the diagonal. The only possibilities are symmetry along the diagonal line itself, as in the square (see Fig. 3.5,(a)); or an arrangement of smaller triangles inside the rectangle as in Práger's construction with three hemiequilateral triangles. In this case, the upper right-angle of the rectangle is cut into three angles of $\pi/6$. Attempts to replace this decomposition by $n > 3$ triangles with angles $\pi/2n$ will not tile a rectangle (see Fig. 3.5,(b)).

These limitations are related to the Method of Images used to construct Green's functions for differential equations in Euclidean domains. The classification of admissible domains for the Method of Images goes back to Keller (1953), wherein our three 'special triangles', *viz.* equilateral, hemiequilateral and 45–90–45, are identified as the only triangular domains for which the method is applicable.

3.8 Appendix: Non-convergence for spherical cutoffs

Let us return to the Fourier series on the 45-90-45 triangle T from Section 3.1. There, we ordered the eigenfunctions $u_{m,n}$ by the condition $0 < m < n$ and summed in the range $n \leqslant N$. As we remarked in the introduction, a more natural choice would be to sum over all $u_{m,n}$ with eigenvalue $\lambda_{m,n} \leqslant N$. Since $\lambda_{m,n} = \pi(m^2 + n^2)$, this is essentially equivalent to summing within the quarter of the disk $m^2 + n^2 \leqslant N^2$ of radius N in the first quadrant. Proposition 3.1.3 immediately extends to show that these partial sums are related to the spherical cut-offs of the traditional Fourier series on the square $[0, 1]^2$, which, as follows from Fefferman's classical result on the ball multiplier, do not in general converge in L^p.

Given a counterexample f on the square, we would need to find a function g on the triangle such that $\mathscr{P}g = f$ on $[0, 1]^2$. It is therefore clear that only a counterexample that is antisymmetric along the diagonal will serve this purpose.

However, Fefferman's result is non-constructive, and we still lack an explicit example of a function on the torus whose spherical partial sums do not converge in L^p. Thus, we cannot assume *a priori* that such counterexamples are available on the triangle.

Nevertheless, we can carefully replicate Fefferman's argument, coupled with a transference result, to draw the conclusion that spherical cut-offs of Fourier series on T will in general not converge in L^p. For this section we are indebted to the the presentation of Fefferman's result in Stein (1993). For a discussion of transference, see Davis and Chang (1987) or, in greater detail, Grafakos (2014b).

The proof requires three ingredients: (1) a transference theorem, (2) Meyer's Lemma, and (3) a carefully constructed Besicovitch set.

The transference theorem exploits the intuition that a Fourier transform is the limit of a Fourier series with an increasingly large period. It is then argued that, since L^p convergence of Fourier series is equivalent to the uniform boundedness of the partial sum operators, this uniform bound it equivalent to the boundedness of a Fourier multiplier. For the spherical partial sums, this means the boundedness of the operator

$$T_B f(x) := \int_{|\xi|<1} \widehat{f}(\xi) \mathrm{e}^{2\pi \mathrm{i} x\cdot\xi} \,\mathrm{d}\xi.$$

The 'increasing period' of the Fourier series can be seen as rescaling the square so that it 'converges' to the entire space $\mathbb{R}^2$. But we need to keep the antisymmetry constraint along the diagonal. Thus, we will be concerned with the space $L^p_-(\mathbb{R}^2)$: the L^p closure of the smooth functions f of compact support that are antisymmetric about the line $y = -x$.

Theorem 3.8.1 (Antisymmetric Transference)**.** *Suppose the spherical cut-off operators S_N^T are uniformly bounded in $N \geqslant 1$ from $L^p(T)$ to $L^p(T)$. Then, there is a constant $C_p > 0$ such that*

$$\|T_B f\|_{L^p(\mathbb{R}^2)} \leqslant C_p \|f\|_{L^p(\mathbb{R}^2)} \quad \textit{for all} \quad f \in L^p_-(\mathbb{R}^2). \tag{3.8.1}$$

This inequality is used to prove a vector-valued inequality known as Meyer's Lemma. The key idea here is that, on small enough scales, the disk at $u \in S^1$ 'looks like' the half plane $\{\xi : \xi \cdot u > 0\}$. Thus, denoting by S^u the multiplier operator of this half plane

$$S^u f(x) = \int_{\xi\cdot u>0} \widehat{f}(\xi) \mathrm{e}^{2\pi \mathrm{i} x\cdot\xi} \,\mathrm{d}\xi,$$

we have our second ingredient.

Theorem 3.8.2 (Meyer's Lemma for L^p_-)**.** *Assume that Eq. (3.8.1) holds. Then,*

for all $u_1, \dots, u_k \in S^1$ *and all* $f_1, \dots, f_k \in L^p_-$,

$$\left\| \left(\sum_j |S^{u_j} f_j|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^2)} \leqslant C_p \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^2)} . \tag{3.8.2}$$

The proof is identical to that of the original Meyer's Lemma, since it only involves rescaling the ball in Fourier space, and multiplying each f_j by a complex scalar, neither of which operations alter the antisymmetry of the f_j.

The crucial next step is finding a set of functions that violate the bound (3.8.2). This is where the third ingredient, the Besicovitch set, comes in: a collection of congruent rectangles R_j that overlap to form a small total area, but whose translates $\widetilde{R}_j$ two lengths along the longest side remain disjoint—see Fig. 3.6 below. (In the next theorem, '$\sqcup$' denotes a disjoint union.)

Theorem 3.8.3 (Besicovitch Set)**.** *For any* $\varepsilon > 0$ *there is a finite collection of congruent rectangles* R_j *such that* $\left| \bigcup_j R_j \right| < \varepsilon$ *but* $\left| \bigsqcup_j \widetilde{R}_j \right| = 1$.

Proof. See Theorem 1 on p. 435 of Stein (1993). □

A simple pointwise calculation shows that there is an absolute constant $c > 0$ such that

$$\left| S^{u_j} \chi_{R_j} \right| \geqslant c \chi_{\widetilde{R}_j} \quad \text{for all} \quad j, \tag{3.8.3}$$

where $u_j \in S^1$ is the unit direction of the longest side of R_j, and thus the collection of 'rectangles' (more precisely: their indicator functions) violates Eq. (3.8.2), as required.

To prove Eq. (3.8.3), consider the rectangle $R = (-\frac{1}{2}, \frac{1}{2}) \times (-2^{-N+1}, -2^{-N-1})$, so that

$$\chi_R(x_1, x_2) = \chi_{(-\frac{1}{2}, \frac{1}{2})}(x_1) \cdot \chi_{(-2^{-N+1}, -2^{-N-1})}(x_2).$$

If $u = (1, 0)$, we obtain

$$S^u \chi_R(x_1, x_2) = (S^+ \chi_{(-\frac{1}{2}, \frac{1}{2})})(x_1) \cdot \chi_{(-2^{-N+1}, -2^{-N-1})}(x_2), \tag{3.8.4}$$

where S^+ denotes the 1-dimensional operator

$$S^+ f(x_1) := \int_0^\infty \widehat{f}(\xi) \mathrm{e}^{2\pi \mathrm{i} x_1 \xi} \, \mathrm{d}\xi.$$

Now, taking $f = \chi_{(-\frac{1}{2},\frac{1}{2})} \in L^2(\mathbb{R})$, we have

$$\begin{aligned} S^+ f(x_1) &= \lim_{\varepsilon \to 0^+} \int_0^\infty \widehat{f}(\xi) \mathrm{e}^{2\pi \mathrm{i}(x_1 + \mathrm{i}\varepsilon)\xi} \, \mathrm{d}\xi \\ &= \lim_{\varepsilon \to 0^+} \int_{-\infty}^\infty \left(\int_0^\infty \mathrm{e}^{-2\pi \mathrm{i} y \xi} \mathrm{e}^{2\pi \mathrm{i}(x_1 + \mathrm{i}\varepsilon)\xi} \, \mathrm{d}\xi \right) f(y) \, \mathrm{d}y \\ &= \frac{1}{2\pi \mathrm{i}} \lim_{\varepsilon \to 0^+} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{y - x_1 - \mathrm{i}\varepsilon} \, \mathrm{d}y. \end{aligned}$$

Realising the denominator, we can write

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{y - x_1 - \mathrm{i}\varepsilon} \, \mathrm{d}y = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{y - x_1}{(y - x_1)^2 - \varepsilon^2} \, \mathrm{d}y + \mathrm{i} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\varepsilon}{(y - x_1)^2 - \varepsilon^2} \, \mathrm{d}y.$$

Using the basic estimates

$$|z| \geqslant |\mathrm{Re}(z)| \quad \text{and} \quad \frac{1}{|x - y|} \geqslant \frac{1}{|x| + |y|},$$

we deduce that, for $|x_1| > \frac{1}{2}$,

$$\left| S^+ \chi_{(-\frac{1}{2},\frac{1}{2})}(x_1) \right| \geqslant \frac{1}{2\pi} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{|y| + |x_1|} \, \mathrm{d}y \geqslant \frac{1}{8\pi \, |x_1|}. \tag{3.8.5}$$

Combining Eq. (3.8.4) with Eq. (3.8.5), and recalling that $\widetilde{R}$ denotes the rectangle $(1, \frac{1}{2}) \times (-2^{-N+1}, -2^{-N-1})$, it immediately follows that

$$|S^u \chi_R(x_1, x_2)| \gtrsim \chi_{\widetilde{R}}(x_1, x_2).$$

One can then deduce Eq. (3.8.3) for general directions u_j by rotations and translations of the R_j.

Let us now turn to the construction of the Besicovitch set, which proceeds by dissecting a triangle and shifting the resulting triangles parallel to their base so as to create overlap. Thus, all the mass lies on one side of the diagonal and we can duplicate the whole thing to create a *disjoint* reflection with respect to this line (see Fig. 3.6, where we rotated the picture). This new figure has double the area, but introduces no additional overlap. Hence, if we denote the reflection of R_j by R_{-j}, we may take

$$f_j = \chi_{R_j} - \chi_{R_{-j}} \in L^p_-, \quad \text{for all} \quad j \geqslant 1$$

and similarly conclude that Eq. (3.8.2), and therefore Eq. (3.8.1), fails.

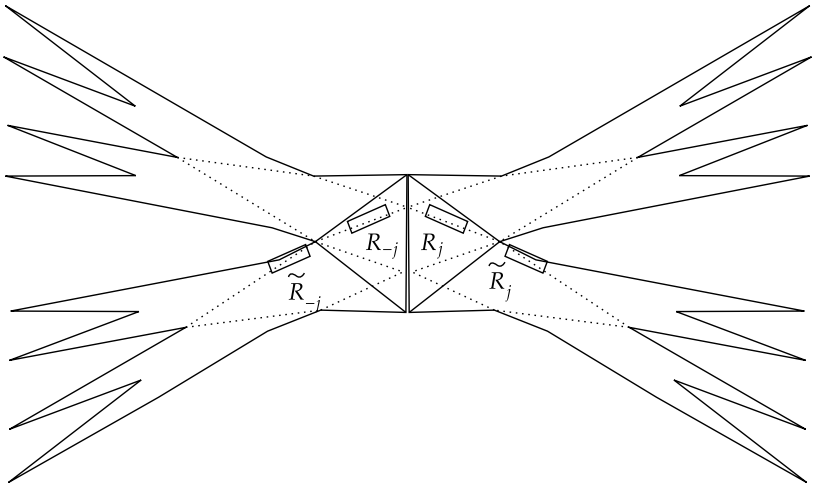


Figure 3.6: Duplication of the rectangles in the Besicovitch set.

The situation for the hemi- and equilateral triangles is much more complicated, since we cannot immediately scale the rectangles in Figs. 3.2 and 3.4 to $\mathbb{R}^2$. The inner rhomboid $T_2 \cup T_3 \cup T_4 \cup T_5$ can be scaled, while preserving the two diagonals as symmetry lines, but the 'fringe' triangles T_1 and T_6 would be, so to speak, 'lost at infinity'.

Chapter 4

Bessel–Fourier series on a disc

μή μου τοὺς κύκλους τάραττε!

—Greek saying, attributed to Archimedes

We next investigate series arising from the eigenfunctions of the Dirichlet Laplacian on the disc. As in the torus case, separation of variables is possible, leading to eigenfunctions that factor into a radial and an angular component. The structure of the eigenvalues, however, precludes a simple factorisation of the summability kernel in a way similar to Eq. (2.2.3) for the Dirichlet kernel.

The relevant eigenfunctions are of the form

$$\mathrm{e}^{2\pi \mathrm{i} n\theta} J_{|n|}(j_{|n|,m} r) \quad \text{for} \quad (r,\theta) \in [0,1]^2, (n,m) \in \mathbb{Z} \times \mathbb{N}$$

corresponding to the respective eigenvalues $-4\pi^2 n^2 - (j_{|n|,m})^2$. Here $J_{|n|}$ denotes a Bessel function of the first kind and $j_{|n|,m}$ its non-negative zeros (see Watson, 1995). There is no explicit formula for the zeros $j_{|n|,m}$, but the following asymptotic is known (see, e.g., Young, 1942, p. 293):

$$j_{|n|,m} \sim \pi m - \frac{\pi\,|n|}{2} + \frac{\pi}{4} \quad \text{as} \quad m \to \infty.$$

Numerical tabulation of the zeros (Davies, 1995, p. 131) shows that there is little relation between the labelling of the eigenvalues by the indices (n,m) and their ordering:

$$j_{0,1} < j_{1,1} < j_{2,1} < j_{0,2} < j_{3,1} < \dots.$$

This is in contrast to the situation with Fourier eigenvalues (ignoring multiplicities):

$$0 < 0^2 + 1^2 < 1^2 + 1^2 < 0^2 + 2^2 < 1^2 + 2^2 < \dots.$$

Most of the literature has focused on the convergence of the Bessel–Fourier series of the form

$$\sum_{m=1}^{\infty} a_m J_\nu(j_{\nu,m} r)$$

for a fixed Bessel function J_ν (see, for instance Watson, 1995; Wing, 1950; Young, 1942; MacRobert, 1921; Benedek and Panzone, 1973). Our focus will instead be on the convergence of the double series

$$\sum_{n\in\mathbb{Z}} \sum_{m\in\mathbb{N}} a_{m,n} \mathrm{e}^{2\pi \mathrm{i} n\theta} J_{|n|}(j_{|n|,m} r) \tag{4.0.1}$$

in the norm topology of $L^p(\mathbb{D}) = L^p(r\,\mathrm{d}r\,\mathrm{d}\theta)$.

We have already remarked that, if we denote the *Fourier* partial sums from by S_N, then the following statements are equivalent (see Proposition 2.2.2 above):

(i) For any $f \in L^p$, $S_N f \to f$ in L^p.

(ii) The operators $S_N : L^p \to L^p$ are uniformly bounded, i.e.

$$\sup_{N\in\mathbb{N}} \|S_N\|_{L^p\to L^p} < \infty.$$

And similarly for the cubic truncations $S_{[N]}$. One of the challenges in obtaining a similar result for the series in (4.0.1) is identifying the appropriate relationship between the N and M limits for the partial sums

$$S_{N,M} f := \sum_{|n|\leqslant N} \sum_{m\leqslant M} a_{m,n} \mathrm{e}^{2\pi \mathrm{i} n\theta} J_{|n|}(j_{|n|,m} r).$$

Another critical issue is the density of the partial sums—the analogues of the trigonometric polynomials from Fourier Analysis—which is required for the 'useful' implication from uniform boundedness to convergence, i.e. (ii) implies (i) above. This issue will be discussed in Section 4.5.

We begin this chapter by deriving the eigenfunctions for the Laplacian on the disc (Section 4.1), while in Section 4.2 we dispose of the easy L^2 convergence of the Bessel–Fourier series (4.0.1). Section 4.3 collects some facts about the asymptotics of Bessel functions and their zeros, which will be needed in later sections. Section 4.4 describes existing results for the 1-dimensional case, while in Section 4.5 we summarise a result of Balodis and Córdoba (1999) regarding the norm convergence of the 2-dimensional Bessel–Fourier series in the mixed norm space $L^p_{\mathrm{rad}}(L^2_{\mathrm{ang}})$ on the disc for the range $\frac{4}{3} < p < 4$. We then describe how to modify their result to obtain

$L^p(\mathbb{D})$-norm convergence in a 'similar' subspace for the restricted range $2 \leqslant p < 4$. Chapter 5 contains a new result showing that for any (non-negative) $a_m \to 0$, there is an $f \in L^1(x\,\mathrm{d}x, [0,1])$ whose Bessel–Fourier coefficients decay slower than a_m. We conjecture that this could be improved to the assumption that $a_m = o(\sqrt{m})$ and show that, if this conjecture holds, then we can construct an $f \in L^1(\mathbb{D})$ whose 2-dimensional Bessel–Fourier series diverges for *any possible choices of* M and N in Eq. (4.0.1).

A summary of Section 4.5 has appeared in *Comptes Rendus – Mathématique* as Acosta Babb (2023a).

4.1 The Poisson problem and Bessel–Fourier series

Consider the Poisson problem on the disc

$$\mathbb{D} := \{(r\cos(2\pi t), r\sin(2\pi t)) : (r,t) \in [0,1]^2\},$$

with the measure $r\,\mathrm{d}r\,\mathrm{d}t$:

$$\begin{cases} \Delta u = f & \text{on } \mathbb{D}; \\ u = 0 & \text{on } \partial\mathbb{D}. \end{cases}$$

One way to solve this problem is to expand f as a series using the eigenfunctions of Δ on $\mathbb{D}$. To find these eigenfunctions, u, say, we apply separation of variables $u(r,t) = R(r)T(t)$ and obtain the two single-variable eigenvalue problems

$$T''(t) + n^2 T(t) = 0 \quad \text{for} \quad n \in \mathbb{Z}, \tag{4.1.1}$$

$$r^2R''(r) + rR'(r) + (r^2\lambda^2 - n^2)R(r) = 0 \quad \text{subject to} \quad R(1) = 0. \tag{4.1.2}$$

The solutions to Eq. (4.1.1) are the periodic Fourier modes

$$T_n(t) = \mathrm{e}^{2\pi \mathrm{i} n t}, \quad \text{where} \quad n \in \mathbb{Z}^2.$$

The solutions to Eq. (4.1.2) are the Bessel functions of the first kind

$$R_{n,m}(r) = J_n(j_{n,m}r), \quad \text{where} \quad J_n(j_{n,m}) = 0 \quad \text{for all} \quad n \in \mathbb{Z}, m \in \mathbb{N}.$$

The latter equation ensures that the boundary conditions $R_{n,m}(1) = 0$ are met. For convenience, we will for now rename

$$J_{n,m}(r) := J_{|n|}(j_{|n|,m}r) \quad \text{for all} \quad n \in \mathbb{Z}, m \in \mathbb{N},$$

since for $n \in \mathbb{N}$

$$J_{-n}(x) \equiv (-1)^n J_n(x).$$

Thus, $J_{\pm n}$ are linearly dependent solutions to the radial equation and there is no loss of generality in keeping only the functions of positive order $n \geqslant 1$.

We also assume that the functions $J_{n,m}$ are L^2 normalised, i.e.

$$\|J_{n,m}\|_{L^2([0,1],r\,\mathrm{d}r)} = 1 \quad \text{for all} \quad n \in \mathbb{Z}, m \in \mathbb{N}.$$

For future reference, the normalisation constant is explicitly

$$\int_0^1 J_n(j_{n,m}r)^2 r\,\mathrm{d}r = \frac{1}{2}J_n'(j_{n,m})^2. \tag{4.1.3}$$

This can be computed from the differential equation (4.1.2) by noting that

$$r\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}J_n(j_{n,m}r)}{\mathrm{d}r}\right) = (n^2 - (j_{n,m}r)^2)J_n(j_{n,m}r),$$

which implies

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}J_n(j_{n,m}r)}{\mathrm{d}r}\right)^2 = (n^2 - (j_{n,m}r)^2)\frac{\mathrm{d}}{\mathrm{d}r}(J_n(j_{n,m}r)^2)$$

and so Eq. (4.1.3) follows by integrating over $[0,1]$ and applying the boundary conditions.

Note that an alternative convention is to normalise with a prefactor of

$$\frac{2}{J_{n+1}(j_{n,m})^2}.$$

(See, e.g., Watson, 1995 and Tolstov and Silverman, 1976.)

More properties of Bessel functions will be derived in Section 4.3. For a detailed treatment of Bessel functions, the reader is directed to Watson (1995).

We are interested in expressing an arbitrary function $f \in L^p(\mathbb{D})$ as a norm-convergent series

$$f(r,t) = \sum_{n=-\infty}^{\infty}\sum_{m=1}^{\infty} c_{n,m}(f)J_{n,m}(r)\mathrm{e}^{2\pi\mathrm{i}nt}, \tag{4.1.4}$$

where

$$c_{n,m}(f) := \frac{2}{J_n'(j_{n,m})^2}\int_0^1\int_0^1 f(\rho,\theta)J_n(j_{n,m}\rho)\mathrm{e}^{-2\pi\mathrm{i}n\theta}\rho\,\mathrm{d}\rho\,\mathrm{d}\theta.$$

For $p = 2$, this is readily achieved by exploiting the product structure afforded

by the polar separation of variables:

$$L^2(\mathbb{D}) = L^2([0,1]^2, r\,\mathrm{d}r\,\mathrm{d}t) \cong L^2([0,1]\times[0,1], r\,\mathrm{d}r\times\,\mathrm{d}t).$$

4.2 Convergence in L^2

For the angular part, we invoke the familiar L^2-theory of Fourier expansions. Specifically, $\{\mathrm{e}^{2\pi\mathrm{i}nt}\}_{n\in\mathbb{Z}}$ is an orthonormal basis for $L^2([0,1],\,\mathrm{d}t)$. As for the radial part, Hochstadt (1967) has shown, using the theory of compact operators, that the set $\{\sqrt{r}J_{n,m}(r)\}_{m\in\mathbb{N}}$ is an orthonormal basis for $L^2([0,1],\,\mathrm{d}r)$ for each $n\in\mathbb{Z}$. In particular, $\{J_{n,m}\}_{m\in\mathbb{N}}$ is an orthonormal basis for $L^2([0,1], r\,\mathrm{d}r)$. These properties obtain since Bessel's differential equation (4.1.2) is an example of a Sturm–Liouville system. (For the classical theory, see Conway, 2007, §6 in Chap. II.)

We can combine these results to obtain an orthonormal basis for $L^2(\mathbb{D})$ as follows.

Proposition 4.2.1. *The set* $\{J_{n,m}(r)\mathrm{e}^{2\pi\mathrm{i}nt} : n\in\mathbb{Z}, m\in\mathbb{N}\}$ *is an orthonormal basis for* $L^2([0,1]^2, r\,\mathrm{d}r\,\mathrm{d}t)$.

Proof. Recall that an orthonormal set $\mathscr{E}\subset\mathscr{H}$ in a Hilbert space $\mathscr{H}$ is a basis if, and only if,

$$(f\perp e \text{ for all } e\in\mathscr{E}) \implies f=0.$$

It is well known that the sets $\{\mathrm{e}^{2\pi\mathrm{i}nt}\}_{n\in\mathbb{Z}}$ and $\{J_{n,m}\}_{m\in\mathbb{N}}$ are orthonormal in their respective L^2 spaces. (This can be verified, e.g., from the differential equations (4.1.1) and (4.1.2).) By Fubini's Theorem, they are therefore orthonormal in the product space.

Fix an $f\in L^2([0,1]^2, r\,\mathrm{d}r\,\mathrm{d}t)$, and suppose that

$$\int_0^1\int_0^1 f(r,t)J_{n,m}(r)\mathrm{e}^{2\pi\mathrm{i}nt}r\,\mathrm{d}r\,\mathrm{d}t = 0 \quad \text{for all} \quad n\in\mathbb{Z}, m\in\mathbb{N}.$$

Since $\{J_{n,m}\}_{m\in\mathbb{N}}$ is an orthonormal basis for $L^2([0,1], r\,\mathrm{d}r)$, it follows that

$$\int_0^1 f(r,t)\mathrm{e}^{2\pi\mathrm{i}nt}\,\mathrm{d}t = 0 \quad r\text{-a.e.} \quad \text{for all} \quad n\in\mathbb{Z},$$

and similarly we conclude that $f(r,t)=0$ a.e. in $\mathbb{D}$. □

4.3 Asymptotics for Bessel functions and their zeros

In order to consider the L^p convergence of the Bessel–Fourier series, we need to develop some basic properties of Bessel functions. A full treatment can be found in Watson (1995); a more modern summary is available as Appendix B to Grafakos (2014b).

For $\nu \in \mathbb{N}_0$, there are several (equivalent) representations of the Bessel functions J_ν:

(i) *As an infinite series*:

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{1}{\Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu}.$$

Note that this representation is in fact valid for $\operatorname{Re}(\nu) > -1/2$.

(ii) *As an oscillatory integral*:

$$J_\nu(x) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{ix\sin t} \mathrm{e}^{-i\nu t}\, \mathrm{d}t.$$

Note that this representation works for $\nu \in \mathbb{N}_0$ only.

(iii) *The Poisson Representation Formula:*

$$J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+\frac{1}{2})\Gamma(1/2)} \int_{-1}^{1} \mathrm{e}^{ixt}(1-t^2)^{\nu-1/2}\, \mathrm{d}t, \tag{4.3.1}$$

valid again for all $\operatorname{Re}(\nu) > -1/2$.

A basic result for our purposes is the behavior of $J_\nu(x)$ as $x \to \infty$, which is asymptotically a cosine. We reproduce the proof given by Grafakos (2014b), both for the sake of completeness and to make the formula a little clearer in the case $\nu = 0$, since we will need these very precise estimates for the error in approximating J_0 below in Chapter 5.

Lemma 4.3.1. *For all $r > 0$ and $\nu \geqslant 0$,*

$$J_\nu(r) = \sqrt{\frac{2}{\pi}} \frac{\cos(r - \pi\nu/2 - \pi/4)}{\sqrt{x}} + R_\nu(r), \tag{4.3.2}$$

where $R_\nu(r) \lesssim_\nu r^{-3/2}$ whenever $r \geqslant 1$.

When $\nu = 0$, we have the estimate

$$|R_0(r)| \lesssim \int_0^2 \mathrm{e}^{-rt}\sqrt{t}\,\mathrm{d}t + \int_2^\infty \frac{\mathrm{e}^{-rt}}{t}\,\mathrm{d}t \tag{4.3.3}$$

for all $r > 0$.

Proof. This proof follows that in Appendix B.8 to Grafakos (2014b).

Using the Poisson Representation Formula for J_ν, we can compute the remainder in Eq. (4.3.2) to be

$$R_\nu(r) = \frac{\sqrt{2\pi}r^\nu}{\Gamma(\nu+\frac{1}{2})}\mathrm{e}^{\mathrm{i}(r-\frac{\pi\nu}{2})+\frac{\pi}{4}}\sum_{j=0}^{1}\int_0^\infty \mathrm{e}^{-rt}t^{\nu+\frac{1}{2}}\left[\left(1+\frac{1+(-1)^j}{2}\frac{\mathrm{i}t}{2}\right)^{\nu-\frac{1}{2}}-1\right]\frac{\mathrm{d}t}{t}, \tag{4.3.4}$$

where we have two integrals, one with a $1+\frac{\mathrm{i}t}{2}$ and one with a $1-\frac{\mathrm{i}t}{2}$ in the integrand.

Next, we make use of the following estimates

$$\begin{aligned}\left|(1\pm\tfrac{\mathrm{i}t}{2})^{\nu-1/2}-1\right| &\lesssim 2^{\nu/2}(\nu+1)t && \text{when } 0<t<2\\ \left|(1\pm\tfrac{\mathrm{i}t}{2})^{\nu-1/2}-1\right| &\lesssim 2^{\nu/2}t^{\nu-\frac{1}{2}} && \text{when } 2\leqslant t<\infty,\end{aligned}$$

We split the integrals over the ranges $0<t<2$ and $2\leqslant t<\infty$, apply the inequalities above in turn, and change variables to $s = rt$:

$$\begin{aligned}|R_\nu(r)| &\lesssim \frac{r^\nu 2^{\nu/2}}{\Gamma(\nu+\frac{1}{2})}\left((\nu+1)\int_0^2 \mathrm{e}^{-rt}t^{\nu+\frac{1}{2}}t\frac{\mathrm{d}t}{t}+\int_2^\infty \mathrm{e}^{-rt}t^{\nu-\frac{1}{2}}t^{\nu+\frac{1}{2}}\frac{\mathrm{d}t}{t}\right) && (4.3.5)\\ &= \frac{r^\nu 2^{\nu/2}}{\Gamma(\nu+\frac{1}{2})}\left((\nu+1)\int_0^{2r} \mathrm{e}^{-s}s^{\nu+\frac{1}{2}}\frac{\mathrm{d}s}{r^{\nu+3/2}}+\int_{2r}^\infty \mathrm{e}^{-s}s^{2\nu-1}\frac{\mathrm{d}s}{r^{2\nu}}\right)\\ &\lesssim \frac{2^{\nu/2}(\nu+1)}{\Gamma(\nu+\frac{1}{2})}\int_0^{2r}\mathrm{e}^{-s}s^{\nu+1/2}\frac{\mathrm{d}s}{r^{3/2}}+\frac{2^{\nu/2}}{\Gamma(\nu+\frac{1}{2})}\int_{2r}^\infty \mathrm{e}^{-s}s^{2\nu-1}\frac{\mathrm{d}s}{r^\nu}. && (4.3.6)\end{aligned}$$

Observe that the first integral in the last line is bounded by a gamma function:

$$\int_0^{2r}\mathrm{e}^{-s}s^{\nu-1/2}\,\mathrm{d}s \leqslant \int_0^\infty \mathrm{e}^{-s}s^{3/2+\nu-1}\,\mathrm{d}s \equiv \Gamma\left(\nu+\frac{3}{2}\right).$$

For the second integral in Eq. (4.3.6), note that $\mathrm{e}^s \geqslant s^{\nu+2}$, so

$$\int_{2r}^\infty \mathrm{e}^{-s}s^{2\nu-1}\frac{\mathrm{d}s}{r^\nu} \leqslant \int_{2r}^\infty s^{\nu-3}\frac{\mathrm{d}s}{r^\nu} \lesssim r^{-2}.$$

Thus, we obtain

$$|R_\nu(r)| \lesssim \frac{2^{\nu/2}(\nu+1)\Gamma(\nu+\frac{3}{2})}{\Gamma(\nu+\frac{1}{2})} r^{-3/2} + O(r^{-2}). \tag{4.3.7}$$

For $r \geqslant 1$, $r^{-2} \leqslant r^{-3/2}$, and we have the desired estimate of $R_\nu = O(r^{-3/2})$.

Eq. (4.3.3) follows from (4.3.5) upon setting $\nu = 0$.

□

As can be seen from Eq. (4.3.7), the constant in the remainder *depends on* ν and grows like 2^ν as $\nu \to \infty$.

Using stationary phase methods, Stein (1993) proved the error bound

$$|J_\nu(x)| \lesssim x^{-1/3} \quad \text{for} \quad x \geqslant 1$$

uniformly in x *and* ν, and that this decay is the best possible if we want it to remain independent of ν. (See Stein, 1993, Chapter VIII, §5.2.)

We will also use the following information about the zeros of J_0.

Lemma 4.3.2. *Let* $\nu \geqslant 0$. *The zeros* $j_{\nu,m}$ *of* J_ν *obey the asymptotics*

$$j_{\nu,m} = \pi m + \frac{\pi\nu}{2} - \frac{\pi}{4} + o(1) \quad \textit{as} \quad m \to \infty. \tag{4.3.8}$$

For the zeros of J_0, *we have*

$$j_{0,m} > m \quad \textit{for all} \quad m \geqslant 1. \tag{4.3.9}$$

Furthermore,

$$j_{0,m} J_1(j_{0,m})^2 \to \frac{2}{\pi} \quad \textit{as} \quad m \to \infty. \tag{4.3.10}$$

In particular,

$$\frac{|J_1(j_{0,m})|}{\sqrt{j_{0,m}}} \asymp \frac{1}{j_{0,m}}. \tag{4.3.11}$$

Proof. The asymptotics (4.3.8) may be found in §9.5.12 of Abramowitz and Stegun (1965). Combining them with Eq. (4.3.2) for $\nu = 1$, we have

$$\begin{aligned} j_{0,m} J_1(j_{0,m})^2 &= \frac{2}{\pi} \cos(j_{0,m} - \pi/2 - \pi/4)^2 + O\left(\frac{1}{j_{0,m}}\right) \\ &= \frac{2}{\pi} \cos(\pi m - \pi + o(1))^2 + O\left(\frac{1}{j_{0,m}}\right) \\ &\to \frac{2}{\pi} \quad \text{as} \quad m \to \infty. \end{aligned}$$

This proves Eq. (4.3.10), and Eq. (4.3.11) follows immediately.

On the other hand, (4.3.9) is easily obtained from the following inequality (Hethcote, 1970, Theorem 3)

$$j_{0,m} \geqslant \pi m - \frac{\pi}{4} \quad \text{for all} \quad m \geqslant 1. \qquad \square$$

4.4 L^p convergence of 1-dimensional Bessel–Fourier series

For the investigation of L^p convergence, a more precise analysis of the eigenfunctions is required.

Recall from Eq. (4.1.3) and the orthogonality of Bessel functions J_ν, that

$$\int_0^1 J_\nu(j_{\nu,m}x) J_\nu(j_{\nu,k}x) x \,\mathrm{d}x = \frac{1}{2} J'_\nu(j_{\nu,m})^2 \delta_{m,k}$$

for $m, k \geqslant 0$. This suggests two possible eigenfunction decompositions, for each fixed ν:

(i) $w_m = c\sqrt{x} J_\nu(j_{\nu,m}x)$ in $L^p([0,1])$ with the usual Lebesgue measure; or

(ii) $u_m = c J_\nu(j_{\nu,m}x)$ in $L^p([0,1], x\,\mathrm{d}x)$.

Note that

$$L^p([0,1]) \subsetneq L^p([0,1], x\,\mathrm{d}x)$$

for all $p > 1$, as $x^{-3/2p} \in L^p(x\,\mathrm{d}x) \setminus L^p$.

Wing (1950) considered both of the above cases, obtaining the following results.

Theorem 4.4.1. *Fix a real $\nu \geqslant -1$.*

(i) If $p > 1$ and $f \in L^p([0,1])$, then

$$\lim_{M\to\infty} \left\| f - \sum_{m=1}^{M} \langle f, w_m \rangle \, w_m \right\|_{L^p([0,1])} = 0,$$

with w_m as in (i) above.

(ii) If $\frac{4}{3} < p < 4$ and $f \in L^p([0,1], x\,\mathrm{d}x)$, then

$$\lim_{M\to\infty} \left\| f - \sum_{m=1}^{M} \langle f, u_m \rangle \, u_m \right\|_{L^p([0,1], x\,\mathrm{d}x)} = 0,$$

with u_m as in (ii) above. Here

$$\langle f, g\rangle = \int_0^1 fg \,\mathrm{d}\mu,$$

where $\mathrm{d}\mu = \mathrm{d}x$ or $\mathrm{d}\mu = x\,\mathrm{d}x$ as appropriate.

Since we are ultimately interested in Bessel–Fourier series on the disc, we will focus on expansions with $x\,\mathrm{d}x$, which correspond to the radial variable in polar coordinates.

Denote the partial sum operator with respect to the 'unweighted' eigenfunctions $u_m = cJ_\nu(j_{\nu,m}x)$ by $S_M^\nu f(x)$, i.e.,

$$S_M^\nu f(x) = \sum_{m=1}^{M} f_m J_\nu(j_{\nu,m}x),$$

where

$$f_m := \frac{2}{J_n'(j_{\nu,m})^2}\int_0^1 f(x)J_\nu(j_{\nu,m}x)x\,\mathrm{d}x.$$

Then

$$S_M^\nu f(x) = \int_0^1 K_M^\nu(x,r)f(r)r\,\mathrm{d}r \quad \text{where} \quad K_M^\nu(x,r) = 2\sum_{m=1}^{M}\frac{J_\nu(j_{\nu,m}x)J_\nu(j_{\nu,m}r)}{J_\nu'(j_{\nu,m})^2}.$$

Wing's proof of Theorem 4.4.1 is based on the following analysis of the kernel $K_M^\nu(x,r)$, obtained by MacRobert (1921).

Proposition 4.4.2. *Let $A_M := \pi(M + \frac{\nu}{2} + \frac{1}{4})$. Then,*

$$K_M^\nu(x,r) = \frac{A_M J_\nu(A_M x)J_\nu'(A_M r) - A_M J_\nu(A_M r)J_\nu'(A_M x)}{2(x-r)} + O\left(\frac{1}{\sqrt{xr}}\left(\frac{1}{x+r} + \frac{1}{2-x-r}\right)\right).$$

For a fixed ν, this estimate essentially reduces the uniform boundedness of the partial sums to the L^p boundedness of the certain singular integrals, given the uniform boundedness (in M and x) of the functions

$$\sqrt{A_M x}\,|J_\nu(A_M x)| \quad \text{and} \quad \sqrt{A_M x}\,\left|J_\nu'(A_M x)\right|,$$

which, in turn, follows from the previously discussed asymptotic formulas (4.3.2)—*provided that ν remains fixed throughout the proof.*

4.5 Mixed norm convergence of the 2-dimensional Bessel–Fourier series

By refining the asymptotic bounds on the Bessel functions, Balodis and Córdoba (1999) obtained convergence of the series (4.1.4) in the space $L^p_{\text{rad}}(L^2_{\text{ang}})$ defined by the norm

$$\|f\|_{p,2} := \left[\int_0^1 \left(\sum_n |f_n(r)|^2\right)^{p/2} r\,\mathrm{d}r\right]^{1/p} \equiv \Big\| \|(f_n(r))\|_{\ell^2} \Big\|_{L^p([0,1],r\,\mathrm{d}r)}, \tag{4.5.1}$$

where $f_n(r)$ are the Fourier coefficients of the angular function $t \mapsto f(r,t)$:

$$f_n(r) := \int_0^1 f(r,\theta)\mathrm{e}^{-2\pi \mathrm{i} n\theta}\,\mathrm{d}\theta.$$

More precisely, they proved the following theorem, where $S_{N,M}f$ denotes the partial sums

$$S_{N,M}f(r,t) := \sum_{n=-N}^{N}\sum_{m=1}^{M} c_{n,m}(f)J_n(j_{n,m}r)\mathrm{e}^{2\pi \mathrm{i} n t}. \tag{4.5.2}$$

Theorem 4.5.1 (Balodis and Córdoba, 1999)**.** *There is a constant $A > 0$ such that, if $M \geqslant AN + 1$, then the operators $S_{N,M}$ are uniformly bounded on $L^p_{\text{rad}}(L^2_{\text{ang}})$ if, and only if, $\frac{4}{3} < p < 4$.*

The norm convergence of the series to f follows by the usual argument. (See Corollary 4.5.3 below.)

To attack the proof of Theorem 4.5.1, they exploit the presence of the Fourier coefficients in the norm (4.5.1). Indeed, $S_{N,M}f$ is a trigonometric polynomial whose n^{th} Fourier mode ($|n| \leqslant N$) is

$$S^n_M f_n(r) \equiv \sum_{m=1}^{M} c_{n,m}(f)J_n(j_{n,m}r), \tag{4.5.3}$$

which is precisely the one-dimensional Bessel–Fourier series summation operator for the radial function $r \mapsto f_n(r)$ in terms of the nth order Bessel function J_n. Thus,

$$\|S_{N,M}f\|_{p,2} = \left[\int_0^1 \left(\sum_{n=-N}^{N} |S^n_M f_n(r)|^2\right)^{p/2} r\,\mathrm{d}r\right]^{1/p} = \|(S^n_M f_n)_n\|_{L^p([0,1],r\,\mathrm{d}r;\ell^2)},$$

so the boundedness of $S_{N,M}$ in $L^p_{\text{rad}}(L^2_{\text{ang}})$ is reduced to a uniform bound for vector-valued inequalities. Such bounds must be independent of the length, $2N + 1$, of the

vector

$$(S_M^n f_{-n}, \dots, S_M^n f_n)\,.$$

Note that $S_M^{-n} = S_M^n$ by our convention that $J_n = J_{|n|}$ for $n \in \mathbb{Z}$. The functions f_n and f_{-n}, however, are distinct in general, as they correspond to distinct Fourier coefficients!

Let us now turn to L^p convergence on the disc, where the relevant norm is

$$\|g\|_{L^p(\mathbb{D})} = \Big\| \|g(r,t)\|_{L^p(\,\mathrm{d}t)} \Big\|_{L^p(r\,\mathrm{d}r)}\,.$$

For $p \neq 2$ we cannot replace the inner 'angular' L^p norm by a sum of Fourier coefficients, (see Proposition 2.1.18). However, for $p > 2$ we have the Reverse Hausdorff–Young Inequality from Theorem 2.1.8. We therefore have

$$\begin{aligned}
\|g\|_{L^p(\mathbb{D})} &= \left[\int_0^1 \|g(r,t)\|^p_{L^p(\mathbb{T},\,\mathrm{d}t)}\, r\,\mathrm{d}r\right]^{1/p} \\
&\leqslant \left[\int_0^1 \left(\sum_k |g_k(r)|^q\right)^{p/q} r\,\mathrm{d}r\right]^{1/p} \\
&=: \|g\|_{p,q}\,. \qquad (4.5.4)
\end{aligned}$$

Using the latter norm, we define the space

$$L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}}) := \Big\{ f \in L^p(\mathbb{D}) : \|f\|_{p,q} < \infty \Big\}\,.$$

Careful inspection of the proofs in Balodis and Córdoba (1999) shows that the space ℓ^2 can be replaced by ℓ^q throughout when

$$\frac{1}{p} + \frac{1}{q} = 1 \quad \text{and} \quad 2 \leqslant p < 4.$$

The kernels of the 1-dimensional summation operators S_M^n in Eq. (4.5.3) are controlled by weighted vector-valued norms on the operators

$$\int_0^1 \frac{f(t)}{2-x-t}\,\mathrm{d}t \quad \text{and} \quad \int_0^1 \frac{f(t)}{x+t}\,\mathrm{d}t, \qquad (4.5.5)$$

as well as the Hilbert Transform (Hf) and the Hardy–Littlewood Maximal Functional $(\mathscr{M}f)$. These operators arise in the course of a micro-local analysis of the kernels S_M^n, the details of which do not matter for our purposes, other than to observe that all vector-valued inequalities involving ℓ^2 can be replaced by analogous

bounds involving ℓ^q.

Once again, for technical reasons, the inequalities concerning the kernel and its various pieces are reduced to weighted estimates using the weight $r^{1-p/2}$, which satisfies the 1-dimensional Muckenhoupt A_p condition if, and only if,

$$-1 < 1 - \frac{p}{2} < p - 1 \quad \text{that is} \quad \frac{4}{3} < p < 4$$

(see Grafakos, 2014b, Example 7.1.7). When this is the case, we have the inequalities

$$\|(\mathscr{M} f_n)_n\|_{L^p(r^{1-p/2};\ell^q)} + \|(Hf_n)_n\|_{L^p(r^{1-p/2};\ell^q)} \lesssim_{p,q} \|(f_n)_n\|_{L^p(r^{1-p/2};\ell^q)}$$

(see Córdoba, 1989, (B) on p. 25). Furthermore, the kernels of the operators in Eq. (4.5.5) are nice enough that both are bounded on the space $L^p([0,1], r^{1-p/2}\,\mathrm{d}r)$ and, since they are positive, they admit vector-valued extensions to ℓ^q too (Grafakos, 2014b, Theorem 5.5.10).

There is one more operator to consider (Balodis and Córdoba, 1999, p. 280):

$$T_M^\nu f(x) := \sqrt{x} f_\nu(A_M^n x) \int_0^1 \sqrt{t} f_\nu(A_M^n t) f(t)\,\mathrm{d}t,$$

with f_ν as in Lemma 1 of Balodis and Córdoba (1999). For the stated range of p, $p/q \geqslant 1$, so the application of Jensen's Inequality on p. 280 of Balodis and Córdoba (1999) goes through just fine, and we obtain the $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$-boundedness of T_M^ν too.

We have therefore shown the following theorem.

Theorem 4.5.2. *The operators $S_{N,M}$ are uniformly bounded on $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$ in the range $2 \leqslant p < 4$, when $M \geqslant AN + 1$ for an absolute constant $A > 0$.*

Corollary 4.5.3. *Let $2 \leqslant p < 4$. Then, if M_k, N_k are sequences of natural numbers such that $M_k \geqslant AN_k + 1$ and $N_k \to \infty$, we have*

$$\lim_{k\to\infty} \|S_{N_k,M_k} f - f\|_{p,q} = 0$$

for all $f \in L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$. In particular,

$$\lim_{k\to\infty} \|S_{N_k,M_k} f - f\|_{L^p(\mathbb{D})} = 0$$

for all $f \in L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$.

Proof. Consider the space

$$S = C_c^\infty[0,1] \otimes \{\text{trigonometric polynomials}\}.$$

For each function $f(r)\mathrm{e}^{2\pi \mathrm{i}lt} \in S$ we have

$$(S_{N,M} f(r,t))^{\wedge}(n) = \sum_{m=1}^{M} c_{l,m}(f) J_n(j_{n,m} r)\delta_{l,n} \quad \text{and} \quad \widehat{f}(r,n) = f(r)\delta_{l,n},$$

so

$$\sum_{n} \left| (S_{N,M} f(r,t))^{\wedge}(n) - \widehat{f}(r,n) \right|^q = \left| \sum_{m=1}^{M} c_{l,m}(f) J_k(j_{l,m} r) - f(r) \right|^q \to 0$$

uniformly as $M \to \infty$ (see, e.g., Watson, 1995, Chapter XVIII). It follows that $S_{M_k,N_k} f \to f$ uniformly for all $f \in S$. Since S is (uniformly) dense in the space $\{f \in C(\mathbb{D}) : f|_{\partial\mathbb{D}} = 0\}$, we see that S is dense in $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$.

For $f \in L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$, find a $g \in S$ such that $\|f - g\|_{p,q} < \varepsilon$. Using the uniform boundedness of the S_{N_k,M_k} in $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$, it follows that

$$\|S_{N_k,M_k} f - f\|_{p,q} \lesssim_p \|f - g\|_{p,q} + \|S_{N_k,M_k} g - g\|_{p,q} < \varepsilon$$

as $k \to \infty$. This proves the result for $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$, and the statement for $\|\cdot\|_{L^p(\mathbb{D})}$ now follows by Eq. (4.5.4). □

This method does not allow us to conclude L^p-norm convergence *for all* $f \in L^p$, but only for the smaller space $L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}}) \subsetneq L^p(\mathbb{D})$. To see this, recall the function

$$g(t) := \sum_{k=2}^{\infty} \frac{\mathrm{e}^{ik\log k}}{\sqrt{k}(\log k)^2} \mathrm{e}^{2\pi \mathrm{i}t},$$

which is continuous on $\mathbb{T}$ but whose Fourier coefficients are not ℓ^q-summable for any $q < 2$ (see the proof of Corollary 2.1.17). If we now set $f(r,t) := J_1(j_{1,1} r) g(t)$, then clearly $f \in L^p(\mathbb{D}) \setminus L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$ by Fubini's Theorem.

The above example, which is a counterexample to Plancherel's theorem in L^p, hints at the underlying issue: we did not obtain L^p *bounds* for the partial sum operators, so we cannot apply the proof of Corollary 4.5.3 to $L^p(\mathbb{D})$ directly.

We can, however, make use of a result of Wing (1950) to show that L^p convergence fails in general when $1 \leqslant p < \frac{4}{3}$—and, by duality, for $p > 4$.

Theorem 4.5.4 (Wing, 1950)**.** *The function $f(r) = r^{-3/2}$ lies in $L^p([0,1], r\,\mathrm{d}r)$ for $1 \leqslant p < \frac{4}{3}$, but its Bessel–Fourier series does not converge to f in $L^p([0,1], r\,\mathrm{d}r)$.*

Thus, the partial sum operators S^{ν}_M are not uniformly bounded on the space $L^p([0,1], r\,\mathrm{d}r)$ for $1 \leqslant p < \frac{4}{3}$, nor, by duality, for $p > 4$.

It follows that the 2-dimensional Bessel–Fourier series of the radial function given by $f(r,t) = r^{-3/2} \in L^p(\mathbb{D})$ fails to converge to f in $L^p(\mathbb{D})$ if $p \leqslant \frac{4}{3}$ or $p \geqslant 4$.

Wing only shows that the Bessel–Fourier series of $f(r) = r^{-3/2}$ is not *absolutely convergent* in $L^p([0,1], r\,\mathrm{d}r)$ by showing that

$$\limsup_{m\to\infty} \|f_m J_0(j_{0,m}r)\|_{L^p([0,1],r\,\mathrm{d}r)} > 0.$$

It could be that the oscillation of the $J_0(j_{0,m}r)$ introduce enough cancellations to guarantee convergence provided that we choose an appropriate method of truncation. We will revisit this question in Chapter 5.

4.6 Concluding remarks

We summarise all the pieces of the convergence puzzle in Table 4.1.

p range	$[1, \frac{4}{3})$	$[\frac{4}{3}, 2)$	2	$(2,4)$	4	$(4,\infty)$
$\|\cdot\|_{L^p(\mathbb{D})}$-convergence	No	?	Yes	$f \in L^p_{\mathrm{rad}}(\ell^q_{\mathrm{ang}})$	?	No

Table 4.1: L^p convergence of Bessel–Fourier series for various ranges of p.

In light of the existing results (Balodis and Córdoba, 1999; Wing, 1950; Benedek and Panzone, 1973), we might offer the following conjecture.

Conjecture 1. For all $\frac{4}{3} < p < 4$ and $f \in L^p(\mathbb{D})$,

$$\lim_{k\to\infty} \|S_{N_k,M_k} f - f\|_{L^p(\mathbb{D})} = 0,$$

for some appropriate choice of $N_k, M_k \in \mathbb{N}$, $k \geqslant 1$. For p outside of this range, convergence fails in general.

However, it is not completely clear that we can expect this. Córdoba (1989) proved that the ball multiplier is bounded on the mixed norm space $L^p_{\mathrm{rad}}(L^2_{\mathrm{ang}})$ in the range $\frac{2n}{n+1} < p < \frac{2n}{n-1}$ that was originally conjectured for $L^p(\mathbb{R}^n)$, but disproved in Fefferman (1971). As we saw above, the $L^p_{\mathrm{rad}}(L^2_{\mathrm{ang}})$ argument works by essentially 'eliminating' the angular eigenfunctions and reducing the problem to bounds on the one-dimensional Bessel-Fourier series. This 'trick' is not available in $L^p(\mathbb{D})$, owing to the failure of Plancherel's Theorem for $p \neq 2$, so obtaining uniform bounds on the operators $S_{N,M}$ is considerably more difficult.

Chapter 5

Decay of Bessel–Fourier coefficients

> *Neque enim quotiens chorda pellitur, unus edi tantum putandus est sonus ... unus sonus sensum pellit vel gravis, vel acutus, quamvis uterque ex pluribus constet.*
>
> —Boethius

In this chapter we will investigate the decay of 1-dimensional Bessel–Fourier coefficients in a manner parallel to that of Section 2.3, where we discussed the case of Fourier series. We will use our results to discuss the potential L^1 divergence of some Bessel–Fourier series on the disc.

Our motivation is Theorem 4.5.4, which exhibits an L^p function whose Bessel–Fourier series does not converge absolutely, but which might still converge due to cancellations, if we pick our truncations right. This creates a problem: how can we construct a function whose Bessel–Fourier series behaves in a particular way without writing down that Bessel–Fourier series? For the series in question cannot converge to the L^p function that we want! We are going to take inspiration from a similar situation in classical Fourier analysis, where one wants to construct an integrable function whose Fourier coefficients vanish arbitrarily slowly.

The Riemann–Lebesgue lemma is a classic result in Fourier analysis, which tells us that the Fourier coefficients of an integrable function, f, decay to 0 at high frequencies, i.e. $\widehat{f}(n) \to 0$ as $|n| \to \infty$.

By increasing the regularity of f, say $f \in C^k$, we can obtain faster rates of decay of the coefficients: $|n|^k \, \widehat{f}(n) \to 0$. But if we merely assume that $f \in L^1$, then the Riemann–Lebesgue lemma is the best possible result we can expect. Indeed, we

can find integrable functions whose Fourier coefficients decay as slowly as we like, as the following theorem shows (see, e.g., Grafakos, 2014b, Theorem 3.3.4).

Theorem 5.0.1. *Let $a_n \geqslant 0$ be a sequence such that $a_n \to 0$ as $|n| \to \infty$. Then, there is a function $f \in L^1([0,1])$ whose Fourier coefficients satisfy $\widehat{f}(n) \geqslant a_n$ for all $n \in \mathbb{Z}$.*

This may seem 'obvious': if we have the coefficients a_n, can we not just set

$$f(x) = \sum_{n=-\infty}^{\infty} a_n \mathrm{e}^{2\pi i n x}?$$

The answer is no, since there is no guarantee that this series will converge to an L^1 function, as the following example, taken from Grafakos (2014b), shows.

Example 5.0.2. Recall the *Fejér kernel* given by

$$F_N(x) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N+1}\right) \mathrm{e}^{2\pi i n x} \equiv \frac{1}{N+1} \frac{\sin(\pi(N+1)x)^2}{\sin(\pi x)^2}, \tag{5.0.1}$$

and consider the function

$$f(x) := \sum_{m=0}^{\infty} \frac{1}{2^m} F_{2^{2^m}}(x).$$

Since $F_N(x) \geqslant 0$ and $\|F_N\|_{L^1([0,1])} = 1$ for all N, this function is clearly well-defined and integrable. However, as Exercises 4.2.3 and 4.2.4 in Grafakos (2014b) show, the partial sums of the Fourier series of f diverge (in the L^1 norm) like $N \log(N)$.

Note that the Fourier coefficients of f are the nasty-looking

$$\widehat{f}(n) = \sum_{m=\lceil \log_2 \log_2 \log_2(|n|) \rceil}^{\infty} \left(\frac{1}{2^m} - \frac{|n|}{2^{2^{2^m}} + 1}\right),$$

which nevertheless tend to 0 as $|n| \to \infty$ by the Riemann–Lebesgue lemma.

In some situations, Fourier series may not be the ideal choice to expand our functions. For instance, as discussed in the last chapter, when working on the disc $\mathbb{D} := \{(r\cos(2\pi\theta), r\sin(2\pi\theta)) : r, \theta \in [0,1]\}$ it is convenient to expand $f\colon \mathbb{D} \to \mathbb{C}$ using eigenfunctions of the Laplacian on $\mathbb{D}$. This led to the (2-dimensional) *Bessel–Fourier series* of f,

$$\sum_{n=-\infty}^{\infty} \sum_{m=1}^{\infty} f_{n,m} J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi i n \theta}, \tag{5.0.2}$$

where J_n are Bessel functions of the first kind (of order $n \in \mathbb{N}$), $j_{n,m} > 0$ ($m \in \mathbb{N}$) are the zeros of J_n, and $f_{n,m}$ are the *Bessel–Fourier coefficients*

$$f_{n,m} := \frac{2}{J_{|n|+1}(j_{|n|,m})^2} \int_0^1 \int_0^1 f(r,\theta) J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi \mathrm{i} n\theta} r \,\mathrm{d}r \,\mathrm{d}\theta.$$

Note the $r\,\mathrm{d}r$ in the measure coming from the use of polar coordinates on $\mathbb{D}$.

If f is radial, i.e. $f(r,\theta) \equiv f(r)$ independently of θ, then the only non-zero (angular) Fourier coefficient occurs for $n = 0$. In this manner we obtain a 1-dimensional Bessel–Fourier series of f of order zero:

$$\sum_{m=1}^{\infty} f_m J_0(j_m x), \tag{5.0.3}$$

where we have set $j_m := j_{0,m}$ for notational convenience, and the Bessel–Fourier coefficients now take on the from

$$f_m := \frac{2}{J_1(j_m)^2} \int_0^1 f(x) J_0(j_m x) x \,\mathrm{d}x.$$

The corresponding 'Riemann–Lebesgue lemma' for the f_m is as follows (Young, 1920).

Theorem 5.0.3. *Fix $0 \leqslant s \leqslant \frac{1}{2}$. If $x^{1/2+s} f(x) \in L^1([0,1])$ and f_m are the Bessel–Fourier coefficients of f, then*

$$\lim_{m\to\infty} \frac{f_m}{j_m^{1/2+s}} = 0.$$

We note that Theorem 5.0.3 is still true for Bessel–Fourier coefficients of higher orders $\nu \in \mathbb{N}$, with $0 \leqslant s \leqslant \nu + \frac{1}{2}$, but these will not concern us here.

Of particular interest are the following special cases:

$$\sqrt{x} f(x) \in L^1 \implies \lim_{m\to\infty} \frac{f_m}{\sqrt{j_m}} = 0$$
$$x f(x) \in L^1 \implies \lim_{m\to\infty} \frac{f_m}{j_m} = 0.$$

We provide a direct proof of the case when $s = 0$ in Theorem 5.0.3 (i.e. when $\sqrt{x} f(x) \in L^1$), which avoids the use of the classical Riemann–Lebesgue lemma; it is a generalisation of that found in Tolstov and Silverman (1976) when $\sqrt{x} f(x) \in L^2$.

Proof of Theorem 5.0.3 when $s = 0$. We use $\nu = 0$ for notational convenience; the

general argument differs only orthographically.

We begin by assuming that $\sqrt{x}f(x) \in L^2$. Then, from Bessel's inequality we have that

$$\sum_{m=1}^{\infty} |f_m|^2 \, \|J_0(j_m)\|^2_{L^2(x\,\mathrm{d}x)} \leqslant \|f\|_{L^2(x\,\mathrm{d}x)}\,.$$

Hence,

$$\lim_{m\to\infty} |f_m| \, \|J_0(j_m)\|_{L^2(x\,\mathrm{d}x)} = 0.$$

Using the inequality

$$\int_0^1 J_0(j_m x)^2 x \,\mathrm{d}x \gtrsim \frac{1}{\sqrt{j_m}},$$

(Tolstov and Silverman, 1976, (1.14)) it follows that

$$\lim_{m\to\infty} \frac{f_m}{\sqrt{j_m}} = 0.$$

Now consider f with $\sqrt{x}f(x) \in L^1$. Fix $\varepsilon > 0$ and find g_ε with $\sqrt{x}g_\varepsilon(x) \in L^2$ such that

$$\int_0^1 |f(x) - g_\varepsilon(x)| \sqrt{x}\,\mathrm{d}x < \varepsilon. \tag{5.0.4}$$

Let us write $g_{\varepsilon,m}$ for the mth Bessel–Fourier coefficient of g_ε. Then, for a large enough $M \in \mathbb{N}$, we have

$$\frac{|g_{\varepsilon,m}|}{\sqrt{j_m}} < \varepsilon \quad \text{for all} \quad m \geqslant M$$

by the previous case. Hence,

$$\frac{|f_m|}{\sqrt{j_m}} \leqslant \varepsilon + \frac{|f_m - g_{\varepsilon,m}|}{\sqrt{j_m}} \quad \text{for all} \quad m \geqslant M. \tag{5.0.5}$$

To bound the remaining term, we use the asymptotics of Bessel functions and their zeros from Lemmas 4.3.1 and 4.3.2:

$$\begin{aligned}
\frac{|f_m - g_{\varepsilon,m}|}{\sqrt{j_m}} &\lesssim \frac{1}{\sqrt{j_m} J_1(j_m)^2} \int_0^1 |f(x) - g_\varepsilon(x)| \, |J_0(j_m x)| \, x \,\mathrm{d}x \\
&\lesssim \sqrt{j_m} \int_0^{1/j_m} |f(x) - g_\varepsilon(x)| \, x \,\mathrm{d}x && (5.0.6) \\
&\quad + \sqrt{j_m} \int_{1/j_m}^1 |f(x) - g_\varepsilon(x)| \, \frac{|\cos(j_m x - \pi/4)|}{\sqrt{j_m x}} x \,\mathrm{d}x && (5.0.7) \\
&\quad + \sqrt{j_m} \int_{1/j_m}^1 |f(x) - g_\varepsilon(x)| \, |R_0(j_m x)| \, x \,\mathrm{d}x. && (5.0.8)
\end{aligned}$$

In Eq. (5.0.6), we have $x \leqslant \frac{\sqrt{x}}{j_m}$ for $0 \leqslant x \leqslant \frac{1}{j_m}$, whence

$$(5.0.6) \lesssim \frac{\sqrt{j_m}}{\sqrt{j_m}} \int_0^1 |f(x) - g_\varepsilon(x)| \sqrt{x}\, \mathrm{d}x \lesssim \varepsilon.$$

For the second term, we simply use $|\cos(j_m x - \pi/4)| \leqslant 1$ and observe that

$$(5.0.7) \lesssim \frac{\sqrt{j_m}}{\sqrt{j_m}} \int_0^1 |f(x) - g_\varepsilon(x)| \sqrt{x}\, \mathrm{d}x \lesssim \varepsilon.$$

Finally, for Eq. (5.0.8), recall the bound $R_0(r) \lesssim r^{-3/2} \leqslant r^{-1/2}$ for $r \geqslant 1$. Since $x > \frac{1}{j_m}$, we have

$$\begin{aligned}(5.0.8) &\lesssim \sqrt{j_m} \int_{1/j_m}^1 |f(x) - g_\varepsilon(x)| \frac{x}{\sqrt{j_m x}}\, \mathrm{d}x \\ &\lesssim \int_0^1 |f(x) - g_\varepsilon(x)| \sqrt{x}\, \mathrm{d}x \lesssim \varepsilon.\end{aligned}$$

Combining these three estimates with Eq. (5.0.5) shows that

$$\frac{|f_m|}{\sqrt{j_m}} \lesssim \varepsilon \quad \text{for all} \quad m \geqslant M,$$

and we are done. □

In light of Theorem 5.0.3, we might suspect that we can find L^1 functions whose coefficients can be made to blow up at any given rate below $\sqrt{m}$ (whenever $\sqrt{x} f(x) \in L^1$) or even m (if $x f(x) \in L^1$). The following example supports our suspicions.

Example 5.0.4. Consider the functions $f^{(\alpha)}(x) = x^{-\alpha}$ for $\alpha > 0$, which satisfy $x f^{(\alpha)}(x) \in L^1$ if, and only if, $\alpha < 2$. Taking $\alpha = 2 - \varepsilon$, the functions $f^{(\alpha)}$ satisfy $x f^{(\alpha)}(x) \in L^1$ for all $\varepsilon > 0$. Their Bessel–Fourier coefficients can be estimated as follows (neglecting constants):

$$\begin{aligned} f_m^{(\alpha)} &= \frac{2}{J_1(j_m)^2} \int_0^1 x^{-2+\varepsilon} J_0(j_m x) x\, \mathrm{d}x \\ &\asymp j_m \int_0^{j_m} j_m^{1-\varepsilon} \frac{J_0(y)}{y^{1-\varepsilon}} \frac{\mathrm{d}y}{j_m} \\ &\asymp j_m^{1-\varepsilon} \left(\int_0^{j_1} \frac{J_0(y)}{y^{1-\varepsilon}}\, \mathrm{d}y + \int_{j_1}^{j_m} \frac{\cos(y - \pi/4)}{y^{3/2-\varepsilon}}\, \mathrm{d}y + \int_{j_1}^{j_m} O(y^{-5/2+\varepsilon})\, \mathrm{d}y \right). \end{aligned}$$

The first integral is clearly finite. The second integral is uniformly bounded as

$j_m \to \infty$ by a similar argument to that in the proof of Lemma 5.3.1 below. The final integral is also uniformly bounded in j_m. Therefore, $f_m^{(\alpha)} = O(m^{1-\varepsilon})$.

In particular, taking $\varepsilon = \frac{1}{2}$, we have $f^{(3/2)}(x) = x^{-3/2}$, so that $xf^{(3/2)}(x) \in L^1$ but $\sqrt{x}f^{(3/2)}(x) \notin L^1$. In this case, the coefficients decay as $\sqrt{m}$.

In this chapter we prove the following analogue of Theorem 5.0.1 for Bessel–Fourier series, which is a partial converse to Theorem 5.0.3.

Theorem 5.0.5. *Let $0 \leqslant \alpha < \frac{1}{2}$ and $a_m \geqslant 0$ be a sequence such that $a_m \to 0$. Then, there is a function f with $x^{\alpha+1}f(x) \in L^1$ whose Bessel–Fourier coefficients of order zero f_m satisfy*

$$0 \leqslant a_m \leqslant \frac{f_m}{j_m^\alpha} \to 0 \quad \text{as} \quad m \to \infty.$$

Furthermore, f can be chosen so that $f_{2m} = 0$ for all $m \in \mathbb{N}$, or so that $f_{2m+1} = 0$ for all $m \in \mathbb{N}$.

We also show how to construct an L^1 function on the disc whose Bessel–Fourier series diverges in norm for any possible truncation of the partial sums, assuming that Theorem 5.0.5 can be improved to coefficients with $a_m = o(\sqrt{j_m})$. Here, the condition $f_{2m+1} = 0$ will be crucial.

Note that Theorem 5.0.5 does not prove that Theorem 5.0.3 is 'sharp'. Indeed,

(i) in Theorem 5.0.3 we need only assume that $x^{\alpha+1/2}f(x) \in L^1$ to get a decay of $f_m = o(j_m^{\alpha+1/2})$, for all $0 \leqslant \alpha \leqslant \frac{1}{2}$;

(ii) in Theorem 5.0.5 we construct an f with $x^{\alpha+1}f(x) \in L^1$ such that $f_m = o(j_m^\alpha)$ for $0 \leqslant \alpha < \frac{1}{2}$.

Our result requires an extra $\sqrt{x}$, loses a factor $\sqrt{j_m}$ in the decay, and leaves the end-point $\alpha = \frac{1}{2}$ open.

Assuming a numerically-plausible conjecture, we can show a stronger result, namely, that for any $0 \leqslant \alpha < \frac{1}{2}$, there is an f with $\sqrt{x}f(x) \in L^1$ whose coefficients decay as $f_m = o(j_m^\alpha)$. We consider this result in Section 5.2. Later, in Section 5.3, we give fully rigorous proofs of Theorem 5.0.5 without assuming any conjectures.

5.1 Two useful identities

In this brief section, we record two useful identities, which will be needed in several calculations in this chapter. The first is

$$J_0'(x) = -J_1(x) \quad \text{for all} \quad x \in \mathbb{R}. \tag{5.1.1}$$

This follows from the power-series representation of J_ν given in Section 4.3:

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{1}{(k+\nu)!} \left(\frac{x}{2}\right)^{2k+\nu}.$$

Note that we have used $\Gamma(k+\nu+1) = (k+\nu)!$, given that $k, \nu \in \mathbb{N}_0$. This series is absolutely convergent on $\mathbb{R}$, and so differentiating term by term yields

$$\begin{aligned} J_0'(x) &= \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \frac{2k}{k!} \frac{x^{2k-1}}{2^{2k}} \\ &= \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{(k+1)!} \frac{2k+2}{(k+1)!} \frac{x^{2k+1}}{2^{2k+2}} \\ &= -\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{1}{(k+1)!} \frac{x^{2k+1}}{2^{2k+1}} = -J_1(x). \end{aligned}$$

The second useful formula is

$$\int_0^x J_0(y) y \, \mathrm{d}y = x J_1(x) \quad \text{for all} \quad x \geqslant 0. \tag{5.1.2}$$

To see this, recall that J_0 satisfies Bessel's differential equation:

$$y^2 J_0'(y) + y J_0'(y) + y^2 J_0(y) = 0.$$

Integrating the differential equation and applying Eq. (5.1.1) therefore leads us to Eq. (5.1.2). Indeed,

$$\begin{aligned} \int_0^x J_0(y) y \, \mathrm{d}y &= -\int_0^x \left(J_0'(y) y + J_0'(y)\right) \mathrm{d}y \\ &= \int_0^x \frac{\mathrm{d}}{\mathrm{d}y} \left(-y J_0'(y)\right) \mathrm{d}y \\ &= \left[y J_1(y)\right]_0^x = x J_1(x). \end{aligned}$$

5.2 'Strong' decay theorems

In this section we consider some stronger results than Theorem 5.0.5, which depend on some numerically plausible conjectures. We do not have proofs of these conjectures, at present; but we will discuss why we think they are true, and how they would yield the stronger decay theorems.

Let us write

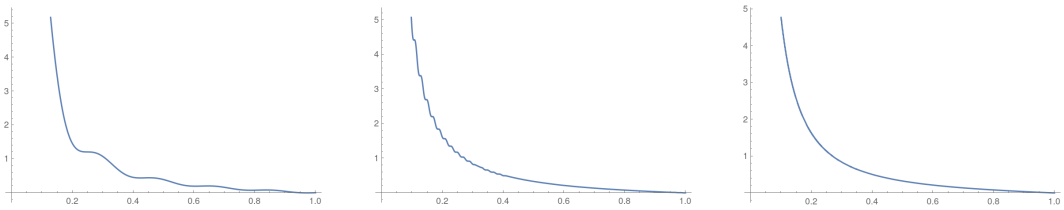
$$S_m^\alpha(x) := \sum_{k=1}^{m} j_k^\alpha J_0(j_k x).$$

We will need the following 'modified Fejér kernel' for the Bessel–Fourier series:

$$K_M^\alpha(x) := \frac{1}{M+1} \sum_{m=1}^{M} S_m^\alpha(x) = \sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) j_m^\alpha J_0(j_m x).$$

We will also set $K_0^\alpha(x) \equiv 1$ when $M = 0$. This kernel will replace the Fejér kernel in the construction of our function with a given decay coefficients—see the outline in Section 2.3.

As we saw in Eq. (5.0.1), the classic Fejér kernel for Fourier series is everywhere non-negative, which makes computing L^1 norms much easier. Numerical simulations suggests that the same is true for K_M^α in the range $0 \leqslant \alpha \leqslant \frac{1}{2}$, i.e. that $K_M^\alpha(x) \geqslant 0$ for all $M \in \mathbb{N}$ and $x \in [0,1]$—see Fig. 5.1.



(a) Plot of $K_{10}^{1/2}(x)$. (b) Plot of $K_{100}^{1/2}(x)$. (c) Plot of $K_{1000}^{1/2}(x)$.

Figure 5.1: Plots of $K_M^{1/2}(x)$ for $0 \leqslant x \leqslant 1$ with $M = 10$, $M = 100$ and $M = 1000$, respectively. (Generated with Mathematica.)

Conjecture 2. For all, $0 \leqslant \alpha \leqslant \frac{1}{2}$, $M \in \mathbb{N}$ and $x \in [0,1]$, $K_M^\alpha(x) \geqslant 0$.

Assuming this conjecture, we can prove the following strengthening of Theorem 5.0.5.

Theorem 5.2.1. *Assume that Conjecture 2 is true. Let $a_m \geqslant 0$ be a sequence such that $a_m \to 0$ and fix $0 \leqslant \alpha < \frac{1}{2}$. Then there is a function f such that $\sqrt{x} f(x) \in L^1$, whose Bessel–Fourier coefficients of order 0 satisfy*

$$0 \leqslant a_m \leqslant \frac{f_m}{j_m^\alpha} \to 0 \quad \text{as} \quad m \to \infty.$$

Proof. By Lemma 2.3.3 there is a decreasing convex sequence $c_m \geqslant a_m$ such that $c_m \to 0$. (Note that we may need to take $c_0 = 2c_1$, say, in order to have a decreasing,

convex sequence defined for $m \geqslant 0$.) We use $(c_m)_{m\geqslant 0}$ to construct the function

$$f(x) := \sum_{l=1}^{\infty}(l+1)(c_l + c_{l+2} - 2c_{l+1})K_l^\alpha(x). \tag{5.2.1}$$

We claim that $\sqrt{x}f(x) \in L^1$.

By assumption (i.e. Conjecture 2), $K_M^\alpha(x) \geqslant 0$, so

$$\left\|\sqrt{x}K_M^\alpha\right\|_{L^1([0,1])} = \int_0^1 K_M^\alpha(x)\sqrt{x}\,\mathrm{d}x.$$

Writing K_M^α out as a sum, and changing variables in the integral we have

$$\begin{aligned}\int_0^1 K_M^\alpha(x)\sqrt{x}\,\mathrm{d}x &= \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right) j_m^\alpha \int_0^1 J_0(j_m x)\sqrt{x}\,\mathrm{d}x \\ &= \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\frac{j_m^\alpha}{j_m^{3/2}}\int_0^{j_m} J_0(y)\sqrt{y}\,\mathrm{d}y. \end{aligned}\tag{5.2.2}$$

We now split the range of integration into two pieces. When $0 \leqslant y \leqslant 1$, we have $J_0(y)\sqrt{y} = O(1)$. When $1 \leqslant y \leqslant j_m$, we can use the asymptotics for J_0 in Lemma 4.3.1. Hence,

$$\frac{j_m^\alpha}{j_m^{3/2}}\int_0^{j_m} J_0(y)\sqrt{y}\,\mathrm{d}y = \frac{j_m^\alpha}{j_m^{3/2}}\int_0^{1} J_0(y)\sqrt{y}\,\mathrm{d}y \tag{5.2.3}$$

$$+ \frac{j_m^\alpha}{j_m^{3/2}}\int_1^{j_m}\sqrt{\frac{2}{\pi}}\cos(y-\pi/4)\,\mathrm{d}y \tag{5.2.4}$$

$$+ \frac{j_m^\alpha}{j_m^{3/2}}\int_1^{j_m} O\left(\frac{1}{y}\right)\mathrm{d}y. \tag{5.2.5}$$

Now, using the asymptotics for j_m in Lemma 4.3.2, we have

$$(5.2.3) \lesssim \frac{1}{m^{3/2-\alpha}}, \quad (5.2.4) \lesssim \frac{1}{m^{3/2-\alpha}} \quad \text{and} \quad (5.2.5) \lesssim \frac{\log m}{m^{3/2-\alpha}} \lesssim \frac{1}{m^{3/2-\alpha/2}},$$

since $m^{-\alpha/2}\log m \to 0$. For $\alpha < 1/2$, all of the above terms are summable. Combining and summing these terms in Eq. (5.2.2), we deduce that the norms $\|\sqrt{x}K_M^\alpha\|_{L^1([0,1])}$ are bounded uniformly in M. Therefore, from Eq. (5.2.1) and Lemma 2.3.4 we see that

$$\left\|\sqrt{x}f\right\|_{L^1([0,1])} \lesssim \sum_{l=0}^{\infty}(l+1)(c_l + c_{l+2} - 2c_{l+1}) = c_0 < \infty,$$

completing the proof of our claim.

A straightforward (but tedious) calculation shows that the Bessel–Fourier coefficients of f are precisely $j_m^\alpha c_m$, i.e.

$$\frac{2}{J_1(j_m)^2}\int_0^1 f(x)J_0(j_m x)x\,\mathrm{d}x = j_m^\alpha c_m \quad \text{for all} \quad m\in\mathbb{N},$$

and this concludes the proof. Indeed, since f is an L^1 function, we can exchange the sum and the integral to compute (following Theorem 3.3.4 in Grafakos 2014b)

$$\begin{aligned}
f_m &= \frac{2}{J_1(j_m)^2}\int_0^1 f(x)J_0(j_m x)x\,\mathrm{d}x \\
&= \sum_{l=0}^{\infty}(l+1)(c_l+c_{l+2}-2c_{l+1})\frac{2}{J_1(j_m)^2}\int_0^1 K_l^\alpha(x)J_0(j_m x)x\,\mathrm{d}x \\
&= \sum_{l=0}^{\infty}(l+1)(c_l+c_{l+2}-2c_{l+1})\sum_{k=1}^{l}\left(1-\frac{k}{l+1}\right)j_m^\alpha\times \\
&\qquad \left[\frac{2}{J_1(j_m)^2}\int_0^1 J_0(j_l x)J_0(j_m x)x\,\mathrm{d}x\right] \\
&= \sum_{l=m}^{\infty}(l+1)(c_l+c_{l+2}-2c_{l+1})\sum_{k=1}^{l}\left(1-\frac{k}{l+1}\right)j_m^\alpha\delta(k-m) \\
&= j_m^\alpha\sum_{l=m}^{\infty}(l+1)(c_l+c_{l+2}-2c_{l+1})\frac{l+1-m}{l+1} \\
&= j_m^\alpha\sum_{l=m}^{\infty}(l+1-m)(c_l+c_{l+2}-2c_{l+1}) \\
&= j_m^\alpha\sum_{s=0}^{\infty}(s+m+1)(c_{s+m}+c_{s+m+2}-2c_{s+m+1}) \\
&= j_m^\alpha c_m \geqslant j_m^\alpha a_m,
\end{aligned}$$

by Lemma 2.3.4. Note that the mth Bessel–Fourier coefficient of K_l^α can only be non-zero for $l\geqslant m$, which is why we truncate the sum to start from $l=m$, and then re-index by $s=l-m\geqslant 0$ in order to apply the lemma. □

Remark. We have reason to suspect that, if $\alpha=\frac{1}{2}$, then we can prove a similar result, this time with $xf(x)\in L^1$.

In this case, the terms obtained from the decomposition in Eqs. (5.2.3) to (5.2.5) are no longer summable. However, if we continue to assume that $K_M^{1/2}(x)$

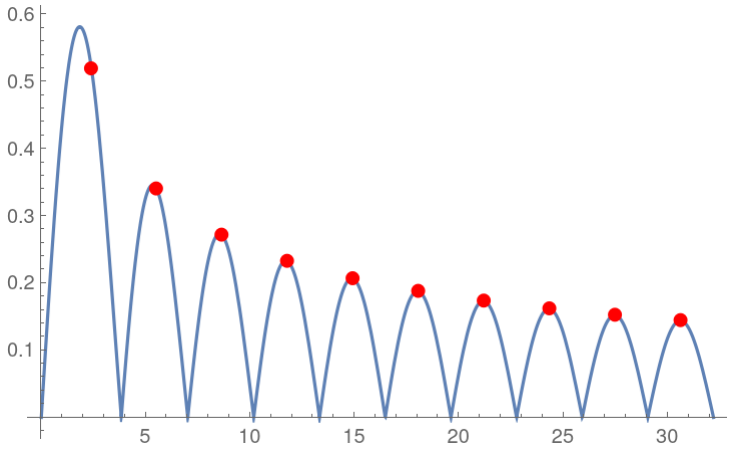


Figure 5.2: The values of $|J_1(j_m)|$ appear monotonically decreasing. (Generated with Mathematica.)

is non-negative, then we can make use of Eq. (5.1.2) to compute

$$\begin{aligned}\int_0^1 K_M^{1/2}(x)x\,\mathrm{d}x &= \sum_{m=1}^{M}\left(1-\frac{m}{M+1}\right)\frac{\sqrt{j_m}}{j_m^2}\int_0^{j_m} J_0(y)y\,\mathrm{d}y\\ &= \sum_{m=1}^{M}\left(1-\frac{m}{M+1}\right)\frac{\sqrt{j_m}}{j_m^2}j_m J_1(j_m).\end{aligned}$$

Now, since by Eq. (5.1.1), $J_1 = -J_0'$, it is not hard to see that the signs of $J_1(j_m)$ alternate, with $J_1(j_{2m}) < 0$ and $J_1(j_{2m+1}) > 0$. On the other hand, we expect that the values of $|J_0'(j_m)| = |J_1(j_m)|$ form a decreasing sequence. Thus,

$$\begin{aligned}\int_0^1 K_M^{1/2}(x)x\,\mathrm{d}x &= \sum_{m=1}^{M}\left(1-\frac{m}{M+1}\right)\frac{\sqrt{j_m}}{j_m^{3/2}}j_m J_1(j_m)\\ &= \sum_{m=1}^{M}\left(1-\frac{m}{M+1}\right)(-1)^{m+1}\,|J_1(j_m)|\\ &= \frac{1}{M+1}\sum_{m=1}^{M}\sum_{n=1}^{m}(-1)^{m+1}\,|J_1(j_m)|\,,\end{aligned}\tag{5.2.6}$$

which should be summable by the Alternating Series Test. It is then clear that, defining f as in Eq. (5.2.1) with $\alpha = \frac{1}{2}$ yields $xf(x) \in L^1$, and the computation of the coefficients f_m goes through just as before.

However, we have not been able to prove that the sequence $(|J_1(j_m)|)_m$ is decreasing—see Fig. 5.2 for why think that this is true.

5.3 'Weak' decay theorems

Conjecture 2 seems likely—see Fig. 5.1—but we do not currently have a proof. However, we have obtained uniform lower bounds on $x^{\alpha+1}K_M^\alpha(x)$ for $0 \leqslant \alpha < \frac{1}{2}$, and this is enough to prove Theorem 5.0.5. The results in this section, although weaker than Theorem 5.2.1, have completely analytical proofs and no longer depend on any conjectures.

The main tool is the following lemma.

Lemma 5.3.1. *For $0 \leqslant \alpha < \frac{1}{2}$, there is a constant $C > 0$ such that*

$$x^{\alpha+1}K_M^\alpha(x) \geqslant -C \quad \text{uniformly in} \quad M \in \mathbb{N} \quad \text{and} \quad x \in [0,1].$$

We can now re-state and prove Theorem 5.0.5.

Theorem 5.3.2 (= Theorem 5.0.5)**.** *Let $0 \leqslant \alpha < \frac{1}{2}$ and $a_m \geqslant 0$ be a sequence such that $a_m \to 0$. Then, there is a function f with $x^{\alpha+1}f(x) \in L^1$ whose Bessel–Fourier coefficients of order zero f_m satisfy*

$$0 \leqslant a_m \leqslant \frac{f_m}{j_m^\alpha} \to 0 \quad \text{as} \quad m \to \infty.$$

Furthermore, f can be chosen so that $f_{2m} = 0$ for all $m \in \mathbb{N}$, or so that $f_{2m+1} = 0$ for all $m \in \mathbb{N}$.

Proof. Define f as in Eq. (5.2.1). This time we will show that $x^{\alpha+1}f(x) \in L^1$.

By Lemma 5.3.1, $x^{\alpha+1}K_M^\alpha(x) + C \geqslant 0$, so

$$\left\|x^{\alpha+1}K_M^\alpha\right\|_{L^1([0,1])} \leqslant C + \int_0^1 (K_M^\alpha(x)x^{\alpha+1} + C)\,\mathrm{d}x = 2C + \int_0^1 K_M^\alpha(x)x^{\alpha+1}\,\mathrm{d}x.$$

Writing K_M^α out as a sum, and changing variables in the integral as before we have

$$\begin{aligned}\int_0^1 K_M^\alpha(x)x\,\mathrm{d}x &= \sum_{m=1}^M \left(1 - \frac{m}{M+1}\right) j_m^\alpha \int_0^1 J_0(j_m x)x^{\alpha+1}\,\mathrm{d}x \\ &= \sum_{m=1}^M \left(1 - \frac{m}{M+1}\right) \frac{j_m^\alpha}{j_m^{\alpha+2}} \int_0^{j_m} J_0(y)y^{\alpha+1}\,\mathrm{d}y. \qquad (5.3.1)\end{aligned}$$

We now split the integral into the ranges $0 \leqslant y \leqslant 1$, $1 \leqslant y \leqslant j_m$ and use the asymptotics from Lemma 4.3.1:

$$\int_0^{j_m} J_0(y)y^{\alpha+1}\,\mathrm{d}y = \int_0^1 J_0(y)y^{\alpha+1}\,\mathrm{d}y \qquad (5.3.2)$$

$$+\int_1^{j_m}\sqrt{\frac{2}{\pi}}\int_0^{j_m}\cos(y-\pi/4)y^{\alpha+1/2}\,\mathrm{d}y \tag{5.3.3}$$

$$+\int_1^{j_m}O\left(y^{\alpha-1/2}\right)\,\mathrm{d}y. \tag{5.3.4}$$

Now, for (5.3.2), we immediately have

$$\frac{1}{j_m^2}\int_0^1 J_0(y)y^{\alpha+1}\,\mathrm{d}y\lesssim\frac{1}{j_m^2},$$

and for (5.3.4)

$$\frac{1}{j_m^2}\int_1^{j_m}O\left(y^{\alpha-1/2}\right)\,\mathrm{d}y\lesssim\frac{1}{j_m^2}\left(j_m^{\alpha+1/2}-1\right)\lesssim\frac{1}{j_m^{3/2-\alpha}}.$$

To get suitable decay in (5.3.3), we have to integrate by parts. Neglecting nuisance factors of $\frac{2}{\pi}$, we have

$$\begin{aligned}\frac{1}{j_m^2}\int_1^{j_m}\cos(y-\pi/4)y^{\alpha+1/2}\,\mathrm{d}y&=\frac{1}{j_m^2}\left[\sin(y-\pi/4)y^{\alpha+1/2}\right]_1^{j_m}\\&\quad-\frac{\alpha+1/2}{j_m^2}\int_1^{j_m}\sin(y-\pi/4)y^{\alpha-1/2}\,\mathrm{d}y\\&\lesssim\frac{j_m^{\alpha+1/2}}{j_m^2}+\frac{1}{j_m^2}+\frac{j_m^{\alpha+1/2}}{j_m^2}+\frac{1}{j_m^2}\\&\lesssim\frac{1}{j_m^{3/2-\alpha}}.\end{aligned}$$

Since $\alpha<1/2$, all of the above terms are summable, and, therefore, so is (5.3.1). In other words, $\left\|x^{\alpha+1}K_M^\alpha\right\|_{L^1([0,1])}=O(1)$ uniformly in M, and so, from Eq. (5.2.1) and Lemma 2.3.4, we see that

$$\left\|x^{\alpha+1}f\right\|_{L^1([0,1])}\lesssim\sum_{l=0}^{\infty}(l+1)(c_l+c_{l+2}-2c_{l+1})=c_0<\infty,$$

completing the proof of our claim.

The rest of the proof is now identical to that of Theorem 5.2.1.

A slight modification of the above arguments shows that we can find an f with $x^{\alpha+1}f(x)\in L^1$ whose only non-zero Bessel–Fourier coefficients are the even-indexed

f_{2m}. It suffices to replace the kernel K_M^α with the 'even kernel'

$$K_M^{\alpha,e}(x) := \sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) j_m^\alpha J_0(j_{2m}x).$$

As can be seen from the proof of Lemma 5.3.1, we still have that $x^{\alpha+1}K_M^{\alpha,e}(x) \geqslant -C$ uniformly in $M \in \mathbb{N}$ and $x \in [0,1]$ (even if $K_M^{\alpha,e}(x)$ may now not be non-negative for all x and M). We construct f as in Eq. (5.2.1) and show that it satisfies $x^{\alpha+1}f(x) \in L^1$, just as before. Then, the same computation of the f_m shows that

$$f_{2m} = j_m^\alpha c_m \quad \text{and} \quad f_{2m+1} = 0 \quad \text{for all} \quad m \in \mathbb{N}. \tag{5.3.5}$$

Indeed,

$$\begin{aligned}(K_M^{\alpha,e})_m &= \sum_{k=1}^{M} \left(1 - \frac{k}{M+1}\right) \frac{2j_m^\alpha}{J_1(j_m)^2} \int_0^1 J_0(j_m x) J_0(j_{2k}x) x \,\mathrm{d}x \\ &= \sum_{k=1}^{M} \left(1 - \frac{k}{M+1}\right) j_m^\alpha \delta_{m,2k},\end{aligned}$$

which vanishes whenever $m > 2M$ or m is odd. Thus,

$$(K_M^{\alpha,e})_m = \begin{cases} \left(1 - \frac{m/2}{M+1}\right) j_{m/2}^\alpha, & \text{if } m \leqslant 2k \text{ is even;} \\ 0, & \text{if } m \text{ is odd or } m > 2k. \end{cases}$$

Eq. (5.3.5) now follows.

By making the obvious adjustments, one can also construct an f satisfying

$$f_{2m} = 0 \quad \text{and} \quad f_{2m+1} = j_m^\alpha c_m \quad \text{for all} \quad m \in \mathbb{N}. \qquad \square$$

Remark. The argument to construct and even-only or odd-only coefficients does not extend in a straightforward way to the case $\alpha = \frac{1}{2}$, discussed in the remark after Theorem 5.2.1. Assuming that the $|J_1(j_m)|$ are decreasing, we need both the odd- and even-indexed terms to ensure that the the series in Eq. (5.2.6) converges.

5.4 Bounds on $x^{\alpha+1}K_M^\alpha(x)$

In this section, we prove Lemma 5.3.1. First we will need another technical lemma.

Lemma 5.4.1. *For every $\alpha > 0$ and $u > 0$, we have*

$$\max_{m>0} \left\{ m^{\alpha} \mathrm{e}^{-mu} \right\} = \left(\frac{\alpha}{\mathrm{e}u} \right)^{\alpha} \lesssim u^{-\alpha}.$$

Proof. Let $h(y) := y^{\alpha} \mathrm{e}^{-yu}$ and differentiate twice to find

$$h'(y) = \alpha y^{\alpha-1} \mathrm{e}^{-yu} + -u y^{\alpha} \mathrm{e}^{-yu},$$

and

$$h'(y) = \left[\alpha(\alpha-1) y^{\alpha-2} - 2\alpha u y^{\alpha-1} + u^2 y^{\alpha} \right] \mathrm{e}^{-uy}.$$

It is then easy to see that the only solution to $h'(y) = 0$ is $y = \frac{\alpha}{u}$ and that

$$h'\left(\frac{\alpha}{u} \right) = -\frac{\alpha^{\alpha-1}}{u^{\alpha-2}} \mathrm{e}^{-\alpha} < 0,$$

so that $\max_{y>0} h(y) = h(\alpha/u) = \left(\frac{\alpha}{\mathrm{e}u} \right)^{\alpha}$. □

On to the proof of Lemma 5.3.1. To begin, note that it is sufficient to prove that $x^{\alpha+1} S_M^{\alpha}(x) \geqslant -C$ uniformly in x and M. Using the asymptotics from Eq. (4.3.3), we have that

$$\begin{aligned} x^{\alpha+1} S_M^{\alpha}(x) &= x^{\alpha+1} \sum_{m=1}^{M} j_m^{\alpha} J_0(j_m x) \\ &= \sqrt{\frac{2}{\pi}} x^{\alpha+1/2} \sum_{m=1}^{M} \frac{\cos(j_m x - \pi/4)}{j_m^{1/2-\alpha}} + \sum_{m=1}^{M} x^{\alpha+1} j_m^{\alpha} R_0(j_m x) \\ &\equiv \sqrt{\frac{2}{\pi}} x^{\alpha+1/2} j_m^{\alpha} \widetilde{S}_M^{\alpha}(x) + R_M^{\alpha}(x), \end{aligned} \tag{5.4.1}$$

where we have defined

$$\widetilde{S}_M^{\alpha} := \sum_{m=1}^{M} \frac{\cos(j_m x - \pi/4)}{j_m^{1/2-\alpha}} \quad \text{and} \quad R_M^{\alpha}(x) := \sum_{m=1}^{M} x^{\alpha+1} R_0(j_m x).$$

It suffices to obtain lower bounds on $x^{\alpha+1/2} \widetilde{S}_M^{\alpha}(x)$ and upper bounds on $|R_M(x)|$ that are uniform in $x \in [0,1]$ and $M \in \mathbb{N}$.

Lower bounds on $\widetilde{S}_M^{\alpha}$. Since $x^{\alpha+1/2} \geqslant 0$, it suffices to obtain lower bounds on the sums $\widetilde{S}_M^{\alpha}$. To this end, we follow Zygmund (2003) and relate the sum to an

integral. We claim that

$$\widetilde{S}_M^{\alpha}(x) = \frac{x/2}{\sin(\pi x/2)} \int_0^{j_M+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u + O(1). \tag{5.4.2}$$

To show this, note that, for each $m \in \mathbb{N}$, we have

$$\begin{aligned}\int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \cos(ux-\pi/4)\,\mathrm{d}u &= \left[\frac{\sin(ux-\pi/4)}{x}\right]_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \\ &= \frac{2\sin(\pi x/2)}{x}\cos(j_m x-\pi/4),\end{aligned}$$

using the sum-to-product formula for a difference of sines

$$\sin(a)-\sin(b) = 2\sin\left(\frac{a-b}{2}\right)\cos\left(\frac{a+b}{2}\right).$$

Hence, integrating over all intervals $[j_m-\frac{\pi}{2}, j_m+\frac{\pi}{2}]$ for $1 \leqslant m \leqslant M$, yields

$$\begin{aligned}\sum_{m=1}^{M}\int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u &= \sum_{m=1}^{M}\int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \cos(ux-\pi/4)\left(\frac{1}{u^{1/2-\alpha}}-\frac{1}{j_m^{1/2-\alpha}}\right)\mathrm{d}u \\ &\quad + \sum_{m=1}^{M}\int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{j_m^{1/2-\alpha}}\,\mathrm{d}u \\ &\leqslant \sum_{m=1}^{M}\sup_{|u-j_m|<\frac{\pi}{2}}\left|\frac{1}{u^{1/2-\alpha}}-\frac{1}{j_m^{1/2-\alpha}}\right| \\ &\quad + \frac{\sin(\pi x/2)}{x/2}\sum_{m=1}^{M}\frac{\cos(j_m x-\pi/4)}{j_m^{1/2-\alpha}}.\end{aligned}$$

By the mean value theorem, for $|u-j_m|<\frac{\pi}{2}$,

$$\frac{1}{u^{1/2-\alpha}}-\frac{1}{j_m^{1/2-\alpha}} = \left(\alpha-\frac{1}{2}\right)\frac{j_m-u}{y^{3/2-\alpha}} \quad \text{with} \quad j_m-\frac{\pi}{2} \leqslant y \leqslant j_m+\frac{\pi}{2},$$

whence we obtain, using Eq. (4.3.9), that

$$\sup_{|u-j_m|<\frac{\pi}{2}}\left|\frac{1}{u^{1/2-\alpha}}-\frac{1}{j_m^{1/2-\alpha}}\right| \lesssim \frac{1}{j_m^{3/2-\alpha}} \lesssim \frac{1}{m^{3/2-\alpha}}.$$

Using this bound we have that

$$\sum_{m=1}^{M} \int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u - \frac{\sin(\pi x/2)}{x/2} \sum_{m=1}^{M} \frac{\cos(j_m x-\pi/4)}{j_m^{1/2-\alpha}} \lesssim \sum_{m=1}^{M} \frac{1}{m^{3/2-\alpha}} = O(1) \tag{5.4.3}$$

since $\alpha < 1/2$.

The next step is to relate

$$\sum_{m=1}^{M} \int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u \quad \text{to} \quad \int_0^{j_M+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u.$$

For this we need to recall the asymptotics for j_m:

$$j_m = \pi m - \frac{\pi}{4} + \varepsilon_m,$$

where (according to McMahon's asymptotic expansions for large zeros)

$$\varepsilon_m = \frac{1}{8\pi m - 2\pi} + O\left(\frac{1}{m^3}\right) \quad \text{as} \quad m \to \infty.$$

(See the formula on p. 505 of Watson 1995.)

Thus, if we sum over all the intervals $[j_m - \frac{\pi}{2}, j_m + \frac{\pi}{2}]$ for $1 \leqslant m \leqslant M$, we will obtain the intergral over $[j_1 - \frac{\pi}{2}, j_M + \frac{\pi}{2}]$, plus a sum of the errors obtained from the overlaps that occur around the midpoints

$$\frac{j_{m+1} - j_m}{2} = \frac{\pi}{2} + \frac{\varepsilon_{m+1} - \varepsilon_m}{2} = \frac{\pi}{2} + O\left(\frac{1}{m^2}\right).$$

These errors are therefore of order

$$\int_{\frac{\pi}{2}-\frac{1}{m^2}}^{\frac{\pi}{2}+\frac{1}{m^2}} \frac{1}{u^{1/2-\alpha}}\,\mathrm{d}u \lesssim \frac{1}{m^2},$$

and so are summable in m. It follows that

$$\sum_{m=1}^{M} \int_{j_m-\frac{\pi}{2}}^{j_m+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u = \int_0^{j_M+\frac{\pi}{2}} \frac{\cos(ux-\pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u + O(1).$$

As $(x/2)/\sin(\pi x/2) > 0$ is bounded above and below on $[0,1]$, Eq. (5.4.2) follows from this display and Eq. (5.4.3).

Since we are interested in the behaviour of this integral as $M \to \infty$, we will replace $j_M + \frac{\pi}{2}$ by M. We now claim that

$$\int_0^M \frac{\cos(ux - \pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u \geqslant -\frac{C_1}{x^{\alpha+1/2}} \quad \text{for some} \quad C_1 > 0. \tag{5.4.4}$$

After a change of variables and by using the identity

$$\cos(a - \pi/4) = \frac{1}{\sqrt{2}}\cos(a) + \frac{1}{\sqrt{2}}\sin(a),$$

we obtain

$$\int_0^M \frac{\cos(ux - \pi/4)}{u^{1/2-\alpha}}\,\mathrm{d}u = \frac{1}{x^{\alpha+1/2}\sqrt{2}}\left(\int_0^{Mx} \frac{\cos(v)}{v^{1/2-\alpha}}\,\mathrm{d}v + \int_0^{Mx} \frac{\sin(v)}{v^{1/2-\alpha}}\,\mathrm{d}v\right).$$

Thus, it remains to bound the indefinite integrals

$$G(y) := \int_0^y \frac{\cos(v)}{v^{1/2-\alpha}}\,\mathrm{d}v \quad \text{and} \quad H(y) := \int_0^y \frac{\sin(v)}{v^{1/2-\alpha}}\,\mathrm{d}v.$$

We deal with the cosine integral, the sine case being easier. Since

$$G'(y) = \frac{\cos(y)}{y^{1/2-\alpha}} \quad \text{and} \quad G'(y) = -\frac{\sin(y)}{y^{1/2-\alpha}} - \left(\alpha - \frac{1}{2}\right)\frac{\cos(y)}{y^{\alpha+1/2}},$$

and $\alpha < 1/2$, we find that the local minima of G of occur precisely for $y_k = \frac{3\pi}{2} + 2\pi k$ (for any $k \in \mathbb{N}_0$). These minima occur between consecutive zeros of the cosine. Since $v^{\alpha-1/2}$ is decreasing in v, the net contribution to the integral over this period is positive, since the negative part of the $\cos(v)$ is weighted against a smaller value of $v^{\alpha-1/2}$—see Fig. 5.3. More formally,

$$\begin{aligned}
G(y_{k+1}) - G(y_k) &= \int_{\frac{3\pi}{2}+2\pi k}^{\frac{3\pi}{2}+2\pi k+2\pi} \frac{\cos(v)}{v^{1/2-\alpha}}\,\mathrm{d}v \\
&= \int_0^{2\pi} \frac{\cos(w + 3\pi/2 + 2\pi k)}{(w + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w \quad (w = v - 3\pi/2 - 2\pi k) \\
&= \int_0^{2\pi} \frac{\sin(w)}{(w + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w \quad (\cos(a + 3\pi/2) = \sin(a)) \\
&= \int_0^{\pi} \frac{\sin(w)}{(w + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w \\
&\quad + \int_{\pi}^{2\pi} \frac{\sin(w)}{(w + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w
\end{aligned}$$

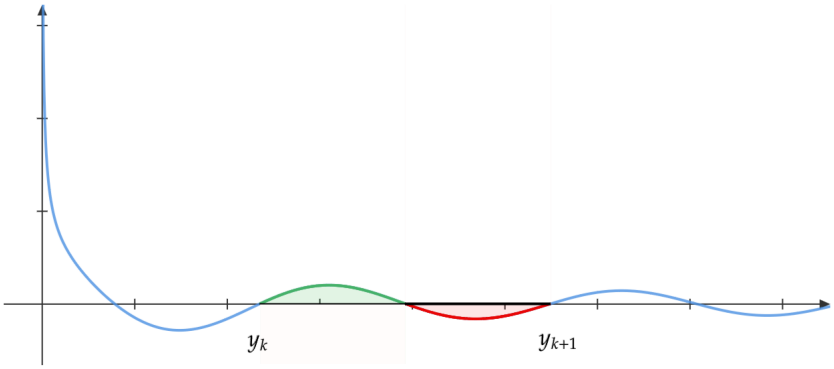


Figure 5.3: The function $\frac{\cos v}{v^{1/2-\alpha}}$. The positive green area outweighs the negative red area for a positive net contribution between y_k and y_{k+1}.

$$\begin{aligned}
&\geqslant \int_0^{\pi} \frac{\sin(w)}{(\pi + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w \\
&\quad + \int_{\pi}^{2\pi} \frac{\sin(w)}{(\pi + 3\pi/2 + 2\pi k)^{1/2-\alpha}}\,\mathrm{d}w \\
&= \frac{1}{(\pi + 3\pi/2 + 2\pi k)^{1/2-\alpha}} \left(\int_0^{\pi} \sin(w)\,\mathrm{d}w + \int_{\pi}^{2\pi} \sin(w)\,\mathrm{d}w \right) \\
&= 0.
\end{aligned}$$

This shows that the local minima of G are increasing. Thus, $G(Mx) \geqslant G\left(\frac{3\pi}{2}\right)$ uniformly in $M \in \mathbb{N}$ and $x \in [0,1]$, and, similarly, $H(Mx) \geqslant H(0) = 0$, proving Eq. (5.4.4).

Combining Eqs. (5.4.2) and (5.4.4) yields

$$x^{\alpha+1/2}\widetilde{S}_M^{\alpha}(x) \geqslant -C_1 - x^{\alpha+1/2}C_0 \geqslant -C/2 \tag{5.4.5}$$

uniformly in $M \in \mathbb{N}$ and $x \in [0,1]$.

Bounds on R_M^{α}. Recall that $R_M^{\alpha}(x) := \sum_{m=1}^{M} x^{\alpha+1} j_m^{\alpha} R(j_m x)$. By Eq. (4.3.9), $\mathrm{e}^{-j_m xt} \leqslant \mathrm{e}^{-mxt}$ for all $x \in [0,1]$ and $t \in (0,\infty)$. On the other hand, $j_m \lesssim m$ for all $m \geqslant 1$. Taking $u = xt/2$ in Lemma 5.4.1, we have

$$m^{\alpha}\mathrm{e}^{-mxt} = \frac{m^{\alpha}}{\mathrm{e}^{\frac{mxt}{2}}}\mathrm{e}^{-\frac{mxt}{2}} \lesssim \frac{\mathrm{e}^{-\frac{mxt}{2}}}{(xt)^{\alpha}}.$$

Hence, from Eq. (4.3.3) we obtain the following bounds for all $x > 0$:

$$\begin{aligned}
|R_M^\alpha(x)| &\leqslant x^{\alpha+1} \int_0^2 \sum_{m=1}^{\infty} m^\alpha \mathrm{e}^{-mxt} \sqrt{t}\, \mathrm{d}t + x^{\alpha+1} \int_2^\infty \sum_{m=1}^{\infty} \frac{m^\alpha \mathrm{e}^{-mxt}}{t}\, \mathrm{d}t \\
&\lesssim \int_0^2 \sum_{m=1}^{\infty} \mathrm{e}^{-\frac{mxt}{2}} \frac{x^{\alpha+1}\sqrt{t}}{(xt)^\alpha}\, \mathrm{d}t + \int_2^\infty \sum_{m=1}^{\infty} \mathrm{e}^{-\frac{mxt}{2}} \frac{x^{\alpha+1}}{x^\alpha t^{\alpha+1}}\, \mathrm{d}t \\
&= \int_0^2 \frac{\mathrm{e}^{-xt/2}}{1-\mathrm{e}^{-xt/2}} \frac{x\sqrt{t}}{t^\alpha}\, \mathrm{d}t + \int_2^\infty \frac{\mathrm{e}^{-xt/2}}{1-\mathrm{e}^{-xt/2}} \frac{x}{t^{\alpha+1}}\, \mathrm{d}t \\
&= \int_0^2 \frac{1}{\mathrm{e}^{xt/2}-1} \frac{x\sqrt{t}}{t^\alpha}\, \mathrm{d}t + \int_2^\infty \frac{1}{\mathrm{e}^{xt/2}-1} \frac{x}{t^{\alpha+1}}\, \mathrm{d}t \\
&\leqslant \int_0^2 \frac{2}{xt} \cdot \frac{x\sqrt{t}}{t^\alpha}\, \mathrm{d}t + \int_2^\infty \frac{2}{xt} \cdot \frac{x}{t^{\alpha+1}}\, \mathrm{d}t \\
&\lesssim \int_0^2 \frac{1}{t^{\alpha+1/2}}\, \mathrm{d}t + \int_2^\infty \frac{1}{t^{\alpha+2}}\, \mathrm{d}t \leqslant C/2.
\end{aligned}$$

uniformly in $x > 0$ and $M \in \mathbb{N}$. Hence, we can take $x \to 0^+$ and obtain uniform bounds on $|R_M^\alpha(x)|$ for $x \in [0,1]$ and $M \in \mathbb{N}$. In conjunction with Eq. (5.4.5), we deduce the result from Eq. (5.4.1).

5.5 Comments for further research

We mentioned at the beginning of the chapter that our Theorem 5.0.5 does not quite give us that Theorem 5.0.3 is 'sharp'. Our result requires an extra $\sqrt{x}$, loses a factor $\sqrt{m}$ in the decay, and leaves the end-point $\alpha = \frac{1}{2}$ open. We saw in the proof that these 'limitations' essentially come from Lemma 5.3.1, since we require higher powers of x to ensure that the remainder terms in the asymptotics for J_0 remain uniformly bounded as $x \to 0^+$.

If, on the other hand, we were able to show that $K_M^\alpha \geqslant 0$ directly, then we obtain stronger Theorem 5.2.1, where we always get $\sqrt{x} f(x) \in L^1$ for any value of $0 \leqslant \alpha < \frac{1}{2}$, and, assuming further that $|J_1(j_m)|$ is decreasing, we can show the end-point case with $xf(x) \in L^1$.

Based on our numerical calculations—see Fig. 5.1—we make the following conjecture.

Conjecture 3. Given any non-negative sequence $a_m \to 0$, there is an $L^1(x\,\mathrm{d}x)$ function f whose Bessel–Fourier coefficients satisfy

$$0 \leqslant a_m \leqslant \frac{f_m}{\sqrt{j_m}} \to 0 \quad \text{as} \quad m \to \infty.$$

In order to prove the L^1 divergence result we are after, we need to make an additional assumption.

Conjecture 4. The function in Conjecture 3 can be chosen such that

$$f_{2m} = 0 \quad \text{and} \quad f_{2m+1} \geqslant \sqrt{j_m} a_m \quad \text{for all} \quad m \in \mathbb{N},$$

or with

$$f_{2m} \geqslant \sqrt{j_m} a_m \quad \text{and} \quad f_{2m+1} = 0 \quad \text{for all} \quad m \in \mathbb{N}.$$

We saw that this is true when $0 \leqslant \alpha < \frac{1}{2}$, so Conjecture 4 is a natural extension to the case $\alpha = \frac{1}{2}$. However, as we saw earlier, we are less certain that such a function can actually be constructed in this case—at least using our methods.

Assuming both Conjecture 3 and Conjecture 4, we can prove the following proposition.

Proposition 5.5.1. *Assume that Conjecture 3 and Conjecture 4 are both true. Then, there is a function $f \in L^1(\mathbb{D})$ such that the partial sums*

$$S_{M,N}(r,\theta) = \sum_{n=-N}^{N} \sum_{m=1}^{M} c_{n,m}(f) J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi \mathrm{i} n \theta}$$

diverge in $L^1(r\,\mathrm{d}r\,\mathrm{d}\theta)$ for any choice of $M = M_k, N = N_k \to \infty$ as $k \to \infty$.

Proof. If the conjectures are true, then by taking $a_m = \frac{1}{\log m}$, we can find a function $g \in L^1([0,1], x\,\mathrm{d}x)$ whose 1-dimensional Bessel–Fourier coefficients satisfy and

$$g_{2m} = 0 \quad \text{and} \quad g_{2m+1} \geqslant \frac{\sqrt{j_{0,m}}}{\log(m)}.$$

Set $f(r,\theta) := g(r)$, so that the 2-dimensional Bessel–Fourier series of f is given by

$$S_{M,N} f(r,\theta) = \sum_{m=1}^{M} g_m J_0(j_{0,m} r),$$

and so

$$\|S_{M,N} f\|_{L^1(\mathbb{D})} \geqslant \left| \int_0^1 S_{M,N} f(r,\theta) r\,\mathrm{d}r \right| = \left| \sum_{m=1}^{M} g_m \int_0^1 J_0(j_{0,m} r) r\,\mathrm{d}r \right|.$$

Using Eq. (5.1.2) we have

$$\int_0^1 S_{M,N}f(r,\theta)r\,\mathrm{d}r = \sum_{m=1}^{M}\frac{g_m}{j_{0,m}^2}\int_0^{j_{0,m}} J_0(y)y\,\mathrm{d}y = \sum_{m=1}^{M}\frac{g_m}{j_{0,m}}J_1(j_{0,m}).$$

Now, since $J_1 = -J_0'$, it is not hard to see that the signs of $J_1(j_{0,m})$ alternate, with $J_1(j_{0,2m}) < 0$ (i.e., J_0 is decreasing at $j_{0,2m}$) and $J_1(j_{0,2m+1}) > 0$ (i.e., J_0 is increasing at $j_{0,2m+1}$). Since $g_{2m} = 0$ for all m, the partial sums are non-negative and therefore, using

$$\frac{|J_1(j_{0,m})|}{\sqrt{j_{0,m}}} \gtrsim \frac{1}{j_{0,m}}$$

from Eq. (4.3.10), we obtain

$$\|S_{N,M}f\|_{L^1(\mathbb{D})} = \sum_{m=1}^{M}\frac{g_{2m+1}}{\sqrt{j_{0,2m+1}}}\frac{|J_1(j_{0,2m+1})|}{\sqrt{j_{0,2m+1}}} \gtrsim \sum_{m=1}^{M}\frac{1}{m\log(2m+1)} \to \infty,$$

as required. □

Chapter 6

Domain perturbation

Whereof hee soon aware,
Each perturbation smooth'd with outward calme.
—Milton

We have looked at the L^p convergence (and divergence) of eigenfunction expansions in very specific settings—*viz.* the torus, some triangles and the disc.

One way in which we could explore the issue of convergence in other planar domains is by perturbation. The idea is to take a planar region Ω and deform it into another region $\widetilde{\Omega}$. We then study how the eigenfunctions of the Laplacian and the projection operators onto eigenspaces change alongside this change to Ω. The hope is that, for sufficiently small perturbations we can carry over convergence results on a domain where they are known (e.g. a square, a triangle) to new domains where the problem is still unsolved.

We now discuss the setup of the perturbation approach in some more detail, following Lamberti and Lanza de Cristoforis (2005).

Let $\Omega \subset \mathbb{R}^2$ be a bounded, open and connected domain in the plane (so, in particular, the Poincaré inequality holds on Ω). We perturb Ω by a map $\varphi\colon \Omega \to \widetilde{\Omega}$ close to the identity in a sense to be made precise later. We are interested in relating the Dirichlet eigenproblem on $\widetilde{\Omega}$ back to that on Ω. We wish to compare the two operators by applying them to functions defined on Ω. Thus, Δ will denote the Laplacian acting on $u \in H_0^1(\Omega)$, while Δ_φ denotes the Laplacian acting on $u \circ \varphi^{-1} \in H_0^1(\widetilde{\Omega})$, and then 'pulled back' via φ to $H_0^1(\Omega)$. Thus, $\Delta - \Delta_\varphi$ is well-defined as an operator acting on functions defined on Ω.

We list the (positive) eigenvalues of $-\Delta$, *including multiplicities*, as

$$0 < \lambda_1 \leqslant \lambda_2 \leqslant \lambda_3 \leqslant \dots.$$

Similarly, we denote the eigenvalues of $-\Delta_\varphi$ by $\widetilde{\lambda}_j$.

Note that, since 0 is not an eigenvalue, the Laplacian and its inverse yield equivalent eigenvalue problems:

$$-\Delta u = \lambda u \quad \text{if and only if} \quad -\Delta^{-1} u = \lambda^{-1} u. \tag{6.0.1}$$

We will therefore concentrate on the latter problem, since the inverse operators are compact and self-adjoint (see Lamberti and Lanza de Cristoforis, 2005, Lemma 3.5) and Section 6.4.

We will need to impose a further technical condition on the eigenspaces we consider. Fix a finite subset $F \subset \mathbb{N}$ and denote by P_F the orthogonal projection in L^2 onto the eigenfunctions u_j associated to the λ_j with $j \in F$. Note, in particular, that $u_j \neq u_{j+1}$ with $\lambda_j = \lambda_{j+1}$ if λ_j has a multiplicity of at least 2.

Definition 6.0.1. We say that F *splits* an eigenvalue λ if there are indices $j, k \in \mathbb{N}$ such that $\lambda = \lambda_j = \lambda_k$ and $j \in F$ but $k \notin F$.

In terms of the projection P_F, this means that $P_F u_j = u_j$ but $P_F u_k = 0$ even though $-\Delta u_j = \lambda u_j$ and $-\Delta u_k = \lambda u_k$. We need to rule out eigenvalue splitting in order to apply Kato's projection formula (see the remark following Theorem 6.1.1 for further details).

In Section 6.1 we set up the necessary notation and discuss the perturbation of the domain in some detail. In Section 6.2 we prove that under certain assumptions on φ, F will not split eigenvalues of Δ_φ if it did not split eigenvalues of Δ. Next, in sections Sections 6.3 and 6.4, we obtain a series of auxiliary L^p estimates on the resolvents $(-\Delta^{-1} - z)^{-1}$ and the operators $\Delta_\varphi^{-1} - \Delta^{-1}$ which are crucial for the proof of our main theorem of the chapter. In Section 6.5 we state and prove our main result for the Dirichlet Laplacian: L^p to L^p boundedness of the projections under perturbations of the domain Ω. Section 3.7 explores some open problems and directions for future research.

This chapter has been uploaded to arXiv and submitted for publication—see Acosta Babb and Robinson (2024).

6.1 Preliminaries

Consider a homeomorphism $\varphi\colon \Omega \to \widetilde{\Omega}$; we wish to study operators on both of these domains. Since these operators act on different function spaces, e.g. $H_0^1(\Omega)$ and $H_0^1(\widetilde{\Omega})$, we cannot compare them directly.

Let $u, v \in H_0^1(\Omega)$. Their L^2 inner product is

$$\langle u, v \rangle_{L^2(\Omega)} = \int_\Omega uv \, \mathrm{d}x. \tag{6.1.1}$$

On the other hand, $u \circ \varphi^{-1}, v \circ \varphi^{-1} \in H_0^1(\widetilde{\Omega})$, with L^2 inner product

$$\begin{aligned} \left\langle u \circ \varphi^{-1}, v \circ \varphi^{-1} \right\rangle_{L^2(\widetilde{\Omega})} &= \int_{\varphi(\Omega)} u(\varphi^{-1}(y)) v(\varphi^{-1}(y)) \, \mathrm{d}y \\ &= \int_\Omega u(x) v(x) \left| \det D\varphi(x) \right| \, \mathrm{d}x =: Q_\varphi[u, v]. \end{aligned} \tag{6.1.2}$$

If we denote the standard inner product on $L^2(\Omega)$ by Q, then we can compare the inner products from Eqs. (6.1.1) and (6.1.2):

$$(Q - Q_\varphi)[u, v] = \int_\Omega uv(1 - |\det D\varphi|) \, \mathrm{d}x \quad \text{for all} \quad u, v \in H_0^1(\Omega). \tag{6.1.3}$$

We begin by recording a key result from Lamberti and Lanza de Cristoforis (2005) which we are going to generalise in this chapter. Let H be a Hilbert space equipped with two different bilinear forms: Q and $\widetilde{Q}$. Suppose that T and $\widetilde{T}$ are two compact operators from H to itself and self-adjoint on (H, Q) and $(H, \widetilde{Q})$, respectively. Let $F \subset \mathbb{N}$ be a finite set of indices and denote by $P_F[Q, T]$ the projection

$$P_F[Q, T]u := \sum_{j \in F} Q[u, u_j] u_j,$$

where u_j are eigenfunctions of T corresponding to the indices $j \in F$.

Theorem 6.1.1 (Lamberti and Lanza de Cristoforis, 2005)**.** *Let $F \subset \mathbb{N}$ be a finite set of indices that does not split eigenvalues of T or $\widetilde{T}$. Then, there is a positive constant $C = C(T, \widetilde{T}, F)$ such that*

$$\left\| P_F[Q, T] - P_F[\widetilde{Q}, \widetilde{T}] \right\|_{H \to H} \leqslant C \left(\left\| Q - \widetilde{Q} \right\|_{H \times H \to \mathbb{R}} + \left\| T - \widetilde{T} \right\|_{H \to H} \right). \tag{6.1.4}$$

Remark. The main tool in the proof is Kato's integral formula for a projection:

$$P_F[Q, T] = \operatorname{Re} \left(-\frac{1}{2\pi \mathrm{i}} \int_{\gamma[F,T]} (T - zI)^{-1} \, \mathrm{d}z \right). \tag{6.1.5}$$

Here $\gamma[F, T]$ is a contour encircling the portion of the spectrum of T that contains the eigenvalues μ_j of T with $j \in F$.

As we can see, the integral formula is 'blind' to multiplicities: it only differ-

entiates the *values* of the eigenvalues, and there is no way of keeping track of exactly which eigenfunctions we want to keep in our projection. This is the reason behind the no-splitting condition, since if our set F were to split an eigenvalue, then Kato's formula would not actually give us the projection $P_F[T, Q]$, but rather a projection onto a larger space which includes *all eigenfunctions* for every eigenvalue in F.

We will use a similar argument in which $T = \Delta^{-1}$ and $\widetilde{T}$ is a pullback to Ω of the inverse Laplacian on $\widetilde{\Omega}$. Thus, there are three Laplacians in play here: the Laplacian $\Delta = \Delta_\Omega$ acting, say, on $C_c^\infty(\Omega)$; the Laplacian $\Delta_{\widetilde{\Omega}}$ acting on $C_c^\infty(\widetilde{\Omega})$; and a transformed Laplacian Δ_φ acting also on $C_c^\infty(\Omega)$ as follows:

$$\Delta_\varphi f(x) := (\Delta_{\widetilde{\Omega}}(f \circ \varphi^{-1}))(\varphi(x)) \quad \text{for all} \quad f \in C_c^\infty(\Omega).$$

That is: the function $f \circ \varphi^{-1}$ is defined on $\widetilde{\Omega}$, which we then differentiate using $\Delta_{\widetilde{\Omega}}$, and finally pull back to Ω by composing the result with φ.

We will denote the eigenfunctions of Δ_Ω and $\Delta_{\widetilde{\Omega}}$ respectively by u_j and $\widetilde{u}_j$. The eigenfunctions of Δ_φ are $u_j^\varphi := \widetilde{u}_j \circ \varphi$ (see Lamberti and Lanza de Cristoforis, 2005, Theorem 3.10).

We are now in a position to distinguish three 'partial sum' operators of interest:

(i) Partial sums on Ω, which are expansions in $L^2(\Omega)$ in terms of the eigenfunctions u_j of Δ_Ω relative to Ω:

$$P_F f = \sum_{j \in F} \langle u_j, f \rangle_\Omega \, u_j \quad \text{for all} \quad f \in L^2(\Omega). \tag{6.1.6}$$

(ii) Partial sums on $\widetilde{\Omega}$, which are expansions in $L^2(\widetilde{\Omega})$ in terms of the eigenfunctions $\widetilde{u}_j$ of $\Delta_{\widetilde{\Omega}}$ relative to $\widetilde{\Omega}$:

$$\widetilde{P}_F g = \sum_{j \in F} \langle \widetilde{u}_j, g \rangle_{\widetilde{\Omega}} \, \widetilde{u}_j \quad \text{for all} \quad g \in L^2(\widetilde{\Omega}).$$

(iii) Transformed partial sums on Ω which are 'pull-backs' of $\widetilde{P}_F$ to $L^2(\Omega)$. These are expansions with respect to the eigenfunctions u_j^φ of Δ_φ in $L^2(\Omega)$, now equipped with the inner product Q_φ:

$$P_F^\varphi f = \sum_{j \in F} Q_\varphi[u_j^\varphi, f] u_j^\varphi \quad \text{for all} \quad f \in L^2(\Omega). \tag{6.1.7}$$

Note that, by Moser Iteration (see, e.g, Gilbarg and Trudinger, 2001, Theorem

8.15), the eigenfunctions u_j of Δ satisfy the bounds

$$\|u_j\|_{L^\infty(\Omega)} \lesssim \lambda_j^{1/p} \|u_j\|_{L^2(\Omega)} = \lambda_j^{1/p} \quad \text{for all} \quad j \in \mathbb{N}.$$

Thus, for any $u \in L^p(\Omega)$,

$$|\langle u_j, u\rangle| \leqslant \|u_j\|_{L^\infty(\Omega)} \|u\|_{L^1(\Omega)} \lesssim_{j,p,|\Omega|} \|u\|_{L^p(\Omega)} < \infty,$$

and so the projection operators P_F are well-defined on $L^p(\Omega)$. With the necessary adjustments, the same is true of P_F^φ.

Most of the work will go into obtaining bounds on $\left\|T - \widetilde{T}\right\|_{L^p(\Omega)}$ which depend on the change of variables map $\varphi\colon \Omega \to \widetilde{\Omega}$, and on the set F. In order to do so, we first need to make sense of Δ^{-1} as a bounded operator on L^p.

6.1.1 The inverse of the Laplacian on L^p

Under certain assumptions on the regularity of Ω, the Laplacian

$$\Delta\colon W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega) \to L^p(\Omega)$$

is invertible; see, for example, Theorem 9.15 in Gilbarg and Trudinger (2001).

Theorem 6.1.2. *Let Ω be a $C^{1,1}$ domain in $\mathbb{R}^2$. Then, given an $f \in L^p(\Omega)$ when $1 < p < \infty$, the Dirichlet problem $\Delta u = f$ in Ω with $u \in W_0^{1,p}(\Omega)$ has a unique solution $u \in W^{2,p}(\Omega)$.*

A similar result also holds for a square, as follows from Theorem 4.4.3.7 in Grisvard (1985) (whose statement we slightly adapt).

Theorem 6.1.3. *Assume that Ω is a polygon with straight boundary edges meeting at right angles. Then, for each $f \in L^p(\Omega)$ there is a unique solution $u \in W^{2,p}(\Omega) \cap W_0^{1,p}(\Omega)$ to the Poisson problem $\Delta u = f$ with zero boundary conditions.*

These are the more 'obvious' examples; for a discussion on more general conditions on $\partial\Omega$ for the $W^{2,p}$-solvability of the Poisson equation with L^p initial data, see Chapter 7 of Maz'ya and Shaposhnikova (1985).

In other words: for squares, rectangles and $C^{1,1}$ domains (among others), the Dirichlet Laplacian is invertible with inverse $\Delta^{-1}\colon L^p \to W^{2,p} \cap W_0^{1,p}$.

For brevity, we will henceforth write X_p for the Banach space $W^{2,p} \cap W_0^{1,p}$ equipped with the $W^{2,p}$ norm.

Definition 6.1.4. Let $\varphi\colon \Omega \to \widetilde{\Omega}$ be a bijection with bounded weak derivatives up to second order. Furthermore, we assume that

$$\inf_{\Omega} |\det D\varphi| > 0.$$

We call the triple $(\Omega, \varphi, \widetilde{\Omega})$ *p-admissible* if φ is as above and $\Delta\colon X_p \to L^p$ is invertible on both Ω and $\widetilde{\Omega}$. We will often simplify this to '$\varphi\colon \Omega \to \widetilde{\Omega}$ is p-admissible' or even 'φ is p-admissible'.

Remark. We require one more order of differentiation than Lamberti and Lanza de Cristoforis (2005), as well as the invertibility of Δ from X_p to L^p, which comes 'for free' on bounded domains in $\mathbb{R}^2$ when working with $\Delta\colon H^1_0 \to H^{-1}$.

Henceforth, we will assume that $(\Omega, \varphi, \widetilde{\Omega})$ is p-admissible.

Definition 6.1.5. Let $Y(\cdot)$ be a function space on Ω and $\widetilde{\Omega}$, and let $\varphi\colon \Omega \to \widetilde{\Omega}$ be a change of coordinates. We define the following operators:

$$\begin{aligned} C_\varphi\colon Y(\widetilde{\Omega}) &\longrightarrow Y(\Omega) \\ v &\longmapsto v \circ \varphi \end{aligned} \qquad \text{and} \qquad \begin{aligned} C_{\varphi^{-1}}\colon Y(\Omega) &\longrightarrow Y(\widetilde{\Omega}) \\ u &\longmapsto u \circ \varphi. \end{aligned}$$

Remark. Note that $(C_\varphi)^{-1} = C_{\varphi^{-1}}$ and vice versa.

We will denote by T the inverse Laplacian

$$T = \Delta^{-1} \equiv (\Delta_\Omega)^{-1}\colon L^p(\Omega) \to X_p(\Omega) \hookrightarrow L^p(\Omega).$$

We will also denote by T_φ the inverse of the pull-back Δ_φ, i.e.

$$T_\varphi = C_\varphi \circ \Delta^{-1}_{\widetilde{\Omega}} \circ C_{\varphi^{-1}}\colon L^p(\Omega) \to L^p(\Omega).$$

Thus, we have the following diagram:

$$\begin{array}{ccccc} L^p(\widetilde{\Omega}) & \xrightarrow{\Delta^{-1}_{\widetilde{\Omega}}} & X_p(\widetilde{\Omega}) & \hookrightarrow & L^p(\widetilde{\Omega}) \\ {\scriptstyle C_{\varphi^{-1}}}\uparrow & & & & \downarrow{\scriptstyle C_\varphi} \\ L^p(\Omega) & \xrightarrow{\Delta^{-1}_{\Omega}} & X_p(\Omega) & \hookrightarrow & L^p(\Omega) \end{array}$$

where T_φ is the 'up–over–down' path from $L^p(\Omega)$ to $L^p(\Omega)$.

We now introduce a slightly modified version of the operator Δ_φ.

Definition 6.1.6. Let $\mathscr{L}_\varphi \colon L^p(\Omega) \to L^p(\Omega)$ be the operator defined by

$$\mathscr{L}_\varphi := C^t_{\varphi^{-1}} \circ \Delta_{\widetilde{\Omega}} \circ C_{\varphi^{-1}},$$

where $C^t_{\varphi^{-1}} \colon L^p(\widetilde{\Omega}) \to L^p(\Omega)$ is the operator defined, for $v \in L^p(\widetilde{\Omega})$, by

$$\left\langle C^t_{\varphi^{-1}} v, w \right\rangle := \left\langle v, C_{\varphi^{-1}} w \right\rangle \equiv \int_{\widetilde{\Omega}} v(y) w(\varphi^{-1}(y)) \, \mathrm{d}y \quad \text{for all} \quad w \in L^q(\Omega).$$

Lemma 6.1.7. *The operator $C^t_{\varphi^{-1}}$ is given by*

$$\left[C^t_{\varphi^{-1}} v\right](x) = |\det D\varphi(x)| \, [C_\varphi v](x) \quad \textit{for all} \quad v \in L^p(\widetilde{\Omega}),$$

with inverse

$$\left[(C^t_{\varphi^{-1}})^{-1} u\right](y) = C^{-1}_\varphi \left(\frac{u(x)}{|\det D\varphi(x)|} \right)(y) \quad \textit{for all} \quad u \in L^p(\Omega).$$

In particular,

$$(C^t_{\varphi^{-1}})^{-1} = C^t_\varphi. \tag{6.1.8}$$

Definition 6.1.8. Define the operator $\mathscr{J}_\varphi \colon L^p(\Omega) \to L^p(\Omega)$ by

$$\mathscr{J}_\varphi u(x) := |\det D\varphi(x)| \, u(x) \quad \text{for all} \quad u \in L^p(\Omega).$$

If $\inf_\Omega |\det D\varphi| > 0$, then

$$\mathscr{J}_\varphi^{-1} u(x) = \frac{u(x)}{|\det D\varphi(x)|}.$$

In particular, we have that

$$C^t_{\varphi^{-1}} = \mathscr{J}_\varphi \circ C_\varphi.$$

Remark. Henceforth, we will use the notational convention $S_\varphi^{-1} \equiv (S_\varphi)^{-1}$ for any symbol S_φ with a φ subscript.

From the above definitions we can therefore conclude the following.

Corollary 6.1.9. *We have $T_\varphi = \mathscr{L}_\varphi^{-1} \circ \mathscr{J}_\varphi$.*

Proof. First, note that $\mathscr{L}_\varphi = C^t_{\varphi^{-1}} \circ \Delta_{\widetilde{\Omega}} \circ C_{\varphi^{-1}}$ implies that

$$\begin{aligned} \mathscr{L}_\varphi^{-1} &= (C_{\varphi^{-1}})^{-1} \circ (\Delta_{\widetilde{\Omega}})^{-1} \circ (C^t_{\varphi^{-1}})^{-1} \\ &= C_\varphi \circ \Delta^{-1}_{\widetilde{\Omega}} \circ C^{-1}_\varphi \circ \mathscr{J}_\varphi^{-1} \end{aligned}$$

$$= \Delta_\varphi^{-1} \circ \mathscr{J}_\varphi^{-1},$$

and so composing with $\mathscr{J}_\varphi$ finishes the proof. □

These rather formal calculations are justified by the admissibility conditions on φ.

Remark. The crucial detail to observe is that we factor the (transformed) inverse Laplacian T_φ into two operators: the inverse 'Laplacian' $\mathscr{L}_\varphi^{-1}\colon L^p(\Omega) \to L^p(\Omega)$ and the embedding $\mathscr{J}_\varphi\colon L^p(\Omega) \to L^p(\Omega)$. This circuitous route, which is inspired by Lamberti and Lanza de Cristoforis (2005), will vastly simplify our proof of Theorem 6.3.5 below. Essentially, we will need to transfer integration by parts from Ω to $\widetilde{\Omega}$. If we were simply to use Δ_φ, we would have

$$\begin{aligned}
\langle \Delta_\varphi u, w\rangle &= \left\langle C_\varphi \Delta_{\widetilde{\Omega}} C_{\varphi^{-1}} u, w\right\rangle \\
&= \int_\Omega (\Delta_{\widetilde{\Omega}}(u\circ\varphi^{-1}))(\varphi(x)) w(x)\,\mathrm{d}x \\
&= \int_{\widetilde{\Omega}} \Delta_{\widetilde{\Omega}}(u\circ\varphi^{-1})(y) w(\varphi^{-1}(y)) \left|\det D\varphi^{-1}(y)\right|\,\mathrm{d}y \\
&= -\int_{\widetilde{\Omega}} D_y(u\circ\varphi^{-1})(y)\cdot D_y\left[(w\circ\varphi^{-1})\left|\det D\varphi^{-1}(y)\right|\right]\,\mathrm{d}y.
\end{aligned}$$

Using the operator $\mathscr{L}_\varphi$ instead will allow us to circumvent differentiating the determinant, which is then reintroduced by $\mathscr{J}_\varphi$ in a more convenient place.

6.2 Continuity and splitting of eigenvalues

In order to apply Kato's projection formula, we must assume that the set F does not split eigenvalues of either operator. In this section, we show that, under reasonable assumptions on φ, it is enough to assume that F does not split eigenvalues of Δ on Ω. For, in this case, continuity results for the eigenvalues with respect to changes of the domain will ensure that F will not split the eigenvalues of $\Delta_{\widetilde{\Omega}}$ on $\widetilde{\Omega}$. The next lemma guarantees that $\Delta_{\widetilde{\Omega}}$ and its pull-back to Ω have the same eigenvalues. Thus, if F does not split eigenvalues of T, then it will not split eigenvalues of T_φ, and we can apply Kato's projection formula. (Recall that $\lambda > 0$ is an eigenvalue of $-\Delta$ if, and only if, $\frac{1}{\lambda} > 0$ is an eigenvalue of $-\Delta^{-1}$.)

Lemma 6.2.1. *Let $\lambda \in \mathbb{R}$ and $v \in L^2(\widetilde{\Omega}) \setminus \{0\}$. Then,*

$$\Delta_{\widetilde{\Omega}}^{-1} v = \lambda^{-1} v \quad \textit{if, and only if,} \quad T_\varphi(C_\varphi v) = \lambda^{-1} C_\varphi v.$$

Proof. Immediate from the definitions; see Theorem 3.10 in Lamberti and Lanza de Cristoforis (2005). □

Lemma 6.2.1 tells us that the eigenvalues of $\Delta_{\widetilde{\Omega}}^{-1}$ remain unchanged after we 'pull them back to Ω'—that is the purpose of T_φ.

We next check that the eigenvalues of Δ^{-1} are close to the eigenvalues of $\Delta_{\widetilde{\Omega}}^{-1}$. This we do via a theorem due to Courant and Hilbert (1989) (see p. 423).

Theorem 6.2.2. *Let $\varphi : \Omega \to \widetilde{\Omega}$ be a bijection with $\inf_{x\in\Omega} |\det D\varphi| > 0$. Then, for any $\eta > 0$ there is an $\varepsilon > 0$ such that if*

$$\|\mathrm{id} - \varphi\|_{W^{1,\infty}(\Omega)} \leqslant \varepsilon,$$

then

$$\left| \frac{\widetilde{\lambda}_j}{\lambda_j} - 1 \right| < \eta \quad \textit{for all} \quad j \in \mathbb{N}.$$

Remark. Strictly speaking, this result applies to $-\Delta$, not $-\Delta^{-1}$. But we have seen that the eigenvalue problems for both operators are equivalent (see Eq. (6.0.1) and the surrounding discussion). The proof of this result uses the variational formulation of the eigenvalue problem. As Lamberti and Lanza de Cristoforis (2005) point out, the 'variational eigenvalues' need not coincide with the true eigenvalues. However, this is the case in bounded domains. (More generally, whenever the embedding $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact; see their discussion on p. 289.)

Lemma 6.2.3. *Let $j, k \in F$ be such that $\lambda_j \neq \lambda_k$. Assume that*

$$\gamma := \inf\{\lambda_l - \lambda_m : l, m \in F \textit{ and } \lambda_l > \lambda_m\} > 0.$$

Then, $\left|\widetilde{\lambda}_j - \widetilde{\lambda}_k\right| > \gamma/2$, provided that $\|\mathrm{id} - \varphi\|_{W^{1,\infty}(\Omega)}$ is small enough.

Proof. Without loss of generality, we may assume that $\lambda_j > \lambda_k$, and set the quantity $\delta := \lambda_j - \lambda_k \geqslant \gamma$. Note that

$$\widetilde{\lambda}_j - \widetilde{\lambda}_k = \lambda_j \cdot \frac{\widetilde{\lambda}_j}{\lambda_j} - \lambda_k \cdot \frac{\widetilde{\lambda}_k}{\lambda_k} = (\lambda_j - \lambda_k) \cdot \frac{\widetilde{\lambda}_j}{\lambda_j} + \lambda_k \left(\frac{\widetilde{\lambda}_j}{\lambda_j} - \frac{\widetilde{\lambda}_k}{\lambda_k} \right).$$

By Theorem 6.2.2,

$$1 - \eta < \frac{\widetilde{\lambda}_j}{\lambda_j} < 1 + \eta, \quad \left| \frac{\widetilde{\lambda}_j}{\lambda_j} - \frac{\widetilde{\lambda}_k}{\lambda_k} \right| < 2\eta \quad \text{and} \quad |\lambda_k| \leqslant \max_{l\in F} |\lambda_l|.$$

Hence,

$$\widetilde{\lambda}_j - \widetilde{\lambda}_k \geqslant (\lambda_j - \lambda_k)(1-\eta) - 2\max_{l\in F}|\lambda_l|\,\eta \geqslant \gamma - \eta(\delta + 2\max_{l\in F}|\lambda_l|),$$

which is larger than $\gamma/2$, provided that $\eta < \frac{\gamma}{\delta+2\max_{l\in F}|\lambda_l|}$. □

Since there are only finitely many indices in F, it follows that, for a fixed F and a sufficiently small perturbation φ, the multiplicities of the eigenvalues of $\Delta_{\widetilde{\Omega}}$ (in F) will not decrease with respect to the multiplicities of the eigenvalues of Δ (in F).

Combining Lemma 6.2.3 with Lemma 6.2.1, we deduce that F does not split eigenvalues for T nor T_φ.

Corollary 6.2.4. *If φ is p-admissible and $\|\mathrm{id} - \varphi\|_{W^{1,\infty}(\Omega)}$ sufficiently small, then F will not split eigenvalues of T nor T_φ.*

6.3 Bounds on $T - T_\varphi$

Our next task is to obtain bounds on the operator norm of $T - T_\varphi$. As a first step, we prove some auxiliary bounds on the operators $C_\varphi\colon L^p(\widetilde{\Omega}) \to L^p(\Omega)$ and $C_\varphi^t\colon L^p(\Omega) \to L^p(\widetilde{\Omega})$.

Lemma 6.3.1. *If φ is p-admissible, then*

$$\|C_\varphi\|_{L^p(\widetilde{\Omega})\to L^p(\Omega)} \leqslant \frac{1}{\inf_\Omega |\det D\varphi|^{1/p}},$$

and, with $1/p + 1/q = 1$,

$$\left\|C_\varphi^t\right\|_{L^p(\Omega)\to L^p(\widetilde{\Omega})} \leqslant \frac{1}{\inf_\Omega |\det D\varphi|^{1/q}}. \tag{6.3.1}$$

Proof. For any $v \in L^p(\widetilde{\Omega})$ we have

$$\begin{aligned}
\|C_\varphi v\|_{L^p(\Omega)}^p &= \int_\Omega |v(\varphi(x))|^p \,\mathrm{d}x \\
&= \int_\Omega |v(\varphi(x))|^p \,\frac{|\det D\varphi|}{|\det D\varphi|}\,\mathrm{d}x \\
&\leqslant \frac{1}{\inf_\Omega |\det D\varphi|} \int_\Omega |v(\varphi(x))|^p \,|\det D\varphi|\,\mathrm{d}x \\
&= \frac{1}{\inf_\Omega |\det D\varphi|} \int_{\widetilde{\Omega}} |v(y)|^p \,\mathrm{d}y,
\end{aligned}$$

by the change of variables formula, and (6.3.1) immediately follows by a simple duality argument. □

Corollary 6.3.2. *If φ is p-admissible, then*

$$\left\|\mathscr{L}_{\varphi}^{-1}\right\|_{L^p(\Omega)\to L^p(\Omega)} \lesssim_{\varphi,p} \frac{1}{\inf_{\Omega}|\det D\varphi|}.$$

Proof. Recall that, by Eq. (6.1.8), we have

$$\mathscr{L}_{\varphi}^{-1} = C_{\varphi}\circ\Delta_{\widetilde{\Omega}}^{-1}\circ C_{\varphi}^{t}.$$

Hence, the result follows from Lemma 6.3.1 and the estimate

$$\left\|\Delta_{\widetilde{\Omega}}^{-1}\right\|_{L^p(\widetilde{\Omega})\to L^p(\widetilde{\Omega})} \leqslant \left\|\Delta_{\widetilde{\Omega}}^{-1}\right\|_{L^p(\widetilde{\Omega})\to X_p(\widetilde{\Omega})} \lesssim_{\varphi,p} 1,$$

by the p-admissibility of $\widetilde{\Omega}$. More precisely, the last implicit constant depends on $\widetilde{\Omega} = \varphi(\Omega)$. □

Next, we need some technical lemmas concerning φ and its derivatives. We will write $\varphi = \mathrm{id} + (f, g)$ to mean

$$\varphi(x, y) = (x + f(x, y), y + g(x, y)).$$

We want to consider φ 'close to the identity' in the sense that $\|\mathrm{id} - \varphi\|_{W^{2,\infty}(\Omega;\mathbb{R}^2)}$ is small. In other words, f, g and their derivatives up to second order have small L^∞ norms.

Lemma 6.3.3. *Suppose that $\|\mathrm{id} - \varphi_k\|_{W^{1,\infty}(\Omega;\mathbb{R}^2)} \to 0$ as $k \to \infty$. Then,*

$$\|1 - \det D\varphi_k\|_{L^\infty(\Omega)} \to 0.$$

Proof. Let

$$D\varphi_k = \begin{pmatrix} a_k & b_k \\ c_k & d_k \end{pmatrix}.$$

Then, by assumption,

$$I - D\varphi_k = \begin{pmatrix} 1 - a_k & b_k \\ c_k & 1 - d_k \end{pmatrix} \to 0 \quad \text{in} \quad L^\infty.$$

Hence,

$$\begin{aligned} 1 - \det(D\varphi_k) &= 1 - a_k d_k + b_k c_k \\ &= (1 - a_k) + a_k(1 - d_k) + b_k c_k \to 0, \end{aligned}$$

as required. □

Remark. Note that we only required $\mathrm{id} - \varphi \in W^{1,\infty}(\Omega;\mathbb{R}^2)$ for this lemma. The next result will require convergence in $W^{2,\infty}$.

Lemma 6.3.4. *Suppose that* $\|\mathrm{id} - \varphi_k\|_{W^{2,\infty}(\Omega;\mathbb{R}^2)} \to 0$ *as* $k \to \infty$ *and write*

$$M_k := D\varphi_k^{-1} D\varphi_k^{-t} \,|\det D\varphi| - I \equiv \begin{pmatrix} a_k & b_k \\ c_k & d_k \end{pmatrix}.$$

Then, as $k \to \infty$,

(i) $\|M_k\|_{L^\infty(\Omega)} \to 0$, *and*

(ii) $\left\| \frac{\partial a_k}{\partial x} + \frac{\partial b_k}{\partial y} \right\|_{L^\infty(\Omega)}, \left\| \frac{\partial c_k}{\partial x} + \frac{\partial d_k}{\partial y} \right\|_{L^\infty(\Omega)} \to 0.$

Proof. Recall that $\varphi_k = \mathrm{id} + (f_k, g_k)$. For simplicity, we will drop the k subscripts on φ, f and g, writing $f_x := \frac{\partial f_k}{\partial x}$, and so on. Thus, e.g., '$f_x \to 0$' will mean '$\frac{\partial f_k}{\partial x} \to 0$ as $k \to \infty$'. Note that our assumption means precisely that f_x, f_y, g_x, g_y tend to 0 in k, as well as f, g and all second partial derivatives.

By Lemma 6.3.3, eventually $\det D\varphi > 0$ a.e., so we may drop the absolute value signs. A simple calculation shows that

$$D\varphi^{-1} D\varphi^{-t} = \frac{1}{(\det D\varphi)^2} \begin{pmatrix} g_x^2 + (1+g_y)^2 & -(f_y + g_x + f_x g_y + f_y g_x) \\ -(f_y + g_x + f_x g_y + f_y g_x) & (1+f_x)^2 + f_y^2 \end{pmatrix}$$

and that

$$\det D\varphi = 1 + f_x + g_y + f_x g_y + f_y g_x.$$

Hence, for example, the top-left entry of M_k is

$$a_k = \frac{g_x^2 + g_y^2 - g_y - f_x - f_x g_y - f_y g_x}{\det D\varphi}.$$

By Lemma 6.3.3, the denominator remains bounded, while the numerator vanishes as $k \to \infty$ by our assumption. Similar calculations for b_k, c_k and d_k establish (i).

For (ii), we differentiate the last display with respect to x to find

$$\frac{\partial a_k}{\partial x} = \frac{p(Df, Dg, D^2 f, D^2 g)}{(\det D\varphi)^2},$$

where p is a homogeneous polynomial in the first and second derivatives of f and g. Thus, as above, $\partial a_k/\partial x \to 0$. We omit the remaining cases. □

With these estimates, we can now control $T - T_\varphi$, which is a key estimate needed for our main result.

Theorem 6.3.5. *If $\varphi\colon \Omega \to \widetilde{\Omega}$ is p-admissible, then*

$$\|T - T_\varphi\|_{L^p(\Omega)\to L^p(\Omega)} \lesssim_{\varphi,p} \frac{1}{\inf_\Omega |\det D\varphi|} \left(\|1 - |\det D\varphi|\|_{L^\infty(\Omega)} + K_\varphi \right),$$

where K_φ is a constant depending on $D\varphi$ and its derivatives that goes to zero when $\|\mathrm{id} - \varphi\|_{W^{2,\infty}(\Omega)} \to 0$.

Proof. Fix $u \in L^p(\Omega)$. Recall that $T_\varphi = \mathscr{L}_\varphi^{-1} \circ \mathscr{J}_\varphi$. Hence, it follows from Corollary 6.3.2 that

$$\begin{aligned} \|Tu - T_\varphi u\|_{L^p(\Omega)} &\leqslant \left\|\mathscr{L}_\varphi^{-1}\right\|_{L^p(\Omega)\to L^p(\Omega)} \left\|\mathscr{J}_\varphi u - \mathscr{L}_\varphi \Delta^{-1} u\right\|_{L^p(\Omega)} \\ &\lesssim_{\varphi,p} \frac{1}{\inf_\Omega |\det D\varphi|} \left\|\mathscr{J}_\varphi u - \mathscr{L}_\varphi \Delta^{-1} u\right\|_{L^p(\Omega)}. \end{aligned} \tag{6.3.2}$$

To bound the last L^p norm, we use duality. For convenience, let us write

$$Au := \mathscr{J}_\varphi u - \mathscr{L}_\varphi \Delta^{-1} u,$$

and compute, for $w \in C_c^\infty(\Omega)$ with $\|w\|_{L^q(\Omega)} \leqslant 1$

$$\begin{aligned} \langle Au, w\rangle &= \langle \mathscr{J}_\varphi u, w\rangle - \left\langle \mathscr{L}_\varphi \Delta^{-1} u, w\right\rangle \\ &= \int_\Omega uw\, |\det D\varphi| \, \mathrm{d}x - \left\langle \mathscr{L}_\varphi \Delta^{-1} u, w\right\rangle. \end{aligned} \tag{6.3.3}$$

We now unravel the definition of $\mathscr{L}_\varphi$, integrate by parts and change variables:

$$\begin{aligned} \left\langle \mathscr{L}_\varphi [\Delta^{-1} u], w\right\rangle &= \left\langle C^t_{\varphi^{-1}} \Delta_{\widetilde{\Omega}} C^{-1}_\varphi [\Delta^{-1} u], w\right\rangle \\ &= \left\langle \Delta_{\widetilde{\Omega}} C_{\varphi^{-1}} [\Delta^{-1} u], C_{\varphi^{-1}} w\right\rangle \\ &= -\int_{\widetilde{\Omega}} D_y \left[\Delta^{-1} u(\varphi^{-1}(y))\right] \cdot D_y \left[w(\varphi^{-1}(y))\right] \, \mathrm{d}y \end{aligned}$$

$$= -\int_{\widetilde{\Omega}} D_x(\Delta^{-1}u)(\varphi^{-1}(y))D\varphi^{-1}(y)D\varphi^{-t}(y)(D_x w)^t(\varphi^{-1}(y))\,\mathrm{d}y$$
$$= -\int_{\Omega} D(\Delta^{-1}u)D\varphi^{-1}(D\varphi^{-1})^t(Dw)^t\,|\det D\varphi|\,\mathrm{d}x,$$

where for a matrix M, $M^{-t} = (M^{-1})^t$, and $Dv \cdot Dw = Dv(Dw)^t$. We will now add and subtract

$$\pm\int_{\Omega} D(\Delta^{-1}u)\cdot Dw\,\mathrm{d}x$$

from (6.3.3) to obtain

$$\langle Au, w\rangle = \int_{\Omega} uw\,|\det D\varphi|\,\mathrm{d}x + \int_{\Omega} D(\Delta^{-1}u)\cdot Dw\,\mathrm{d}x \tag{6.3.4}$$

$$\int_{\Omega} D(\Delta^{-1}u)D\varphi^{-1}D\varphi^{-t}(Dw)^t\,|\det D\varphi|\,\mathrm{d}x - \int_{\Omega} D(\Delta^{-1}u)\cdot Dw\,\mathrm{d}x. \tag{6.3.5}$$

To estimate (6.3.4), we integrate by parts in the second summand to see that

$$\begin{aligned}(6.3.4) &= \int_{\Omega} uw\,|\det D\varphi|\,\mathrm{d}x - \int_{\Omega}\Delta(\Delta^{-1}u)w\,\mathrm{d}x\\ &= \int_{\Omega} uw(|\det D\varphi| - 1)\,\mathrm{d}x\\ &\leqslant \|1 - |\det D\varphi|\|_{L^\infty(\Omega)}\,\|u\|_{L^p(\Omega)}\,\|w\|_{L^q(\Omega)}\,.\end{aligned} \tag{6.3.6}$$

For the second term, note that

$$(6.3.5) = \int_{\Omega} D(\Delta^{-1}u)\left\{D\varphi^{-1}(D\varphi^{-1})^t\,|\det D\varphi| - I\right\}\cdot Dw\,\mathrm{d}x,$$

Writing the expression in braces as

$$M_\varphi = \begin{pmatrix} a & b\\ c & d\end{pmatrix}$$

and integrating by parts yields

$$(6.3.5) = -\int_{\Omega}\left[(a_x + b_y, c_x + d_y)\cdot D(\Delta^{-1}u) + M_\varphi\cdot D^2(\Delta^{-1}u)\right]w\,\mathrm{d}x,$$

where $A\cdot B$ is the dot product of the matrices A and B, seen as vectors in $\mathbb{R}^4$. Then,

$$(6.3.5) \lesssim K_\varphi\left(\left\|D^2(\Delta^{-1}u)\right\|_{L^p(\Omega)} + \left\|D(\Delta^{-1}u)\right\|_{L^p(\Omega)}\right)\|w\|_{L^q(\Omega)}$$

$$\begin{aligned} &\leqslant K_\varphi \left\|\Delta^{-1} u\right\|_{W^{2,p}(\Omega)} \|w\|_{L^q(\Omega)} \\ &\lesssim K_\varphi \|u\|_{L^p(\Omega)} \|w\|_{L^q(\Omega)} . \end{aligned} \tag{6.3.7}$$

The last line follows from the boundedness of $\Delta^{-1}\colon L^p(\Omega) \to X_p(\Omega)$, while K_φ is a constant depending on $D\varphi$ and its derivatives that vanishes in the required manner by Lemma 6.3.4.

Hence, combining the estimates (6.3.6) and (6.3.7) with (6.3.3) and (6.3.2) yields the result. □

Remark. In order to get a stronger continuity result, we must add the additional assumption that

$$\left\|\Delta^{-1}_{\varphi_k(\Omega)}\right\|_{L^p(\varphi_k(\Omega)) \to L^p(\varphi_k(\Omega))}$$

remain uniformly bounded in k. Then, we can conclude that

$$\|T - T_{\varphi_k}\|_{L^p(\Omega)\to L^p(\Omega)} \to 0 \quad \text{as} \quad \|\mathrm{id} - \varphi_k\|_{W^{2,\infty}(\Omega;\mathbb{R}^2)} \to 0.$$

We are working on a proof of this assumption in the case where the domains $\varphi_k(\Omega)$ have $C^{1,1}$ boundaries.

Remark. If $\varphi\colon \Omega \to \widetilde{\Omega}$ is conformal, i.e. angle-preserving, then we can regard it as a holomorphic map from $\mathbb{C}$ to $\mathbb{C}$ whose complex derivative $\varphi'(x+\mathrm{i}y)$ does not vanish at any $(x, y) \in \Omega$. In such a case we have

$$\det D\varphi = \left|\varphi'\right|^2 \quad \text{and} \quad \frac{1}{|\det D\varphi|} D\varphi D\varphi^t = I,$$

whence it follows that

$$D\varphi^{-1} D\varphi^{-t} |\det D\varphi| = I,$$

and so Eq. (6.3.3) simplifies to

$$\begin{aligned} \langle Au, w\rangle &= \int_\Omega uw\, |\det D\varphi| \, \mathrm{d}x + \int_\Omega D[\Delta^{-1}u] \cdot Dw \, \mathrm{d}x \\ &= \int_\Omega uw \left(|\det D\varphi| - 1\right) \mathrm{d}x. \end{aligned}$$

Corollary 6.3.6. *Suppose, in addition, that φ is conformal. Then,*

$$\|T - T_\varphi\|_{L^p(\Omega)\to L^p(\Omega)} \lesssim_{p,\varphi} \frac{1}{\inf_\Omega |\det D\varphi|} \left\|1 - |\det D\varphi|\right\|_{L^\infty(\Omega)} .$$

6.4 Resolvent bounds

We will also need L^p bounds on the resolvents

$$(T-z)^{-1} \equiv (-\Delta^{-1}-z)^{-1} \colon L^p(\Omega) \to X_p(\Omega) \hookrightarrow L^p(\Omega)$$

when z lies on a contour in the complex plane, as in Eq. (6.1.5).

In Lemma 6.4.4, we will see that the spectrum of Δ (and so of Δ^{-1}) does not depend on p (Arendt, 1994). To make our presentation clearer, we will temporarily attach a p to the Laplacian to distinguish

$$\Delta_p^{-1} \colon L^p(\Omega) \to L^p(\Omega) \quad \text{from} \quad \Delta_2^{-1} \colon L^2(\Omega) \to L^2(\Omega).$$

We know what the spectrum of Δ_2, and hence of Δ_2^{-1}, looks like, and we will need the result that

$$\sigma(-\Delta_p^{-1}) = \sigma(-\Delta_2^{-1}).$$

The operators Δ_p^{-1} are generators for a family of uniformly (hence strongly) continuous semigroups $(T_p(t))_{t\geqslant 0}$ defined as

$$T_p(t) := \mathrm{e}^{t\Delta_p^{-1}} \colon L^p(\Omega) \to L^p(\Omega).$$

Proposition 6.4.1. *Let $(T(t))_{t\geqslant 0}$ be C_0 semigroup. Then there are constants $M \geqslant 1$ and $\omega \in \mathbb{R}$ such that*

$$\|T(t)\| \leqslant M\mathrm{e}^{\omega t} \quad \textit{for all} \quad t \geqslant 0.$$

Proof. See Engel et al. (2006, Prop. I.5.5). □

Hence, the following value is well-defined:

$$\omega_0(A) := \inf\left\{\omega \in \mathbb{R} : \exists M_\omega \geqslant 1 \forall t \geqslant 0 (\|T(t)\| \leqslant M_\omega \mathrm{e}^{\omega t})\right\} < \infty.$$

Here A is the generator of the semigroup. In particular, when A is bounded, we have the following.

Proposition 6.4.2. *If A is a bounded operator, then*

$$\omega_0(A) = \sup\{\mathrm{Re}(\lambda) : \lambda \in \sigma(A)\}.$$

Proof. See Engel et al. (2006, Cor. IV.2.4). □

This gives us a useful way to calculate the optimal exponent $\omega_0 = \omega_0(A)$, which is handy for obtaining the following bounds.

Theorem 6.4.3. *If A is a bounded operator, then*

$$\left\|(A-\lambda)^{-1}\right\| \lesssim \frac{1}{\operatorname{Re}(\lambda)-\omega_0} \quad \textit{for all} \quad \operatorname{Re}(\lambda) > \omega_0.$$

Proof. Follows from Theorem II.1.10, (iii) in Engel et al. (2006). □

Let $A_p = -\Delta_p^{-1}\colon L^p(\Omega) \to L^p(\Omega)$. Our first task is to compute the spectrum of A_p. Recall $\Delta_p^{-1}\colon L^p(\Omega) \to X_p(\Omega)$ is bounded. The next lemma tells us that the eigenvalues of A_p are precisely those of $A_2\colon L^2(\Omega) \to L^2(\Omega)$.

Lemma 6.4.4. *The spectrum of $A_p = -\Delta_p^{-1}$ is*

$$\sigma(A_p) = \left\{ \frac{1}{\lambda_k} : \lambda_k \in \sigma(-\Delta_2) \right\} \cup \{0\}. \tag{6.4.1}$$

Proof. It is easy to see that the (non-zero) eigenvalues of A_p are precisely the inverses of $-\Delta$. Given the boundedness of $A_p\colon L^p(\Omega) \to X_p(\Omega)$ and the compact inclusion

$$X_p(\Omega) \subset W_0^{1,p}(\Omega) \Subset L^p(\Omega),$$

the operator $A_p\colon L^p(\Omega) \to L^p(\Omega)$ is compact. Thus, by standard results (e.g. Robinson, 2020, Corollary 24.5), the spectrum of A_p consists precisely of eigenvalues of A_p and 0. This gives us half of (6.4.1).

For the converse inclusion, we note that eigenfunctions of $-\Delta_2$ are bounded, and so lie in all L^p spaces. (See, e.g, Gilbarg and Trudinger, 2001, Theorem 8.15.) □

For convenience, we will write μ_k for $1/\lambda_k$, and set $\mu_\infty := 0$, so that

$$\sigma(A_p) = \{\mu_k : 1 \leqslant k \leqslant \infty\} \quad \text{for all} \quad 1 < p < \infty.$$

Note, however, that $0 = \mu_\infty$ is *not* an eigenvalue of A_p.

Now pick any $N \in \mathbb{N}$ and decompose the spectrum as follows:

$$\sigma(A_p) = \{\mu_k : k \leqslant N\} \cup \{\mu_k : N < k \leqslant \infty\} = \sigma_{\leqslant N} \cup \sigma_{>N}.$$

Note that $\sigma_{\leqslant N}$ is compact and that $0 \in \sigma_{>N}$. Hence, from the functional calculus we have a projection

$$P := \frac{1}{2\pi \mathrm{i}} \int_{\gamma_N} (A_p - z)^{-1}\, \mathrm{d}z$$

where $\gamma_N \subset \mathbb{C}$ is a contour encircling $\sigma_{\leqslant N}$. We can therefore decompose

$$L^p(\Omega) = \mathrm{ran} P \oplus \ker P.$$

Furthermore, the operator A_p also decomposes into $A_p = A_{\leqslant N} \oplus A_{>N}$ where

$$A_{\leqslant N} := A_p|_{\mathrm{ran} P} \quad \text{and} \quad A_{>N} := A_p|_{\ker P},$$

with

$$\sigma(A_{\leqslant N}) = \sigma_{\leqslant N} \quad \text{and} \quad \sigma(A_{>N}) = \sigma_{>N}.$$

In fact, we also have

$$(A_{\leqslant N} - z)^{-1} = (A - z)^{-1}|_{\mathrm{ran} P} \quad \text{and} \quad (A_{>N} - z)^{-1} = (A - z)^{-1}|_{\ker P},$$

and therefore

$$(A_p - z)^{-1} = (A_{\leqslant N} - z)^{-1} \oplus (A_{>N} - z)^{-1}. \tag{6.4.2}$$

Hence we obtain the following bound.

Corollary 6.4.5. *With A_p and N as above, let $z \in \mathbb{C}$ with $\mathrm{Re}(z) > \mu_{N+1}$. Then,*

$$\left\|(A_p - z)^{-1}\right\|_{L^p \to L^p} \lesssim_N \frac{1}{|z - \mu_N|} + \frac{1}{|z - \mu_{N+1}|}.$$

Proof. We estimate each piece in Eq. (6.4.2) separately. First, since $(A_{\leqslant N} - z)^{-1}$ is a finite rank operator, we can express it as

$$(A_{\leqslant N} - z)^{-1} u = \sum_{k \leqslant N} \frac{\langle u, u_k \rangle}{\mu_k - z} u_k$$

on the dense subset $u \in L^p \cap L^2$. Hence,

$$\left\|(A_{\leqslant N} - z)^{-1}\right\|_{L^p \to L^p} = \left\| \sum_{k \leqslant N} \frac{\langle \cdot, u_k \rangle}{\mu_k - z} u_k \right\|_{L^p \to L^p} \lesssim_N \frac{1}{|z - \mu_N|}.$$

For the second part, we apply Theorem 6.4.3 with $\omega_0(A_{>N}) = \mu_{N+1} < \mathrm{Re}(z)$ (Proposition 6.4.2) to obtain

$$\left\|(A_{>N} - z)^{-1}\right\|_{L^p \to L^p} \lesssim_N \frac{1}{|z - \mu_{N+1}|}. \qquad \square$$

6.5 Continuity of projections under domain perturbations

We will now combine the estimates for $T - T_\varphi$ with Kato's integral formula to obtain an L^p continuity theorem for the projections as $\mathrm{id} - \varphi \to 0$ in $W^{2,\infty}$, provided one can establish L^p bounds on the resolvents on $T = \Delta^{-1}$. In the next section we will show that this assumption is satisfied for any Ω for which $\Delta\colon X_p(\Omega) \to L^p(\Omega)$ has a bounded inverse.

Before we state the theorem, we note that the operator P_F is simultaneously the projection operator for Δ and its inverse T, as can be seen from its definition in Eq. (6.1.6). And the same is true for P_F^φ with respecto to Δ_φ and T_φ, as can be seen from Eq. (6.1.7).

Theorem 6.5.1. *Suppose that $\varphi\colon \Omega \to \widetilde{\Omega}$ is p-admissible, and F does not split eigenvalues of T or T_φ. Then,*

$$\left\|P_F - P_F^\varphi\right\|_{L^p(\Omega)\to L^p(\Omega)} \lesssim C(F,\Omega,\varphi)\left(\frac{\|1-|\det D\varphi|\|_{L^\infty(\Omega)} + K_\varphi}{\inf_\Omega |\det D\varphi|}\right), \tag{6.5.1}$$

where K_φ is the constant from Lemma 6.3.4 and the implicit constants depend only on p.

Furthermore, if φ_k is a sequence of p-admissible transformations, such that $\|\mathrm{id} - \varphi_k\|_{W^{2,\infty}(\Omega)} \to 0$, then $\left\|P_F - P_F^{\varphi_k}\right\|_{L^p(\Omega)\to L^p(\Omega)} \to 0$ for each fixed F that does not split eigenvalues of Δ.

Proof. Under the assumptions on Ω and φ, T and T_φ are well-defined compact operators on $L^2(\Omega)$. Label their eigenvalues as $(\mu_j)_{j\in\mathbb{N}}$ and $(\widetilde{\mu}_j)_j$, respectively, and recall that $\mu_j = 1/\lambda_j$, etc.

For any $u \in C_c^\infty(\Omega)$, we have the $L^2(\Omega)$ expansions

$$Tu = \sum_{j\in\mathbb{N}} \mu_j \langle u_j, u\rangle_\Omega\, u_j \quad \text{and} \quad T_\varphi = \sum_{j\in\mathbb{N}} \widetilde{\mu}_j Q_\varphi[u_j^\varphi, u] u_j^\varphi.$$

For any $z \in \mathbb{C}$ distinct from any μ_j and $\widetilde{\mu}_j$, we have (still in L^2)

$$(T-z)^{-1}u = \sum_{j\in\mathbb{N}} \frac{\langle u_j, u\rangle_\Omega}{\mu_j - z} u_j \quad \text{and} \quad (T_\varphi - z)^{-1} = \sum_{j\in\mathbb{N}} \frac{Q_\varphi[u_j^\varphi, u]}{\widetilde{\mu}_j - z} u_j^\varphi.$$

Now select a contour γ in the complex plane enclosing precisely the μ_j and $\widetilde{\mu_j}$ with $j \in F$ and no others. This is possible by Theorem 6.2.2.

By Kato's projection formula,

$$P_F u - P_F^\varphi u = \frac{1}{2\pi \mathrm{i}} \int_\gamma \left\{ (T-z)^{-1} - (T_\varphi - z)^{-1} \right\} u \, \mathrm{d}z$$
$$= \frac{1}{2\pi \mathrm{i}} \int_\gamma (T_\varphi - z)^{-1} (T_\varphi - T)(T-z)^{-1} u \, \mathrm{d}z. \tag{6.5.2}$$

We want to replace the first resolvent, $(T_\varphi - z)^{-1}$, with a truncated version:

$$R_F^\varphi(z) f := \sum_{j \in F} \frac{Q_\varphi[u_j^\varphi, u]}{\widetilde{\mu}_j - z} u_j^\varphi.$$

Indeed, we claim that

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma (T_\varphi - z)^{-1} f \, \mathrm{d}z = \frac{1}{2\pi \mathrm{i}} \int_\gamma R_F^\varphi(z) f \, \mathrm{d}z \quad \text{for all} \quad f \in L^2(\Omega). \tag{6.5.3}$$

It suffices to check that (Conway, 2007, VII,§4)

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma Q_\varphi[(T_\varphi - z)^{-1} f, g] \, \mathrm{d}z = \frac{1}{2\pi \mathrm{i}} \int_\gamma Q_\varphi[R_F^\varphi(z) f, g] \, \mathrm{d}z$$

for all f and g in $L^2(\Omega)$.

For $f \in L^2(\Omega)$, we can write $f = \sum_j Q_\varphi[u_j^\varphi, f] u_j^\varphi$ and apply the bounded operator $(T_\varphi - z)^{-1}$:

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma Q_\varphi[(T_\varphi - z)^{-1} f, g] \, \mathrm{d}z = \frac{1}{2\pi \mathrm{i}} \int_\gamma \sum_j \frac{Q_\varphi[(T_\varphi - z)^{-1} f, g]}{\widetilde{\mu}_j - z} Q_\varphi[u_j^\varphi, g] \, \mathrm{d}z$$
$$= \sum_j Q_\varphi[u_j^\varphi, f] Q_\varphi[u_j^\varphi, g] \left(\frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{1}{\widetilde{\mu}_j - z} \, \mathrm{d}z \right).$$

By the classic Cauchy integral formula, the integral in brackets is 1 if $j \in F$ and 0 otherwise. Hence, the last expression is equal to

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma Q_\varphi \left[\sum_{j \in F} \frac{Q_\varphi[u_j^\varphi, u_j]}{\widetilde{\mu}_j - z} u_j^\varphi, g \right] \mathrm{d}z,$$

and the claim follows.

The next step is to obtain L^p bounds on the operator norm of $R_F^\varphi(z)$. Taking

$f \in C_c^\infty$, which is dense in both L^2 and L^p, we have

$$\left\|R_F^\varphi(z) f\right\|_{L^p(\Omega)} \leqslant \sum_{j \in F} \frac{\left|Q_\varphi[u_j^\varphi, f]\right|}{|\widetilde{\mu}_j - z|} \left\|u_j^\varphi\right\|_{L^p(\Omega)} .$$

The change of variables formula immediately shows that (for $1/p + 1/q = 1$)

$$\left\|u_j^\varphi\right\|_{L^p(\Omega)} \leqslant \frac{\|\widetilde{u}_j\|_{L^p(\widetilde{\Omega})}}{\inf_\Omega |\det D\varphi|^{1/p}}$$

and

$$\left|Q_\varphi[u_j^\varphi, f]\right| \leqslant \frac{\|\det D\varphi\|_{L^\infty(\Omega)} \|\widetilde{u}_j\|_{L^q(\widetilde{\Omega})}}{\inf_\Omega |\det D\varphi|^{1/q}} \|f\|_{L^p(\Omega)} .$$

Hence, it follows that

$$\left\|R_F^\varphi(z)\right\|_{L^p(\Omega) \to L^p(\Omega)} \leqslant \frac{1}{\min_F |\widetilde{\mu}_j - z|} \frac{\|\det D\varphi\|_{L^\infty(\Omega)}}{\inf_\Omega |\det D\varphi|} \sum_{j \in F} \|\widetilde{u}_j\|_{L^p(\widetilde{\Omega})} \|\widetilde{u}_j\|_{L^q(\widetilde{\Omega})} .$$

Let us write

$$C(F, \varphi) := \frac{\|\det D\varphi\|_{L^\infty(\Omega)}}{\inf_\Omega |\det D\varphi|} \sum_{j \in F} \|\widetilde{u}_j\|_{L^p(\widetilde{\Omega})} \|\widetilde{u}_j\|_{L^q(\widetilde{\Omega})} . \tag{6.5.4}$$

Then, applying (6.5.3) to (6.5.2), and recalling the bound

$$\left\|(T - z)^{-1}\right\|_{L^p \to L^p} \leqslant C(T, z)$$

for all $z = \gamma(t)$ (Corollary 6.4.5), we obtain

$$\left\|P_F u - P_F^\varphi u\right\|_{L^p(\Omega)} \lesssim C(F, \varphi) \int_\gamma \frac{C(T, z)}{\min_F |\widetilde{\mu}_j - z|} \, \mathrm{d}z \, \|T - T_\varphi\|_{L^p(\Omega) \to L^p(\Omega)} \|u\|_{L^p(\Omega)} .$$

We can package the first two terms into a constant $C(F, T, \varphi)$. On the other hand, Theorem 6.3.5 gave us the estimate

$$\|T - T_\varphi\|_{L^p(\Omega) \to L^p(\Omega)} \lesssim \frac{1}{\inf_\Omega |\det D\varphi|} \left(\left\| 1 - |\det D\varphi| \right\|_{L^\infty(\Omega)} + K_\varphi \right),$$

and Eq. (6.5.1) immediately follows.

The continuity statement is an immediate consequence of Lemma 6.3.3 and Lemma 6.3.4. □

Corollary 6.3.6 gives us the following simplification when φ is conformal.

Corollary 6.5.2. *Suppose, in addition, that φ is conformal. Then,*

$$\left\|P_F - P_F^{\varphi}\right\|_{L^p(\Omega)\to L^p(\Omega)} \lesssim C(F)\left\|1 - |\det D\varphi|\right\|_{L^\infty(\Omega)} \times E_\varphi,$$

where E_φ remains bounded as $\mathrm{id} - \varphi \to 0$ in $W^{2,\infty}$.

Remark. When $p = 2$, the term

$$\sum_{j\in F} \|u_j\|_{L^p} \|u_j\|_{L^q}$$

appearing in Eq. (6.5.4) can be improved to 1 (see Lamberti and Lanza de Cristoforis, 2005, eq. (13)). Indeed, for any compact operator T,

$$\left\|(T-z)^{-1} f\right\|_{L^2}^2 = \sum_j \frac{|\langle u_j, f\rangle|^2}{|\mu_j - z|^2} \leqslant \frac{1}{\min|\mu_j - z|^2}\|f\|_{L^2}^2 .$$

In their paper, Lamberti and Lanza de Cristoforis (2005) claim to obtain this bound from the following inequality (Taylor and Lay, 1986, VI.3, Theorem 3.1):

$$\left\|(T-z)^{-1}\right\| \leqslant \frac{1}{d(z, V(T))},$$

where $d(z, V(T))$ is the distance of $z \in \gamma$ to the *numerical range* of the operator T,

$$V(T) := \{(Tx, x) : \|x\| = 1\}.$$

However, in a Hilbert Space (Halmos, 1982, p. 116), $V(T)$ is the closed convex hull of the spectrum of T. Since T is compact, 0 lies in the closure of the spectrum, and so the contour γ as described will necessarily intersect the numerical range when it crosses the real axis. Our alternative calculation has the advantage of generalising to L^p for the *truncated* operators $R_F^{\varphi}(z)$, at the cost of increasing the size of the constants.

Under suitable assumptions on φ, one can use P_F^{φ} to transfer bounds on $P_F\colon L^p(\Omega) \to L^p(\Omega)$ to bounds on $\widetilde{P}_F\colon L^p(\widetilde{\Omega}) \to L^p(\widetilde{\Omega})$. This 'transference' result is encapsulated in the following lemma.

Lemma 6.5.3 (Transference from Ω to $\widetilde{\Omega}$)**.** *Let $\varphi\colon \Omega \to \widetilde{\Omega}$ be (weakly) differentiable,*

invertible and satisfy the bounds

$$0 < \inf_{\Omega} |\det D\varphi| \leqslant \sup_{\Omega} |\det D\varphi| < \infty.$$

Suppose that we have bounds of the form

$$\|P_F\|_{L^p(\Omega)\to L^p(\Omega)} \leqslant A$$

and

$$\left\|P_F - P_F^{\varphi}\right\|_{L^p(\Omega)\to L^p(\Omega)} \leqslant B.$$

Then, we have the bound

$$\left\|\widetilde{P}_F\right\|_{L^p(\widetilde{\Omega})\to L^p(\widetilde{\Omega})} \leqslant \left(\frac{\sup|\det D\varphi|}{\inf|\det D\varphi|}\right)^{1/p}(A+B).$$

Proof. Fix $f \in L^p(\Omega)$ and let $g := f\circ\varphi^{-1} \in L^p(\widetilde{\Omega})$. Then, by a change of variables in the integral defining the inner product on $\widetilde{\Omega}$, we have

$$\widetilde{P}_F g = \sum_{j\in F}\left(\int_{\Omega} \widetilde{u}_j(\varphi(x)) f(x)\,|\det D\varphi(x)|\,\mathrm{d}x\right)\widetilde{u}_j = \sum_{j\in F} Q_{\varphi}[\widetilde{u}_j\circ\varphi, f]\widetilde{u}_j.$$

Taking $L^p(\widetilde{\Omega})$ norms and changing variables again:

$$\begin{aligned}
\left\|\widetilde{P}_F f\right\|^p_{L^p(\widetilde{\Omega})} &= \int_{\widetilde{\Omega}} \left|\sum_{j\in F} Q_{\varphi}[\widetilde{u}_j\circ\varphi, f]\widetilde{u}_j(y)\right|^p \mathrm{d}y \\
&= \int_{\Omega} \left|\sum_{j\in F} Q_{\varphi}[\widetilde{u}_j\circ\varphi, f]\widetilde{u}_j(\varphi(x))\right|^p |\det D\varphi(x)|\,\mathrm{d}x \\
&\leqslant \|\det D\varphi(x)\|_{L^\infty(\Omega)} \left\|\sum_{j\in F} Q_{\varphi}[u_j^{\varphi}, f]u_j^{\varphi}\right\|^p_{L^p(\Omega)},
\end{aligned}$$

whence we arrive at

$$\left\|\widetilde{P}_F g\right\|_{L^p(\widetilde{\Omega})} \leqslant \|\det D\varphi(x)\|^{1/p}_{L^\infty(\Omega)} \left\|P_F^{\varphi} f\right\|_{L^p(\Omega)}.$$

The result now follows by applying the triangle inequality to $P_F^{\varphi} f = P_F f + P_F^{\varphi} f - P_F f$ and noticing that

$$\|f\|^p_{L^p(\Omega)} \leqslant \sup_{\widetilde{\Omega}} \left|\det D\varphi^{-1}\right| \|g\|^p_{L^p(\widetilde{\Omega})}. \qquad \square$$

6.6 Conclusions and open problems

Our approach has the following important limitation: the projections P_F map onto eigenspaces which do not split eigenvalues. Eigenvalue splitting is not allowed because of Kato's projection formula, as discussed in the remark immediately after Theorem 6.1.1.

The following example motivates our interest in relaxing the notion of splitting.

Example 6.6.1. Let $\Omega = [0,1]^2$, a unit square; then the eigenvalues of the Dirichlet Laplacian are $\pi^2(m^2+n^2)$ for (m,n) running over $\mathbb{N}^2$. Fix any $N \in \mathbb{N}$. The following results are classic (see Grafakos, 2014b):

(i) If we take $F_N = \{(m,n) : 0 < m,n \leqslant N\}$, then P_{F_N} is the partial sum operator corresponding to 'cubic cutoff' truncations of a double sine-series on the torus. For any value of $1 < p < \infty$, these projections are uniformly bounded in N, which is equivalent to saying that the partial sums $P_{F_N}u$ converge to u for any $u \in L^p$.

(ii) If we take $B_N = \{(m,n) : m^2 + n^2 \leqslant N^2\}$, then P_{B_N} is the partial sum operator corresponding to 'spherical cutoff' truncations of a double sine-series on the torus. These partial sums do not converge in any L^p space other than $p = 2$.

It would be interesting to transfer the convergence results for (i) to other domains obtained by perturbing the square by some suitable φ, as described here. However, by definition, the sets B_N will not split eigenvalues, while the sets F_N will often split eigenvalues. Take, for instance, the pairs $(5,5)$ and $(1,7)$, both corresponding to the eigenvalue $50\pi^2$. Then $(5,5) \in F_6$ but $(1,7) \notin F_6$.

Denote this 'bad region' by Γ_N, i.e.

$$\Gamma_N = \{(m,n) \in F_N : m^2 + n^2 = k^2 + l^2 \text{ for some } (k,l) \notin F_N\}.$$

Then F_N splits eigenvalues if, and only, if $\Gamma_N \neq \varnothing$ (see Fig. 6.1).

It is rather surprising that this 'small' region Γ_N lying between the circle and the square is responsible for such dramatic differences in the convergence of the partial sums. Indeed, Fefferman's proof shows that curvature is a key part of the failure of L^p convergence.

Thus, two open questions of special interest remain:

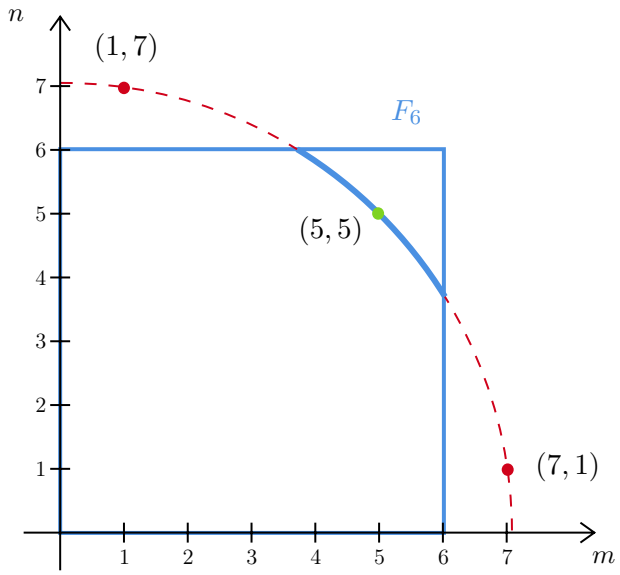


Figure 6.1: The 'box' F_6 splits the eigenvalue $\lambda = 50\pi^2$. The region Γ_6 is denoted by the solid blue arc.

(i) Can we eliminate the 'no-splitting' requirement in order to apply our results to the square cutoffs?

(ii) Can we improve the constant $C(F)$ in order to get uniform estimates on the operator norms?

Regarding (ii), we observe that in some simple examples, there is *no* constant $C(F)$, and so we can immediately recover L^p convergence on $\widetilde{\Omega}$ from L^p convergence on Ω.

Example 6.6.2. Let $\Omega = [0, \pi]^2$ with eigenfunctions $u_{m,n} = \sin(mx)\sin(ny)$, and $\varphi(x, y) = (ax, by)$ mapping onto $\widetilde{\Omega} = [0, \pi a] \times [0, \pi b]$. The eigenfunctions on the new domain are $\widetilde{u}_{m,n} = \sin(mx/a)\sin(ny/b)$. A simple calculation then shows that

$$\left\| P_F u - P_F^{\varphi} u \right\|_{L^p(\Omega)} \leqslant |1 - ab| \, \|P_F u\|_{L^p(\Omega)} \,,$$

and so we can recover L^p boundedness of the *square* projections.

Chapter 7

Conclusion

> 'Begin at the beginning,' the King said, very gravely, 'and go on till you come to the end: then stop.'
>
> —Lewis Carroll

We have certainly come to the end of this thesis, but will disobey the King's injunction to stop, for there remain many open questions to stimulate further research.

We have looked closer at the issue of convergence for some of the most natural domains and began the study of a more general procedure of perturbation of domains.

Our analysis in Chapter 3 shows that, for those triangles amenable to the method of reflection—*viz.* the 45-90-45 isosceles, the hemi-equilateral and the equilateral triangles—L^p convergence results carry over cleanly from the square for cubic truncations of the partial sums. We even saw that Fefferman's proof of the Ball Multiplier Theorem can be adapted to yield a similar negative result for $p \neq 2$ on the 45-90-45 triangle.

Lamé's theorem, on the other hand, raises a more serious and unsettling issue: for other triangles, we cannot expect to have trigonometric eigenfunctions at all!

Thus, the open questions that remain are

(i) Can we show that spherical truncations diverge for the equilateral triangle?

(ii) Can we find *some* truncation method that yields L^p convergence for other types of triangular domains?

The next natural candidate was the disc, whose eigenfunctions are the '*nearly* trigonometric' Bessel functions. We were unable to establish L^p convergence results on the disc, but were lead to a very interesting investigation of the decay rates of

Bessel–Fourier coefficients. Our results in Chapters 4 and 5, admittedly, are partial, but leave us with some exciting questions:

(i) Can we show that the kernels K_M^α are positive, as the numerics clearly suggest?

(ii) Can we improve Theorem 5.0.5 to $\alpha = \frac{1}{2}$? And can we find a way of constructing a function f whose coefficients decay like $o(\sqrt{m})$, but such that $f_{2m} = 0$ for all m?

And, of course, the main question:

(iii) Can we show convergence of the 2-dimensional Bessel–Fourier series in the $L^p(\mathbb{D}, r\,\mathrm{d}r\,\mathrm{d}\theta)$ for $4/3 < p < 4$?

The results in Chapter 6 are encouraging, as we were able to generalise the work of Lamberti and Lanza de Cristoforis (2005) from the Hilbert space setting H_0^1 to L^p, requiring no weak derivatives either. However, our results leave us with some of the questions which the researchers from Padova elicited in us:

(i) Can we obtain convergence results truncation methods that might split eigenvalues, such as the cubic truncations that work so well for the square and (some) triangles?

(ii) Can we eliminate the dependence on N of the constant in Theorem 6.5.1, so that we may use these continuity results to obtain convergence results?

Another question which we toyed with while working on this problem was:

(iii) Can we obtain an L^p continuity result for the *eigenfunctions* of the Dirichlet Laplacian as we deform Ω?

However, it is doubtful that such results would follow from an analysis like that of Chapter 6, since we deal with projections, which fix a normalisation of the eigenfunctions u_n, and so we probably cannot say much about $\|u_n\|_{L^p(\Omega)}$ given information about $\left\|P_{\{n\}}\right\|_{L^p(\Omega)\to L^p(\Omega)}$.

Appendix A

Translations of the epigraphs

Latin translations are my own. The quotation from Proclus, was taken from Glenn R. Morrow's 1970 translation of the *Commentary*, published by Princeton University Press. The aphorism from Goethe was taken from E. Stopp's 1998 (Penguin) translation of the *Maxims and Reflections*.

Acknowledgments From Cicero's speech in defence of Marcus Plancius:

> There is nothing that I would value more highly than to be, and be seen as being grateful. For this one virtue is not only the greatest, but also the mother of all other virtues. (XXXIII.80)

Chapter 1 From Euler's treatise *Institutiones Calculi Differentialis cum eius Vsu in Analysi Finitorum ac Doctrina Seriorum* (*Foundations of Differential Calculus with Applications to Finite Analysis and the Theory of Series*):

> Thus, these difficulties and apparent contradictions can be completely avoided if we give the term *sum* a different meaning from that usually attributed to it. (Chapter III)

Notations One of Goethe's *Maximen und Reflexionen* (Maxims and Reflections), which we reproduce with apologies to certain types of Frenchmen:

> Mathematicians are like a certain type of Frenchman: when you talk to them they translate it into their own language, and then it soon turns into something completely different. (1279)

Chapter 2 From Virgil's *Aeneid*, book I. After the storm stirred up by Juno has been placated by Neptune, the Trojan hero tries to cheer up his few surviving companions, by reminiscing about all the other difficulties they have surmounted hitherto, and muses

> Perhaps some day it will please these things to remember. (I.203)

Chapter 3 From Proclus' commentary to Euclid's *Elements*, commenting on Definition V ('A surface is what has length and breadth only'),

> The triangle is the premier of all rectilinear figures, as everyone can see, because it is determined by the number three and formed by it.

Chapter 4 Valerius Maximus tells us that, during the siege of Syracuse, Archimedes' last words to the Roman soldier who would become his murderer were: *noli, obsecro, istum disturbare* ('do not, I beseech you, disturb this'); see his *Facta et Dicta Memorabilia* (*Memorable Doings and Sayings*), VIII.7. The more common saying that made it into later Greek conjectured that 'this' was some circles, imagining Archimedes to have said

> Stay away from my circles!

Chapter 5 From Boethius' treatise *De Institutione Musica* (*On the Foundations of Music*):

> Nor must it be thought that a chord, each time that it is pressed, gives a single sound ... A single sound impinges on our senses, low or high, even though each is in fact composed of many. (I.3)

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